

Generalization error bounds for deep models

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Chapter 1

Introduction

This blueprint formalizes the paper *Why and When Deep is Better than Shallow: Implementation-Agnostic State-Transition Model of Deep Learning*. Each hidden layer is a self-map of a metric state space $\mathcal{X}$, the depth- k hidden class $B(k, F)$ is the word ball of the semigroup generated by a hidden-layer class F , and the depth- k hypothesis class is $\mathcal{H}_k = H \circ B(k, F)$ for an output-layer class H . The blueprint is generated from the `@[blueprint]` annotations in the Lean sources; the informal plan is kept in `PLAN.md`.

The general-purpose tools developed for this formalization (results that belong in `Mathlib`, and tools for `lean-rademacher`: contraction for arbitrary index sets, one-sided uniform-deviation bounds, Dudley’s entropy integral for sub-Gaussian processes, Hilbert-space Hoeffding bounds, Gaussian comparison and the Bernoulli–Sudakov minoration) now live in `lean-rademacher` (`FoML.ToMathlib`, `FoML.ToFoML`) and are used here as a dependency; they are not part of this blueprint.

Chapter 2

Setting

2.1 Word balls and the semigroup generated by the hidden layers

Definition 1. Let $F \subseteq \mathcal{X}^{\mathcal{X}}$ be a hidden-layer class. The depth- k hidden class is the word ball $B(k, F) = \{f_m \circ \dots \circ f_1 : 0 \leq m \leq k, f_i \in F\}$, with $B(0, F) = \{\text{id}\}$. It is defined recursively by $B(k+1, F) = B(k, F) \cup F \circ B(k, F)$.

Definition 2. The words of length exactly m : $F^m = \{f_m \circ \dots \circ f_1 : f_i \in F\}$, with $F^0 = \{\text{id}\}$ and $F^{m+1} = F \circ F^m$.

Definition 3. The semigroup (monoid) generated by F is $\langle F \rangle = \bigcup_{k \geq 0} B(k, F)$.

Theorem 4. $B(0, F) = \{\text{id}\}$.

Proof. □

Theorem 5. $B(k+1, F) = B(k, F) \cup \{g \circ f : g \in F, f \in B(k, F)\}$.

Proof. □

Theorem 6. $\text{id} \in B(k, F)$ for every k .

Proof. Induction on k : $\text{id} \in B(0, F)$ and $B(k, F) \subseteq B(k+1, F)$. □

Theorem 7. The word balls are increasing: $B(k, F) \subseteq B(k+1, F)$, hence $B(k, F) \subseteq B(l, F)$ for $k \leq l$.

Proof. □

Theorem 8. If $g \in F$ and $f \in B(k, F)$ then $g \circ f \in B(k+1, F)$.

Proof. □

Theorem 9. $B(k, F) \subseteq \langle F \rangle$ for every k .

Proof. □

Theorem 10. If $F \subseteq G$ then $B(k, F) \subseteq B(k, G)$.

Proof. Induction on k , using that $\circ$ -images are monotone in both arguments. $\square$

Theorem 11. *If $f \in B(k, F)$ and $g \in B(l, F)$ then $g \circ f \in B(l + k, F)$.*

Proof. Induction on l (the length of the word g). If $g = \text{id}$ this is $f \in B(k, F)$; if $g = a \circ b$ with $a \in F$, $b \in B(l, F)$ then $g \circ f = a \circ (b \circ f)$ with $b \circ f \in B(l + k, F)$ by induction. $\square$

Theorem 12. *$B(k + 1, F) = B(k, F) \cup F^{k+1}$: the word ball of radius $k + 1$ is the word ball of radius k together with the words of length exactly $k + 1$.*

Proof. Induction on k : $F \circ B(k + 1, F) = F \circ B(k, F) \cup F \circ F^{k+1}$ and $F \circ B(k, F) \subseteq B(k + 1, F)$. $\square$

Theorem 13. *$|F^m| \leq |F|^m$ (as extended natural numbers).*

Proof. Induction on k : F^{k+1} is the image of $F \times F^k$ under composition. $\square$

Theorem 14. *$|B(k, F)| \leq \sum_{m=0}^k |F|^m$ (as extended natural numbers).*

Proof. Induction on k using $B(k + 1, F) = B(k, F) \cup F^{k+1}$ and $|F^{k+1}| \leq |F|^{k+1}$. $\square$

Theorem 15. *If F is finite with $|F| = r \geq 2$ then $|B(k, F)| \leq \sum_{m=0}^k r^m \leq r^{k+1}$.*

Proof. Combine the previous bound with the geometric sum estimate $\sum_{m \leq k} r^m = (r^{k+1} - 1)/(r - 1) < r^{k+1}$ for $r \geq 2$. $\square$

2.2 Uniform and empirical metrics

Definition 16. The uniform distance on $\mathcal{X}^{\mathcal{X}}$ is $d_{\infty}(f, g) = \sup_{x \in \mathcal{X}} d(f(x), g(x)) \in [0, \infty]$.

Definition 17. The real-valued uniform distance $d_{\infty}(f, g)$, with the convention that it is 0 when $d_{\infty}(f, g) = \infty$.

Theorem 18. *$d(f(x), g(x)) \leq d_{\infty}(f, g)$ for every x .*

Proof. $\square$

Theorem 19. *Right composition is 1-Lipschitz for d_{∞} : $d_{\infty}(a \circ f, b \circ f) \leq d_{\infty}(a, b)$.*

Proof. $\square$

Theorem 20. *Left composition with a K -Lipschitz map f is K -Lipschitz for d_{∞} : $d_{\infty}(f \circ a, f \circ b) \leq K d_{\infty}(a, b)$.*

Proof. Pointwise, $d(f(a(x)), f(b(x))) \leq K d(a(x), b(x)) \leq K d_{\infty}(a, b)$. $\square$

Definition 21. The pseudo-metric space $(\mathcal{X}^{\mathcal{X}}, d_{\infty})$ of self-maps equipped with the uniform distance (a type synonym of $\mathcal{X}^{\mathcal{X}}$; in Lean it is Mathlib's ' $\mathbf{X} \rightarrow \mathbf{X}$ ', the function space with the uniform structure).

Definition 22. d_{∞} is a pseudo-metric on $\mathcal{X}^{\mathcal{X}}$ (it may take the value ∞). This is Mathlib's instance on ' $\mathbf{X} \rightarrow \mathbf{X}$ '.

Definition 23. The (identity) map $\mathcal{X}^{\mathcal{X}} \rightarrow (\mathcal{X}^{\mathcal{X}}, d_{\infty})$ viewing a self-map as a point of the pseudo-metric space.

Theorem 24. Viewing f in $(\mathcal{X}^{\mathcal{X}}, d_{\infty})$ does not change its values.

Proof. □

Theorem 25. On $(\mathcal{X}^{\mathcal{X}}, d_{\infty})$ the extended distance is d_{∞} .

Proof. □

Theorem 26. d_{∞} is the distance of $(\mathcal{X}^{\mathcal{X}}, d_{\infty})$: $\text{edist}(f, g) = d_{\infty}(f, g)$ for $f, g \in \mathcal{X}^{\mathcal{X}}$.

Proof. □

Theorem 27. $d(f(x), g(x)) \leq d_{\infty}(f, g)$ for $f, g \in (\mathcal{X}^{\mathcal{X}}, d_{\infty})$.

Proof. □

Definition 28. For a sample $S = (x_1, \dots, x_n)$ the empirical distance is $d_S(f, g) = (\frac{1}{n} \sum_{i=1}^n d(f(x_i), g(x_i))^2)^{1/2}$.

Definition 29. For real-valued u, v the empirical L^2 distance is $\|u - v\|_S = (\frac{1}{n} \sum_{i=1}^n |u(x_i) - v(x_i)|^2)^{1/2}$.

Definition 30. The empirical sup norm $\|u\|_{S, \infty} = \max_{i \leq n} |u(x_i)|$ (equal to 0 when $n = 0$).

Definition 31. The empirical diameter of a class F is $\text{diam}_S(F) = \sup_{f, g \in F} d_S(f, g)$.

Theorem 32. $d_S(f, f) = 0$.

Proof. □

Theorem 33. $d_S(f, g) = d_S(g, f)$.

Proof. □

Theorem 34. $d_S(f, g) \geq 0$.

Proof. □

Theorem 35. Minkowski's inequality for the Euclidean norm on $\mathbb{R}^n$: $(\sum_i (u_i + v_i)^2)^{1/2} \leq (\sum_i u_i^2)^{1/2} + (\sum_i v_i^2)^{1/2}$.

Proof. This is the triangle inequality in the Euclidean space $\ell^2(\{1, \dots, n\})$. □

Theorem 36. The empirical distance satisfies the triangle inequality: $d_S(f, h) \leq d_S(f, g) + d_S(g, h)$. Hence d_S is a pseudometric on $\mathcal{X}^{\mathcal{X}}$.

Proof. Pointwise $d(f(x_i), h(x_i)) \leq d(f(x_i), g(x_i)) + d(g(x_i), h(x_i))$, then apply Minkowski's inequality to the vectors of pointwise distances. □

Definition 37. The pseudometric space $(\mathcal{X}^{\mathcal{X}}, d_S)$ of self-maps equipped with the empirical distance of the sample S (a type synonym of $\mathcal{X}^{\mathcal{X}}$).

Definition 38. d_S is a pseudometric on $\mathcal{X}^{\mathcal{X}}$.

Theorem 39. On $(\mathcal{X}^{\mathcal{X}}, d_S)$ the distance is d_S .

Proof. □

Theorem 40. *If $d_\infty(f, g) < \infty$ then $d(f(x), g(x)) \leq d_\infty(f, g)$ as real numbers.*

Proof. □

Theorem 41. *The empirical distance is dominated by the uniform distance: $d_S(f, g) \leq d_\infty(f, g)$ for every sample S .*

Proof. If $d_\infty(f, g) = \infty$ there is nothing to prove. Otherwise each term satisfies $d(f(x_i), g(x_i))^2 \leq d_\infty(f, g)^2$, so the average is at most $d_\infty(f, g)^2$ and we take square roots. □

2.3 Covering and packing numbers

Definition 42. The metric entropy of A at scale ε is $\log N(A, \varepsilon)$, where $N(A, \varepsilon)$ is the (internal) covering number of A by closed ε -balls (with the convention $\log \infty = \log 0 = 0$ in Lean).

2.4 Hypothesis classes and implementation error

Definition 43. For $H \subseteq \mathbb{R}^X$ and $F \subseteq \mathcal{X}^X$, $H \circ F = \{h \circ f : h \in H, f \in F\}$.

Definition 44. The depth- k hypothesis class is $\mathcal{H}_k = H \circ B(k, F)$.

Definition 45. The implementation error of an implementation map ι on a class $\mathcal{H}$ is $\varepsilon_{\text{imp}} = \sup_{f \in \mathcal{H}} \|f - \iota f\|_\infty$.

Theorem 46. *If $F \subseteq F'$ then $H \circ F \subseteq H \circ F'$.*

Proof. □

Theorem 47. $H \circ \{\text{id}\} = H$.

Proof. □

Theorem 48. $\mathcal{H}_0 = H \circ \{\text{id}\} = H$.

Proof. □

Theorem 49. *The hypothesis classes are increasing in depth: $\mathcal{H}_k \subseteq \mathcal{H}_{k+1}$.*

Proof. □

2.5 Loss, risk, and empirical minimizers

Definition 50. A loss is a function $\ell : \mathbb{R} \times \mathcal{Y} \rightarrow [0, b]$ which is β_ℓ -Lipschitz in its first argument: $|\ell(a, y) - \ell(a', y)| \leq \beta_\ell |a - a'|$.

Definition 51. The empirical risk on a sample $D = ((x_1, y_1), \dots, (x_n, y_n))$ is $\hat{L}[f] = \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i)$.

Definition 52. The (population) risk under a distribution P on $\mathcal{X} \times \mathcal{Y}$ is $L[f] = \mathbb{E}_{(X, Y) \sim P} \ell(f(X), Y)$.

Definition 53. f is an η -empirical minimizer over $\mathcal{H}$ if $f \in \mathcal{H}$ and $\hat{L}[f] \leq \inf_{g \in \mathcal{H}} \hat{L}[g] + \eta$.

Definition 54. The model error (approximation term) of $\mathcal{H}$ relative to a target class $\mathcal{C}$ is $\varepsilon_{\text{model}} = \inf_{f \in \mathcal{H}} L[f] - \inf_{c \in \mathcal{C}} L[c]$.

2.6 Rademacher complexity

Definition 55. The empirical Rademacher complexity of $G \subseteq \mathbb{R}^{\mathcal{X}}$ on the sample $S = (x_1, \dots, x_n)$ is $\hat{\mathfrak{R}}_S(G) = \mathbb{E}_\sigma \sup_{g \in G} \frac{1}{n} \sum_{i=1}^n \sigma_i g(x_i) = 2^{-n} \sum_{\sigma \in \{\pm 1\}^n} \sup_{g \in G} \frac{1}{n} \sum_{i=1}^n \sigma_i g(x_i)$ (no absolute value). In Lean this is FoML's ‘`empiricalRademacherComplexity_without_abs`’ for the class indexed by G itself.

Definition 56. The (population) Rademacher complexity is $\mathfrak{R}_n(G) = \mathbb{E}_{S \sim P^{\otimes n}} \hat{\mathfrak{R}}_S(G)$. In Lean we take FoML's ‘`rademacherComplexity`’, i.e. the expectation over $S \sim P^{\otimes n}$ of the absolute version $\mathbb{E}_\sigma \sup_{g \in G} \left| \frac{1}{n} \sum_i \sigma_i g(x_i) \right| \geq \hat{\mathfrak{R}}_S(G)$; this is the quantity for which FoML's deviation bounds are stated, and it dominates the paper's $\mathfrak{R}_n(G)$.

Theorem 57. *If $G_1 \subseteq G_2$, $G_1 \neq \emptyset$ and the Rademacher averages $\{\frac{1}{n} \sum_i \sigma_i g(x_i) : g \in G_2\}$ are bounded above for every sign pattern σ , then $\hat{\mathfrak{R}}_S(G_1) \leq \hat{\mathfrak{R}}_S(G_2)$.*

Proof. Compare the suprema termwise for each sign pattern σ : every element of G_1 is an element of G_2 . $\square$

Theorem 58. *If $|g(x_i)| \leq C$ for all $g \in G$ and $i \leq n$, then the one-sided empirical Rademacher complexity is dominated by the absolute one: $\mathfrak{R}_S(G) \leq \mathbb{E}_\sigma \sup_{g \in G} \left| \frac{1}{n} \sum_i \sigma_i g(x_i) \right|$.*

Proof. $\square$

2.7 The entropy integral

Definition 59. The entropy integral of A up to scale D is $\mathbb{V}(D, A) = \int_0^D \sqrt{\log N(A, \varepsilon)} d\varepsilon$. In the paper, $\mathbb{V}_k(S) = \int_0^{D_k(S)} \sqrt{\log N(B(k, F), d_S, \varepsilon)} d\varepsilon$ with $D_k(S) = \text{diam}_S B(k, F)$ is $\mathbb{V}(\text{diam}_S B(k, F), B(k, F))$ for the empirical pseudometric d_S .

2.8 Assumptions and auxiliary definitions of the main theorems

Definition 60. A class $\mathcal{H} \subseteq \mathbb{R}^{\mathcal{X}}$ is (sup-norm) separable if it has a countable subset $\mathcal{D} \subseteq \mathcal{H}$ which is dense for the uniform norm: for every $f \in \mathcal{H}$ and $\varepsilon > 0$ there is $g \in \mathcal{D}$ with $\sup_x |f(x) - g(x)| \leq \varepsilon$.

Definition 61. For a sample $S = (x_1, \dots, x_n)$, an output-layer class H and a hidden map f , the hidden-indexed process is $Z_f(\sigma) = \sup_{h \in H} \frac{1}{n} \sum_{i=1}^n \sigma_i h(f(x_i))$, $\sigma \in \{\pm 1\}^n$.

Definition 62. Sub-Gaussian output-layer increments. Let $\mathfrak{F}$ be a hidden-layer class and A_H, L constants. For all $f, g \in \mathfrak{F}$ and $t > 0$,

$$\mathbb{P}_\sigma(|Z_f - Z_g| > t) \leq 2 \exp\left(-\frac{nt^2}{2A_H^2 L^2 d_S(f, g)^2}\right),$$

where $\mathbb{P}_\sigma$ is the uniform distribution on $\{\pm 1\}^n$; and when $d_S(f, g) = 0$ the requirement is $Z_f = Z_g$ (for every σ), which is the limiting interpretation of the display (in Lean the display with $d_S(f, g) = 0$ reads $\dots \leq 2 \exp(0)$, hence the separate conjunct).

Definition 63. For a Hilbert space $\mathcal{H}$ and a feature map $\Phi : \mathcal{X} \rightarrow \mathcal{H}$, the norm-bounded linear output-layer class is $H_\Phi = \{x \mapsto \langle w, \Phi(x) \rangle : \|w\| \leq 1\}$.

Definition 64. Output-layer realization of hidden geometry. For a sample S , an output-layer class H , a hidden class B_k and constants κ, R_{out} : there is, for each $g \in B_k$, an output layer $h_g \in H$ such that the map $\Psi_k(g) = h_g \circ g$ satisfies $\|\Psi_k(g) - \Psi_k(g')\|_S \geq \kappa d_S(g, g')$ for $g, g' \in B_k$ and $\|\Psi_k(g)\|_{S, \infty} \leq R_{\text{out}}$ for $g \in B_k$.

Definition 65. The pseudo-emetric space $(\mathbb{R}^{\mathcal{X}}, \|\cdot\|_\infty)$ of real-valued functions with the uniform distance $\|u - v\|_\infty = \sup_x |u(x) - v(x)| \in [0, \infty]$ (a type synonym of $\mathbb{R}^{\mathcal{X}}$; in Lean Mathlib's ' $\mathbf{X} \rightarrow \cdot$ ').

Definition 66. $\|\cdot\|_\infty$ is a pseudo-emetric on $\mathbb{R}^{\mathcal{X}}$.

Theorem 67. On $(\mathbb{R}^{\mathcal{X}}, \|\cdot\|_\infty)$ the extended distance is $\sup_x \text{edist}(u(x), v(x))$.

Proof.

□

Definition 68. Uniform output-layer regularity. The output-layer class $H \subseteq \mathbb{R}^{\mathcal{X}}$ has finite uniform covering numbers $N(H, \|\cdot\|_\infty, u) < \infty$ for all $u > 0$, and there are constants B_H, L_H with $\|h\|_\infty \leq B_H$ and $|h(x) - h(x')| \leq L_H d(x, x')$ for all $h \in H$.

Definition 69. Uniform transition covering. The hidden-layer class $F \subseteq \mathcal{X}^{\mathcal{X}}$ has finite uniform covering numbers $N(F, d_\infty, v) < \infty$ for all $v > 0$.

Definition 70. The (identity) map $\mathbb{R}^{\mathcal{X}} \rightarrow (\mathbb{R}^{\mathcal{X}}, \|\cdot\|_\infty)$ viewing a real-valued function as a point of the pseudo-emetric space.

Theorem 71. $\text{edist}(u, v) = \sup_x \text{edist}(u(x), v(x))$ in $(\mathbb{R}^{\mathcal{X}}, \|\cdot\|_\infty)$.

Proof.

□

Definition 72. The root entropy of the output-layer class at scale u : $\mathcal{E}_H(u) = \sqrt{\log N^{\text{ext}}(H, \|\cdot\|_\infty, u)}$ (with $u \mapsto \max(u, 0)$ and $\log \infty = \log 0 = 0$ in Lean).

Definition 73. The root entropy of the hidden class at scale v : $\mathcal{E}_F(v) = \sqrt{\log N^{\text{ext}}(F, d_\infty, v)}$.

Definition 74. The pseudometric space $(\mathbb{R}^{\mathcal{X}}, \|\cdot\|_S)$ of real-valued functions equipped with the empirical L^2 distance $\|u - v\|_S = \left(\frac{1}{n} \sum_i |u(x_i) - v(x_i)|^2\right)^{1/2}$ of the sample S (a type synonym of $\mathbb{R}^{\mathcal{X}}$).

Definition 75. $\|\cdot\|_S$ is a pseudometric on $\mathbb{R}^{\mathcal{X}}$. In Lean this is FoML's 'empiricalPMet S' (whose distance is FoML's 'empiricalDist S').

Definition 76. The (identity) map $\mathbb{R}^{\mathcal{X}} \rightarrow (\mathbb{R}^{\mathcal{X}}, \|\cdot\|_S)$ viewing a real-valued function as a point of the pseudometric space.

Theorem 77. On $(\mathbb{R}^{\mathcal{X}}, \|\cdot\|_S)$ the distance is FoML's 'empiricalDist S'.

Proof.

□

Theorem 78. On $(\mathbb{R}^{\mathcal{X}}, \|\cdot\|_S)$ the distance is $\|u - v\|_S$.

Proof.

□

Definition 79. The set of points reached by the sample through the hidden class: $\mathcal{R}_S(F) = \{f(x_i) : f \in F, 1 \leq i \leq n\} \subseteq \mathcal{X}$.

Theorem 80. $f(x_i) \in \mathcal{R}_S(F)$ for $f \in F$.

Proof. □

Definition 81. The uniform pushed-forward covering number of the output-layer class: $N_{S,F}(H, u) = \sup_{f \in F} N^{\text{ext}}(H, \|\cdot\|_{f \circ S}, u)$, the largest external covering number of H in the empirical metric of a pushed-forward sample $f \circ S = (f(x_1), \dots, f(x_n))$, $f \in F$. (Any uniform bound $N^{\text{ext}}(H, \|\cdot\|_{f \circ S}, u) \leq \bar{N}(u)$ for all $f \in F$ dominates it.)

Definition 82. Sample output-layer regularity. For a sample S and a hidden class F , the output-layer class $H \subseteq \mathbb{R}^{\mathcal{X}}$ has finite uniform pushed-forward covering numbers $N_{S,F}(H, u) < \infty$ for all $u > 0$, and there are constants B_H, L_H such that every $h \in H$ satisfies $|h(x)| \leq B_H$ and $|h(x) - h(x')| \leq L_H d(x, x')$ for all $x, x' \in \mathcal{R}_S(F)$ (boundedness and the Lipschitz estimate are only required on the points reached by the sample; compare `ass:ent-readout`, where both hold on all of $\mathcal{X}$ and H is covered in the sup norm).

Definition 83. Sample transition covering. The hidden-layer class $F \subseteq \mathcal{X}^{\mathcal{X}}$ has finite covering numbers in the empirical metric: $N^{\text{ext}}(F, d_S, v) < \infty$ for all $v > 0$.

Definition 84. The root entropy of the output-layer class at scale u on the sample: $\mathcal{E}_{H,S}(u) = \sqrt{\log N_{S,F}(H, u)}$ (with $u \mapsto \max(u, 0)$ and $\log \infty = \log 0 = 0$ in Lean).

Definition 85. The root entropy of the hidden class at scale v on the sample: $\mathcal{E}_{F,S}(v) = \sqrt{\log N^{\text{ext}}(F, d_S, v)}$.

Definition 86. The identity map $(\mathcal{X}^{\mathcal{X}}, d_{\infty}) \rightarrow (\mathcal{X}^{\mathcal{X}}, d_S)$.

Definition 87. The depth-dependent part of the estimation term is

$$\text{var}(k, n) := \frac{12A_H L}{\sqrt{n}} \mathbf{V}_k(S), \quad \mathbf{V}_k(S) = \int_0^{D_k(S)} \sqrt{\log N(B(k, F), d_S, \varepsilon)} d\varepsilon,$$

with $D_k(S) = \text{diam}_S B(k, F)$.

Chapter 3

Implementation-agnostic generalization bounds

3.1 Implementation-free bias–variance decomposition

Theorem 88. *If ℓ is β_ℓ -Lipschitz in its first argument (with $\beta_\ell \geq 0$), f, g are measurable and $\sup_x |f(x) - g(x)| \leq c$, then $|L[f] - L[g]| \leq \beta_\ell c$.*

Proof. Both loss integrands are bounded and measurable, hence integrable; the difference of the integrals is the integral of the pointwise difference, which is bounded by $\beta_\ell |f(x) - g(x)| \leq \beta_\ell c$. $\square$

Theorem 89. *If ℓ is β_ℓ -Lipschitz in its first argument (with $\beta_\ell \geq 0$) and $\sup_x |f(x) - g(x)| \leq c$ with $c \geq 0$, then $|\hat{L}[f] - \hat{L}[g]| \leq \beta_\ell c$ for every sample.*

Proof. Termwise bound $|\ell(f(x_i), y_i) - \ell(g(x_i), y_i)| \leq \beta_\ell c$ and average. $\square$

Theorem 90. Deterministic excess-risk bound. *Fix a sample $\mathcal{D}$ and suppose $|L[f] - \hat{L}[f]| \leq \Delta_0$ for every $f \in \mathcal{H}$. If $d_T(\iota f, f) \leq \varepsilon_{\text{imp}}$ and $|L[\iota f] - L[f]| \leq \beta_L d_T(\iota f, f)$ for all $f \in \mathcal{H}$ ($\beta_L \geq 0$), then every η -empirical minimizer $\hat{f} \in \mathcal{H}$ satisfies, with $\hat{h} = \iota(\hat{f})$, $L[\hat{h}] - \inf_{\mathcal{C}} L \leq \beta_L \varepsilon_{\text{imp}} + \varepsilon_{\text{model}} + \eta + 2\Delta_0$.*

Proof. $L[\hat{h}] \leq L[\hat{f}] + \beta_L \varepsilon_{\text{imp}}$ and, for every $f \in \mathcal{H}$, $L[\hat{f}] \leq \hat{L}[\hat{f}] + \Delta_0 \leq \hat{L}[f] + \eta + \Delta_0 \leq L[f] + \eta + 2\Delta_0$; take the infimum over f . $\square$

Theorem 91. Deterministic gap bound. *Fix a sample $\mathcal{D}$ and suppose $|L[f] - \hat{L}[f]| \leq \Delta_0$ for every $f \in \mathcal{H}$. Under the implementation hypotheses ($d_T(\iota f, f) \leq \varepsilon_{\text{imp}}$, $|L[\iota f] - L[f]| \leq \beta_L d_T(\iota f, f)$, $|\hat{L}[\iota f] - \hat{L}[f]| \leq \beta_{\hat{L}} d_T(\iota f, f)$), every $f \in \mathcal{H}$ satisfies $L[\iota f] - \hat{L}[\iota f] \leq (\beta_L + \beta_{\hat{L}})\varepsilon_{\text{imp}} + \Delta_0$.*

Proof. $\square$

Theorem 92. *The one-sided empirical Rademacher complexity of the class $\{(x, y) \mapsto f(x) : f \in \mathcal{H}\}$ on the labelled sample $\mathcal{D}$ equals $\hat{\mathfrak{R}}_S(\mathcal{H})$ on the inputs $S = (x_1, \dots, x_n)$.*

Proof. $\square$

Theorem 93. One-sided Rademacher complexity of the loss class. Let $\mathcal{H} \neq \emptyset$ be pointwise bounded and $\ell : \mathbb{R} \times \mathcal{Y} \rightarrow \mathbb{R}$ be β_ℓ -Lipschitz in its first argument ($\beta_\ell \geq 0$). Then on every sample $\mathcal{D} = ((x_i, y_i))_i$ with $S = (x_i)_i$,

$$\hat{\mathfrak{R}}_{\mathcal{D}}(\ell \circ \mathcal{H}) \leq \beta_\ell \hat{\mathfrak{R}}_S(\mathcal{H}) \quad \text{and} \quad \hat{\mathfrak{R}}_{\mathcal{D}}(-\ell \circ \mathcal{H}) \leq \beta_\ell \hat{\mathfrak{R}}_S(\mathcal{H}),$$

where $\hat{\mathfrak{R}}$ is the one-sided (no absolute value) empirical Rademacher complexity. *Proof:* the one-sided contraction `lem:contraction-without-abs` applied to $\psi(z, u) = \pm \ell(u, y)$, which is β_ℓ -Lipschitz in u (no vanishing condition at $u = 0$ is needed).

Proof. □

Theorem 94. Uniform deviation for a sup-separable Lipschitz-loss class. Let $n \geq 1$, $\mathcal{H}$ be a sup-norm separable, pointwise bounded class of measurable functions, $\ell : \mathbb{R} \times \mathcal{Y} \rightarrow [0, b]$ measurable and β_ℓ -Lipschitz in its first argument ($b > 0$, $\beta_\ell \geq 0$), $\delta \in (0, 1)$. Then with probability at least $1 - \delta$ over $\mathcal{D} \sim P^{\otimes n}$,

$$\sup_{f \in \mathcal{H}} |L[f] - \hat{L}[f]| \leq 2\beta_\ell \hat{\mathfrak{R}}_S(\mathcal{H}) + 3b\sqrt{\frac{2\log(4/\delta)}{n}}.$$

Proof: apply the two-sided bound `lem:two-sided-tail-empirical` (one-sided symmetrization for $\ell \circ \mathcal{H}$ and $-\ell \circ \mathcal{H}$) with $C(\mathcal{D}) = \beta_\ell \hat{\mathfrak{R}}_S(\mathcal{H})$ from `lem:bv-loss-rademacher` and $\varepsilon = b\sqrt{2\log(4/\delta)/n}$, so that $4\exp(-n\varepsilon^2/(2b^2)) = \delta$; the topology on $\mathcal{H}$ is the uniform-convergence topology, which is separable and first countable by `lem:sup-dense-separableSpace`, and evaluations are continuous.

Proof. □

Theorem 95. General bias–variance decomposition (excess risk). Let $\mathcal{H}$ be a sup-norm separable, pointwise bounded class of measurable functions $\mathcal{X} \rightarrow \mathbb{R}$, $\mathcal{C}$ a nonempty class of measurable benchmark functions, and $\ell : \mathbb{R} \times \mathcal{Y} \rightarrow [0, b]$ measurable and β_ℓ -Lipschitz in its first argument ($b > 0$, $\beta_\ell \geq 0$). Let ι be an implementation map, d_T a nonnegative function (pseudo-metric) and assume, for constants $\beta_L, \beta_{\hat{L}} \geq 0$ and $\varepsilon_{\text{imp}} \geq 0$ (any upper bound for $\sup_{f \in \mathcal{H}} d_T(\iota f, f)$), that for every $f \in \mathcal{H}$: $d_T(\iota f, f) \leq \varepsilon_{\text{imp}}$, $|L[\iota f] - L[f]| \leq \beta_L d_T(\iota f, f)$ and $|\hat{L}[\iota f] - \hat{L}[f]| \leq \beta_{\hat{L}} d_T(\iota f, f)$ for every sample. Let $n \geq 1$, $\eta \geq 0$ and $\delta \in (0, 1)$. Then with probability at least $1 - \delta$ over $\mathcal{D} \sim P^{\otimes n}$, every η -empirical minimizer $\hat{f} \in \mathcal{H}$ satisfies, with $\hat{h} := \iota(\hat{f})$,

$$L[\hat{h}] - \inf_{c \in \mathcal{C}} L[c] \leq \beta_L \varepsilon_{\text{imp}} + \varepsilon_{\text{model}} + \eta + 4\beta_\ell \hat{\mathfrak{R}}_S(\mathcal{H}) + 6b\sqrt{\frac{2\log(4/\delta)}{n}}.$$

The Rademacher constant $4\beta_\ell$ is the paper’s; the deviation constant is explicit ($Cb\sqrt{\log(1/\delta)/n}$ in the paper, with unspecified universal C , becomes $6b\sqrt{2\log(4/\delta)/n}$, from `lem:bv-uniform-deviation-onesided`). The pointwise boundedness of $\mathcal{H}$ makes $\hat{\mathfrak{R}}_S(\mathcal{H})$ a genuine supremum.

Proof. On the complement of the bad event of `lem:bv-uniform-deviation-onesided`, apply `lem:bv-deterministic` with $\Delta_0 = 2\beta_\ell \hat{\mathfrak{R}}_S(\mathcal{H}) + 3b\sqrt{2\log(4/\delta)/n}$. □

Theorem 96. General bias–variance decomposition (generalization gap). Under the hypotheses of `thm:bv-general`, with probability at least $1 - \delta$ over $\mathcal{D} \sim P^{\otimes n}$, every $f \in \mathcal{H}$ (in particular every η -empirical minimizer $\hat{f}$) satisfies, with $h := \iota(f)$,

$$L[h] - \hat{L}[h] \leq (\beta_L + \beta_{\hat{L}})\varepsilon_{\text{imp}} + 2\beta_\ell \hat{\mathfrak{R}}_S(\mathcal{H}) + 3b\sqrt{\frac{2\log(4/\delta)}{n}}.$$

(The paper states this for the η -empirical minimizer only, with $2\beta_\ell \hat{\mathfrak{R}}_S(\mathcal{H}) + Cb\sqrt{\log(1/\delta)/n}$; its proof gives it for every $f \in \mathcal{H}$, which is what we state. The deviation constant is explicit, see `thm:bv-general`.)

Proof. □

Theorem 97. If $\varepsilon_{\text{imp}} = \sup_{f \in \mathcal{H}} \|f - \iota f\|_\infty \leq \varepsilon$ with $\varepsilon \geq 0$, then $|\iota f(x) - f(x)| \leq \varepsilon$ for all $f \in \mathcal{H}$, $x \in \mathcal{X}$.

Proof. □

Theorem 98. Implementation-free bias-variance decomposition. Fix $k \geq 0$, $\eta \geq 0$, $n \geq 1$. Assume $\mathcal{H}_k = H \circ B(k, F)$ consists of measurable functions, is sup-norm separable and pointwise bounded, $\mathcal{C}$ is a nonempty class of measurable benchmarks, the loss $\ell : \mathbb{R} \times \mathcal{Y} \rightarrow [0, b]$ is measurable and β_ℓ -Lipschitz in its first argument ($b > 0$, $\beta_\ell \geq 0$), the implementation map ι produces measurable functions, and $\varepsilon_{\text{imp}}(k) = \sup_{f \in \mathcal{H}_k} \|f - \iota f\|_\infty \leq \varepsilon_{\text{imp}}$ for a real $\varepsilon_{\text{imp}} \geq 0$. Then for every $\delta \in (0, 1)$, with probability at least $1 - \delta$ over $\mathcal{D} \sim P^{\otimes n}$, every η -empirical minimizer $\hat{f} \in \mathcal{H}_k$ satisfies, with $\hat{h} := \iota(\hat{f})$,

$$L[\hat{h}] - \inf_{c \in \mathcal{C}} L[c] \leq \beta_\ell \varepsilon_{\text{imp}} + \varepsilon_{\text{model}}(k) + \eta + 4\beta_\ell \hat{\mathfrak{R}}_S(\mathcal{H}_k) + 6b\sqrt{\frac{2\log(4/\delta)}{n}}.$$

This is `thm:bv-general` with $d_T(f, g) = \|f - g\|_\infty$ and $\beta_L = \beta_{\hat{L}} = \beta_\ell$ (the loss is β_ℓ -Lipschitz); the Rademacher constant $4\beta_\ell$ is the paper's and the deviation constant is explicit (as in `thm:bv-general`); $\mathcal{H}_k$ is assumed pointwise bounded.

Proof. Apply the general theorem with $d_T(f, g) = \sup_x |f(x) - g(x)|$; the Lipschitz transfer lemmas give $\beta_L = \beta_{\hat{L}} = \beta_\ell$. □

Theorem 99. Implementation-free generalization gap. Under the hypotheses of `thm:bv`, with probability at least $1 - \delta$ over $\mathcal{D} \sim P^{\otimes n}$, every $f \in \mathcal{H}_k$ (in particular every η -empirical minimizer) satisfies, with $h := \iota(f)$,

$$L[h] - \hat{L}[h] \leq 2\beta_\ell \varepsilon_{\text{imp}} + 2\beta_\ell \hat{\mathfrak{R}}_S(\mathcal{H}_k) + 3b\sqrt{\frac{2\log(4/\delta)}{n}}.$$

(The Rademacher constant $2\beta_\ell$ is the paper's; the deviation constant is explicit, as in `thm:bv-general-gap`.)

Proof. □

3.2 Hidden-output decomposition

Theorem 100. Let $F \neq \emptyset$ and fix a sign pattern σ . If for every $f \in F$ the Rademacher averages $\{\frac{1}{n} \sum_i \sigma_i h(f(x_i)) : h \in H\}$ are bounded above, and $f \mapsto Z_f(\sigma)$ is bounded above on F , then $\sup_{u \in H \circ F} \frac{1}{n} \sum_i \sigma_i u(x_i) = \sup_{f \in F} Z_f(\sigma)$.

Proof. □

Theorem 101. For a single hidden map f_0 with bounded Rademacher averages, $\hat{\mathfrak{R}}_S(H \circ \{f_0\}) = \mathbb{E}_\sigma Z_{f_0}(\sigma)$.

Proof. □

Theorem 102. *Hidden-output decomposition under a sub-Gaussian increment condition.* Let $n \geq 1$, let $F \subseteq \mathcal{X}^X$ be totally bounded in d_S , let $f_0 \in F$ be an arbitrary anchor, and suppose Assumption `ass:sg-increment-main` holds on $\mathfrak{F} \supseteq F$ with constants $A_H, L > 0$. (We also assume that the suprema defining $Z_f(\sigma)$, $f \in F$, are finite, i.e. the Rademacher averages over H are bounded above; this is implicit in the paper.) Then

$$\hat{\mathfrak{R}}_S(H \circ F) \leq \hat{\mathfrak{R}}_S(H \circ \{f_0\}) + \frac{12A_H L}{\sqrt{n}} \int_0^{\text{diam}_S(F)} \sqrt{\log N(F, d_S, \varepsilon)} d\varepsilon,$$

where $\hat{\mathfrak{R}}_S(H \circ \{f_0\}) = \mathbb{E}_\sigma Z_{f_0}(\sigma)$ (`lem:emp-rademacher-comp-singleton`). This generalizes the paper, which assumes $\text{id} \in F$ and anchors at $f_0 = \text{id}$, giving $\hat{\mathfrak{R}}_S(H)$ on the right-hand side (`thm:hidden-decomp-id`); the anchor is arbitrary because the Dudley bound `thm:dudley-subgaussian-finite-space` is anchored at any point. We assume in addition that the entropy integrand $\varepsilon \mapsto \sqrt{\log N(F, d_S, \varepsilon)}$ is integrable on $[0, \text{diam}_S(F)]$; the paper's bound is trivially true with right-hand side $+\infty$ otherwise, whereas Lean's Bochner integral of a non-integrable function is 0.

Proof. □

Theorem 103. *Hidden-output decomposition, anchored at the identity (the paper's form).* Under the hypotheses of `thm:hidden-decomp` with $\text{id} \in F$,

$$\hat{\mathfrak{R}}_S(H \circ F) \leq \hat{\mathfrak{R}}_S(H) + \frac{12A_H L}{\sqrt{n}} \int_0^{\text{diam}_S(F)} \sqrt{\log N(F, d_S, \varepsilon)} d\varepsilon.$$

Proof: `thm:hidden-decomp` with $f_0 = \text{id}$ and $H \circ \{\text{id}\} = H$.

Proof. □

Theorem 104. If Assumption `ass:sg-increment-main` holds on the semigroup $\langle F_0 \rangle$ with constants (A_H, L) , then for every depth $k \geq 0$ with $B(k, F_0)$ totally bounded in d_S ,

$$\hat{\mathfrak{R}}_S(\mathcal{H}_k) \leq \hat{\mathfrak{R}}_S(H) + \frac{12A_H L}{\sqrt{n}} \int_0^{\text{diam}_S(B(k, F_0))} \sqrt{\log N(B(k, F_0), d_S, \varepsilon)} d\varepsilon,$$

with the same constants A_H, L (assuming, as in `thm:hidden-decomp`, that the entropy integrand of $B(k, F_0)$ is integrable on $[0, \text{diam}_S(B(k, F_0))]$).

Proof. $\text{id} \in B(k, F_0) \subseteq \langle F_0 \rangle$ and $\mathcal{H}_k = H \circ B(k, F_0)$. □

Theorem 105. If $h(f(x_i)) = h(g(x_i))$ for all $h \in H$ and $i \leq n$, then $Z_f(\sigma) = Z_g(\sigma)$ for every σ .

Proof. □

Theorem 106. If $d_S(f, g) = 0$ then $d(f(x_i), g(x_i)) = 0$ for every $i \leq n$.

Proof. □

Theorem 107. $n d_S(f, g)^2 = \sum_{i=1}^n d(f(x_i), g(x_i))^2$.

Proof. □

Theorem 108. If φ is Lipschitz into a metric space and $d_S(f, g) = 0$, then $\varphi(f(x_i)) = \varphi(g(x_i))$ for every $i \leq n$.

Proof. □

Theorem 109. *If φ is L -Lipschitz then $\sum_{i=1}^n d(\varphi(f(x_i)), \varphi(g(x_i)))^2 \leq L^2 n d_S(f, g)^2$.*

Proof. □

Theorem 110. *A counting fraction on $\{\pm 1\}^n$ is at most 1: $\#\{\sigma : p(\sigma)\}/2^n \leq 1$.*

Proof. □

Theorem 111. $\frac{1}{n} \sum_k \sigma_k \langle w, u_k \rangle = \langle w, \frac{1}{n} \sum_k \sigma_k u_k \rangle$.

Proof. □

Theorem 112. ***Hilbert output layers satisfy the increment condition, given the vector Hoeffding inequality.** Let $\mathcal{H}$ be a real inner product space and $\Phi : \mathcal{X} \rightarrow \mathcal{H}$ be L -Lipschitz. Assume the Rademacher tail bound `lem:rademacher-hilbert-tail` holds in $\mathcal{H}$ for samples of size n : $\mathbb{P}_\sigma(\|\sum_i \sigma_i v_i\| > t) \leq 2 \exp(-t^2/(2 \sum_i \|v_i\|^2))$. Then Assumption `ass:sg-increment-main` holds for $H = H_\Phi$ with $A_H = 1$ and the same L , for every hidden-layer class $\mathfrak{F}$. *Proof:* $|Z_f - Z_g| \leq \sup_{\|w\| \leq 1} |\langle w, \frac{1}{n} \sum_i \sigma_i v_i \rangle| = \|\frac{1}{n} \sum_i \sigma_i v_i\|$ with $v_i = \Phi(f(x_i)) - \Phi(g(x_i))$, and $\sum_i \|v_i\|^2 \leq L^2 n d_S(f, g)^2$.*

Proof. □

Theorem 113. ***Hilbert output layers satisfy the increment condition.** Let $\mathcal{H}$ be a real Hilbert space and $\Phi : \mathcal{X} \rightarrow \mathcal{H}$ be L -Lipschitz. Then Assumption `ass:sg-increment-main` holds for $H = H_\Phi$ with $A_H = 1$ (and the same L), for every hidden-layer class $\mathfrak{F}$.*

Proof. Combine `prop:hilbert-sg-of-tail` with the vector Hoeffding inequality `lem:rademacher-hilbert-tail` (the only unproved ingredient). □

Theorem 114. ***One-dimensional Hilbert output layers.** Let $\Phi : \mathcal{X} \rightarrow \mathbb{R}$ be L -Lipschitz and $H_\Phi = \{x \mapsto w \Phi(x) : |w| \leq 1\}$. Then Assumption `ass:sg-increment-main` holds for H_Φ with $A_H = 1$ and the same L (this is `prop:hilbert-sg` for $\mathcal{H} = \mathbb{R}$, fully proved from the real Hoeffding inequality).*

Proof. □

Theorem 115. ***Finite Lipschitz scalar output layers.** Let $H = \{h_1, \dots, h_m\}$ be a finite class of real-valued functions on $\mathcal{X}$, each L -Lipschitz. Then Assumption `ass:sg-increment-main` holds with $A_H = (1 + \log m / \log 2)^{1/2}$, for every hidden-layer class $\mathfrak{F}$.*

Proof. $|Z_f - Z_g| \leq \max_j |S_j|$ with $S_j = \frac{1}{n} \sum_i \sigma_i (h_j(f(x_i)) - h_j(g(x_i)))$; real Hoeffding for each S_j (`lem:rademacher-real-tail`), a union bound over the m functions, and `lem:union-bound-absorb` to absorb m into A_H . □

3.3 A conditional Sudakov-type converse

Theorem 116. *If $G \neq \emptyset$ and for every sign pattern σ the Rademacher averages $\{\frac{1}{n} \sum_i \sigma_i g(x_i) : g \in G\}$ are bounded above, then $\hat{\mathfrak{R}}_S(G) \geq 0$.*

Proof. Apply `lem:rademacher-avg-sup-nonneg` to the family $(g(x_i))_i, g \in G$. □

Theorem 117. *If $|g(x_i)| \leq C$ for all $g \in G$ and $i \leq n$, then for every sign pattern σ the Rademacher averages $\{\frac{1}{n} \sum_i \sigma_i g(x_i) : g \in G\}$ are bounded above (by $|C|$).*

Proof. □

Theorem 118. Conditional Sudakov-type lower bound. *There is a universal constant $c > 0$ such that the following holds. Let $(\mathcal{X}, d)$ be a pseudometric space, $S = (x_1, \dots, x_n)$ a sample with $n \geq 1$, H an output-layer class, $B \subseteq \mathcal{X}^{\mathcal{X}}$ an arbitrary hidden class, and suppose Assumption `ass:readout-realization-main` holds for B with constants $\kappa, R_{\text{out}} > 0$. Then for every $\varepsilon > 0$,*

$$\hat{\mathfrak{R}}_S(H \circ B) \geq c \min \left\{ \kappa \varepsilon \sqrt{\frac{\log M(B, d_S, 2\varepsilon)}{n}}, \frac{\kappa^2 \varepsilon^2}{R_{\text{out}}} \right\},$$

equivalently the bound with $\sup_{\varepsilon > 0}$ on the right. (In Lean $\log M$ is read as 0 when $M = \infty$, which only weakens the inequality.) This generalizes the paper, which states the bound for the word ball $B = B(k, F)$ (`thm:sudakov-type-wordball`); the word-ball structure is not used in the proof, and $B = \emptyset$ is allowed (both sides are then 0). In Lean we add the hypothesis that, for every sign pattern, the Rademacher averages over $H \circ B$ are bounded above (equivalently, that the supremum defining $\hat{\mathfrak{R}}_S(H \circ B)$ is finite); without it the Lean statement is false because an unbounded supremum evaluates to 0. The proof is complete modulo the Bernoulli–Sudakov minoration `thm:bernoulli-sudakov`. Proof: take a maximal 2ε -packing $g_1, \dots, g_M$ of B for d_S ($M = M(B, d_S, 2\varepsilon)$; if $M = \infty$ the right-hand side is 0 in Lean and the claim is `lem:emp-rademacher-nonneg`); the transported vectors $u_j = (h_{g_j}(g_j(x_i)))_i$ are $2\kappa\varepsilon$ -separated in $\|\cdot\|_S$ and bounded by R_{out} , so `thm:bernoulli-sudakov` gives $\mathbb{E}_\sigma \max_j \frac{1}{n} \sum_i \sigma_i u_{j,i} \geq c \min\{2\kappa\varepsilon \sqrt{\log M/n}, 4\kappa^2 \varepsilon^2 / R_{\text{out}}\}$, and the left-hand side is at most $\hat{\mathfrak{R}}_S(H \circ B)$ since $\{h_{g_j} \circ g_j\} \subseteq H \circ B$.

Proof. □

Theorem 119. Conditional Sudakov-type lower bound for word balls (the paper’s form). *There is a universal constant $c > 0$ such that, for F a hidden-layer class, $k \geq 0$ and Assumption `ass:readout-realization-main` for $B_k = B(k, F)$ with constants $\kappa, R_{\text{out}} > 0$ (and the boundedness hypothesis of `thm:sudakov-type`), for every $\varepsilon > 0$,*

$$\hat{\mathfrak{R}}_S(\mathcal{H}_k) \geq c \min \left\{ \kappa \varepsilon \sqrt{\frac{\log M(B_k, d_S, 2\varepsilon)}{n}}, \frac{\kappa^2 \varepsilon^2}{R_{\text{out}}} \right\}.$$

Proof: `thm:sudakov-type` with $B = B(k, F)$, since $\mathcal{H}_k = H \circ B(k, F)$.

Proof. □

Theorem 120. Conditional Sudakov-type lower bound for a bounded hypothesis class. *Under the hypotheses of `thm:sudakov-type`, the boundedness hypothesis holds as soon as $\mathcal{H}_k$ is uniformly bounded on the sample, $|g(x_i)| \leq C$ for all $g \in \mathcal{H}_k$ and $i \leq n$; hence the same lower bound.*

Proof. □

Theorem 121. (Substitution step.) *There is a universal constant $c > 0$ such that under Assumption `ass:readout-realization-main` for B_k and the boundedness hypothesis of `thm:sudakov-type`, for every $\varepsilon_0 > 0$ and every $q \geq 0$ with $q \leq \log M(B_k, d_S, 2\varepsilon_0)$, $\hat{\mathfrak{R}}_S(\mathcal{H}_k) \geq c \min\{\kappa \varepsilon_0 \sqrt{q/n}, \kappa^2 \varepsilon_0^2 / R_{\text{out}}\}$.*

Proof. Monotonicity of $q \mapsto \kappa \varepsilon_0 \sqrt{q/n}$ in `thm:sudakov-type-wordball`. $\square$

Theorem 122. *Rates under fixed-scale packing lower bounds (exponential).* There is a universal constant $c > 0$ such that under Assumption `ass:readout-realization-main` for B_k : if $M(B_k, d_S, 2\varepsilon_0) \geq e^{\alpha k}$ for some $\varepsilon_0, \alpha > 0$, then

$$\hat{\mathfrak{R}}_S(\mathcal{H}_k) \geq c \min \left\{ \kappa \varepsilon_0 \sqrt{\alpha k/n}, \kappa^2 \varepsilon_0^2 / R_{\text{out}} \right\}.$$

Proof. $\alpha k = \log e^{\alpha k} \leq \log M$. $\square$

Theorem 123. *Rates under fixed-scale packing lower bounds (polynomial).* There is a universal constant $c > 0$ such that under Assumption `ass:readout-realization-main` for B_k : if $k \geq 1$ and $M(B_k, d_S, 2\varepsilon_0) \geq k^\beta$ for some $\varepsilon_0, \beta > 0$, then

$$\hat{\mathfrak{R}}_S(\mathcal{H}_k) \geq c \min \left\{ \kappa \varepsilon_0 \sqrt{\beta \log k/n}, \kappa^2 \varepsilon_0^2 / R_{\text{out}} \right\}.$$

Proof. $\beta \log k = \log k^\beta \leq \log M$. $\square$

Theorem 124. For $\kappa, \varepsilon_0, R_{\text{out}} > 0$, $n \geq 1$ and $q \geq 0$, the first branch of the minimum is the smaller one as soon as $n \geq R_{\text{out}}^2 q / (\kappa^2 \varepsilon_0^2)$: $\kappa \varepsilon_0 \sqrt{q/n} \leq \kappa^2 \varepsilon_0^2 / R_{\text{out}}$.

Proof. $\square$

Theorem 125. *Matching depth dependence (i).* There is a universal constant $c > 0$ such that under Assumption `ass:readout-realization-main` for B_k (constants κ, R_{out}), the boundedness hypothesis of `thm:sudakov-type` and $\varepsilon_0 > 0$: if $M(B_k, d_S, 2\varepsilon_0) \geq e^{\alpha k}$ for some $\alpha > 0$, then $\hat{\mathfrak{R}}_S(\mathcal{H}_k) \geq c \kappa \varepsilon_0 \sqrt{\alpha k/n}$ whenever $n \geq R_{\text{out}}^2 \alpha k / (\kappa^2 \varepsilon_0^2)$.

Proof. Substitute $\varepsilon = \varepsilon_0$; the threshold on n makes the first branch of the minimum the smaller one. $\square$

Theorem 126. *Matching depth dependence (ii).* There is a universal constant $c > 0$ such that under Assumption `ass:readout-realization-main` for B_k (constants κ, R_{out}), the boundedness hypothesis of `thm:sudakov-type`, $k \geq 1$ and $\varepsilon_0 > 0$: if $M(B_k, d_S, 2\varepsilon_0) \geq k^\beta$ for some $\beta > 0$, then $\hat{\mathfrak{R}}_S(\mathcal{H}_k) \geq c \kappa \varepsilon_0 \sqrt{\beta \log k/n}$ whenever $n \geq R_{\text{out}}^2 \beta \log k / (\kappa^2 \varepsilon_0^2)$.

Proof. $\square$

3.4 Deterministic entropy decomposition

Theorem 127. *Composition of covers.* Let every $h \in H$ be L_H -Lipschitz, let C_H be an r_1 -cover of H for $\|\cdot\|_\infty$ and C_F an r_2 -cover of F for d_∞ (centres anywhere). Then $\{h_a \circ f_b : h_a \in C_H, f_b \in C_F\}$ is an $(r_1 + L_H r_2)$ -cover of $H \circ F$ for $\|\cdot\|_\infty$: $|h(f(x)) - h_a(f_b(x))| \leq |h(f(x)) - h(f_b(x))| + |h(f_b(x)) - h_a(f_b(x))| \leq L_H d_\infty(f, f_b) + \|h - h_a\|_\infty$.

Proof. $\square$

Theorem 128. For L_H -Lipschitz H and radii $r_1, r_2 \geq 0$, $N^{\text{ext}}(H \circ F, \|\cdot\|_\infty, r_1 + L_H r_2) \leq N^{\text{ext}}(H, \|\cdot\|_\infty, r_1) \cdot N^{\text{ext}}(F, d_\infty, r_2)$.

Proof. $\square$

Theorem 129. *Composition covering lemma.* If every $h \in H$ is L_H -Lipschitz then for every $\varepsilon \geq 0$,

$$N^{\text{ext}}(H \circ F, \|\cdot\|_\infty, \varepsilon) \leq N^{\text{ext}}(H, \|\cdot\|_\infty, \varepsilon/2) \cdot N^{\text{ext}}(F, d_\infty, \varepsilon/(2L_H)).$$

(For $L_H = 0$ the second radius is 0 by the convention $x/0 = 0$, and the inequality still holds.)

Proof. □

Theorem 130. $\mathcal{E}_H(u) \geq 0$.

Proof. □

Theorem 131. $\mathcal{E}_F(v) \geq 0$.

Proof. □

Theorem 132. If $c \leq ab$ in $\mathbb{N} \cup \{\infty\}$ with $a, b < \infty$, then $\log c \leq \log a + \log b$ (with $\log 0 = 0$; all three logarithms are ≥ 0).

Proof. □

Theorem 133. *Entropy of the composition class.* Let every $h \in H$ be L_H -Lipschitz with $L_H > 0$, and let $\varepsilon \in \mathbb{R}$ be such that $N^{\text{ext}}(H, \|\cdot\|_\infty, \varepsilon/2)$ and $N^{\text{ext}}(F, d_\infty, \varepsilon/(2L_H))$ are finite. Then

$$\sqrt{\log N^{\text{ext}}(H \circ F, \|\cdot\|_\infty, \varepsilon)} \leq \mathcal{E}_H(\varepsilon/2) + \mathcal{E}_F(\varepsilon/(2L_H)).$$

Proof. □

Theorem 134. FoML's empirical distance is dominated by the uniform distance: $\|u - v\|_S \leq \|u - v\|_\infty$ whenever the latter is finite ($n \geq 1$).

Proof. □

Theorem 135. If $G \subseteq \mathbb{R}^x$ is nonempty with finite uniform covering numbers $N^{\text{ext}}(G, \|\cdot\|_\infty, r) < \infty$ for all $r > 0$, then G is totally bounded for the empirical pseudometric $\|\cdot\|_S$ (FoML's 'EmpiricalFunctionSpace'), $n \geq 1$.

Proof. A finite internal closed $(\varepsilon/2)$ -cover for $\|\cdot\|_\infty$ (which exists since $N(G, 2r) \leq N^{\text{ext}}(G, r) < \infty$) is a finite open ε -cover for $\|\cdot\|_S$. □

Theorem 136. *FoML covering numbers versus uniform external covering numbers.* Let $G \subseteq \mathbb{R}^x$ be nonempty and totally bounded for $\|\cdot\|_S$, and let $0 \leq 2r < x$. Then FoML's open-ball covering number of G for $\|\cdot\|_S$ satisfies $N_S^{\text{open}}(G, x) \leq N(G, \|\cdot\|_\infty, 2r) \leq N^{\text{ext}}(G, \|\cdot\|_\infty, r)$ (internal closed $2r$ -balls for the sup norm with centres in G are contained in open x -balls for $\|\cdot\|_S$).

Proof. □

Theorem 137. If $H \neq \emptyset$ has finite uniform covering numbers at every positive scale, then $u \mapsto \mathcal{E}_H(u)$ is antitone on $(0, \infty)$.

Proof. □

Theorem 138. If $F \neq \emptyset$ has finite d_∞ covering numbers at every positive scale, then $v \mapsto \mathcal{E}_F(v)$ is antitone on $(0, \infty)$.

Proof. □

Theorem 139. $\hat{\mathfrak{R}}_S(\emptyset) = 0$ (the supremum over the empty class is 0 by convention).

Proof. □

Theorem 140. Almost-everywhere comparison of the entropy integrands. Let $H \neq \emptyset$, $F \neq \emptyset$ satisfy Assumptions `ass:ent-readout`, `ass:ent-transition` with $L_H > 0$, $n \geq 1$, and let $N_S(x)$ be FoML's open-ball covering number of $H \circ F$ for $\|\cdot\|_S$. Then for every $\varepsilon > 0$ and Lebesgue-a.e. $x > \varepsilon$,

$$\sqrt{\log N_S(x)} \leq \mathcal{E}_H(x/4) + \mathcal{E}_F(x/(4L_H)).$$

Proof: for every $y < x$, $N_S(x) \leq N^{\text{ext}}(H \circ F, \|\cdot\|_\infty, y/2) \leq N^{\text{ext}}(H, y/4)N^{\text{ext}}(F, y/(4L_H))$; the right-hand side $\psi(y) = \mathcal{E}_H(y/4) + \mathcal{E}_F(y/(4L_H))$ is antitone in y , hence continuous outside a countable set, and letting $y \uparrow x$ at a continuity point gives the claim.

Proof. □

Theorem 141. Deterministic entropy decomposition (FoML form). Under Assumptions `ass:ent-readout` and `ass:ent-transition` with $L_H > 0$, for every sample S of size $n \geq 1$ and every $0 < \varepsilon < B_H/2$,

$$\hat{\mathfrak{R}}_S(H \circ F) \leq 4\varepsilon + \frac{12}{\sqrt{n}} \int_\varepsilon^{B_H/2} \left\{ \mathcal{E}_H(x/4) + \mathcal{E}_F(x/(4L_H)) \right\} dx,$$

where $\mathcal{E}_H(u) = \sqrt{\log N^{\text{ext}}(H, \|\cdot\|_\infty, u)}$ and $\mathcal{E}_F(v) = \sqrt{\log N^{\text{ext}}(F, d_\infty, v)}$ (external covering numbers by closed balls). Proof: FoML's Dudley integral for the class $H \circ F$ with the empirical pseudometric $\|\cdot\|_S$ (`dudley_entropy_integral'`), whose internal open-ball covering number at scale x is at most $N^{\text{ext}}(H \circ F, \|\cdot\|_\infty, y/2)$ for every $y < x$ (`lem:foml-covering-le-external-unifFun`, losing a factor 2 in the radius when passing from external to internal covers), followed by the composition covering lemma `lem:ent-composition-entropy` at scale $y/2$ and the limit $y \uparrow x$ almost everywhere (`lem:entropy-integrand-ae`). Compared with the paper's statement (quoted in `thm:rad.decomp.ent.ent`), the scales $x/2$, $x/(2L_H)$ are replaced by $x/4$, $x/(4L_H)$ (the paper implicitly uses covers with centres in the class, while the assumptions here are stated with external covering numbers), the integration range is $[\varepsilon, B_H/2]$ instead of $[0, 2B_H]$, and there is the additive term 4ε from FoML's Dudley bound.

Proof. □

Theorem 142. Deterministic entropy decomposition. The paper states: under Assumptions `ass:ent-readout` and `ass:ent-transition`, for every sample $S = (x_1, \dots, x_n)$ with $n \geq 1$,

$$\hat{\mathfrak{R}}_S(H \circ F) \leq \frac{12}{\sqrt{n}} \int_0^{2B_H} \left\{ \sqrt{\log N(H, \|\cdot\|_\infty, \varepsilon/2)} + \sqrt{\log N(F, d_\infty, \varepsilon/(2L_H))} \right\} d\varepsilon.$$

The formalized statement is: under Assumptions `ass:ent-readout` and `ass:ent-transition` with $L_H > 0$ and $B_H > 0$, if the entropy integrand $x \mapsto \mathcal{E}_H(x/4) + \mathcal{E}_F(x/(4L_H))$ is integrable on $[0, B_H/2]$, then for every sample S of size $n \geq 1$,

$$\hat{\mathfrak{R}}_S(H \circ F) \leq \frac{12}{\sqrt{n}} \int_0^{B_H/2} \left\{ \mathcal{E}_H(x/4) + \mathcal{E}_F(x/(4L_H)) \right\} dx,$$

where $\mathcal{E}_H(u) = \sqrt{\log N^{\text{ext}}(H, \|\cdot\|_\infty, u)}$ and $\mathcal{E}_F(v) = \sqrt{\log N^{\text{ext}}(F, d_\infty, v)}$ are the root entropies for the external covering numbers by closed balls of the assumptions.

Deviations from the paper. (i) We assume in addition that the entropy integrand is integrable on $[0, B_H/2]$; the paper's bound is trivially true with right-hand side $+\infty$ otherwise (e.g. Lipschitz classes on a two-dimensional domain), whereas Lean's Bochner integral of a non-integrable function is 0, so the statement without this hypothesis is false as transcribed. (ii) FoML's Dudley integral (`dudley_entropy_integral`) uses internal open-ball covers, and converting the external closed-ball covering numbers of the assumptions costs a factor 2 in the radius (`lem:foml-covering-le-external-unifFun`), so the scales are $x/4$ and $x/(4L_H)$ on $[0, B_H/2]$ instead of $\varepsilon/2$, $\varepsilon/(2L_H)$ on $[0, 2B_H]$. Up to these constant rescalings the statement is the paper's.

Proof: let $\varepsilon \rightarrow 0$ in `thm:rad.decomp.ent.ent-foml`, using the integrability hypothesis to compare the integrals over $[\varepsilon, B_H/2]$ and $[0, B_H/2]$.

Proof. □

3.5 Deterministic entropy decomposition on the sample

Theorem 143. $\|u \circ f - v \circ f\|_S = \|u - v\|_{f \circ S}$: the empirical norm of a composition is the empirical norm on the pushed-forward sample $f \circ S = (f(x_1), \dots, f(x_n))$.

Proof. □

Theorem 144. A set C is an ε -cover of A for $\|\cdot\|_S$ iff every $u \in A$ has some $c \in C$ with $\|u - c\|_S \leq \varepsilon$.

Proof. □

Theorem 145. A set C is an ε -cover of A for d_S iff every $f \in A$ has some $c \in C$ with $d_S(f, c) \leq \varepsilon$.

Proof. □

Theorem 146. $N^{\text{ext}}(H, \|\cdot\|_{f \circ S}, u) \leq N_{S, F}(H, u)$ for $f \in F$.

Proof. □

Theorem 147. $u \mapsto N_{S, F}(H, u)$ is antitone.

Proof. □

Theorem 148. $N_{S, F}(H, u) > 0$ when $H, F \neq \emptyset$.

Proof. □

Theorem 149. If $|h(f(x_i)) - h(g(x_i))| \leq c d(f(x_i), g(x_i))$ for every i ($c \geq 0$), then $\|h \circ f - h \circ g\|_S \leq c d_S(f, g)$.

Proof. □

Theorem 150. Composition of covers on the sample. Let every $h \in H$ be L_H -Lipschitz on $\mathcal{R}_S(F)$, let $C_F \subseteq F$ be an r_2 -cover of F for d_S with centres in F , and for every $f_b \in C_F$ let $C_H(f_b)$ be an r_1 -cover of H for $\|\cdot\|_{f_b \circ S}$ (centres anywhere). Then $\{h_a \circ f_b : f_b \in C_F, h_a \in C_H(f_b)\}$ is an $(r_1 + L_H r_2)$ -cover of $H \circ F$ for $\|\cdot\|_S$: $\|h \circ f - h_a \circ f_b\|_S \leq \|h \circ f - h \circ f_b\|_S + \|h \circ f_b - h_a \circ f_b\|_S \leq L_H d_S(f, f_b) + \|h - h_a\|_{f_b \circ S}$, where the first estimate uses the Lipschitz property of h at the reachable points $f(x_i), f_b(x_i)$ (this is where $f_b \in F$ is needed).

Proof. □

Theorem 151. $H \circ F = \emptyset$ iff $H = \emptyset$ or $F = \emptyset$.

Proof. □

Theorem 152. For H L_H -Lipschitz on $\mathcal{R}_S(F)$ and radii $r_1, r_2 \geq 0$, $N^{\text{ext}}(H \circ F, \|\cdot\|_S, r_1 + L_H r_2) \leq N_{S,F}(H, r_1) \cdot N(F, d_S, r_2)$, where $N(F, d_S, \cdot)$ is the internal covering number (centres in F). *Proof:* choose a minimal internal cover C_F of F and, for each $f_b \in C_F$, a minimal external cover of H for $\|\cdot\|_{f_b \circ S}$; apply `lem:comp-cover-sample` and count.

Proof. □

Theorem 153. *Composition covering lemma on the sample.* If every $h \in H$ is L_H -Lipschitz on $\mathcal{R}_S(F)$ then for every $\varepsilon \geq 0$,

$$N^{\text{ext}}(H \circ F, \|\cdot\|_S, \varepsilon) \leq N_{S,F}(H, \varepsilon/2) \cdot N^{\text{ext}}(F, d_S, \varepsilon/(4L_H)).$$

(For $L_H = 0$ the second radius is 0 by the convention $x/0 = 0$.) Compared with the sup-norm lemma `lem:ent-composition-covering`, the radius of F is $\varepsilon/(4L_H)$ instead of $\varepsilon/(2L_H)$: the cover of F must have its centres in F , and $N(F, d_S, 2r) \leq N^{\text{ext}}(F, d_S, r)$.

Proof. □

Theorem 154. $\mathcal{E}_{H,S}(u) \geq 0$.

Proof. □

Theorem 155. $\mathcal{E}_{F,S}(v) \geq 0$.

Proof. □

Theorem 156. *Entropy of the composition class on the sample.* Let every $h \in H$ be L_H -Lipschitz on $\mathcal{R}_S(F)$ with $L_H > 0$, and let $\varepsilon \in \mathbb{R}$ be such that $N_{S,F}(H, \varepsilon/2)$ and $N^{\text{ext}}(F, d_S, \varepsilon/(4L_H))$ are finite. Then

$$\sqrt{\log N^{\text{ext}}(H \circ F, \|\cdot\|_S, \varepsilon)} \leq \mathcal{E}_{H,S}(\varepsilon/2) + \mathcal{E}_{F,S}(\varepsilon/(4L_H)).$$

Proof. □

Theorem 157. If $H, F \neq \emptyset$ and $N_{S,F}(H, u) < \infty$ for every $u > 0$, then $u \mapsto \mathcal{E}_{H,S}(u)$ is antitone on $(0, \infty)$.

Proof. □

Theorem 158. If $F \neq \emptyset$ has finite d_S covering numbers at every positive scale, then $v \mapsto \mathcal{E}_{F,S}(v)$ is antitone on $(0, \infty)$.

Proof. □

Theorem 159. FoML's 'EmpiricalFunctionSpace' of a class G (metric $\|\cdot\|_S$) embeds isometrically into $(\mathbb{R}^{\mathcal{X}}, \|\cdot\|_S)$ by evaluating the index.

Proof. □

Theorem 160. If $G \subseteq \mathbb{R}^{\mathcal{X}}$ has finite empirical covering numbers $N^{\text{ext}}(G, \|\cdot\|_S, r) < \infty$ for all $r > 0$, then G is totally bounded for $\|\cdot\|_S$ (FoML's 'EmpiricalFunctionSpace').

Proof. □

Theorem 161. FoML covering numbers versus empirical external covering numbers. Let $G \subseteq \mathbb{R}^x$ be nonempty and totally bounded for $\|\cdot\|_S$, and let $0 \leq 2r < x$. Then FoML's open-ball covering number of G for $\|\cdot\|_S$ satisfies $N_S^{\text{open}}(G, x) \leq N(G, \|\cdot\|_S, 2r) \leq N^{\text{ext}}(G, \|\cdot\|_S, r)$ (the two metrics coincide; the factor 2 is the price of the internal cover).

Proof. □

Theorem 162. Almost-everywhere comparison of the entropy integrands on the sample. Let $H, F \neq \emptyset$ satisfy `ass:ent-readout-sample` and `ass:ent-transition-sample` with $L_H > 0$, and let $N_S(x)$ be FoML's open-ball covering number of $H \circ F$ for $\|\cdot\|_S$. Then for every $\varepsilon > 0$ and Lebesgue-a.e. $x > \varepsilon$,

$$\sqrt{\log N_S(x)} \leq \mathcal{E}_{H,S}(x/4) + \mathcal{E}_{F,S}(x/(8L_H)).$$

Proof: for every $y < x$, $N_S(x) \leq N^{\text{ext}}(H \circ F, \|\cdot\|_S, y/2) \leq N_{S,F}(H, y/4) N^{\text{ext}}(F, d_S, y/(8L_H))$; the right-hand side is antitone in y , hence continuous outside a countable set, and we let $y \uparrow x$ at a continuity point.

Proof. □

Theorem 163. Deterministic entropy decomposition on the sample (FoML form). Under Assumptions `ass:ent-readout-sample` and `ass:ent-transition-sample` with $L_H > 0$, for every sample S of size $n \geq 1$ and every $0 < \varepsilon < B_H/2$,

$$\hat{\mathfrak{R}}_S(H \circ F) \leq 4\varepsilon + \frac{12}{\sqrt{n}} \int_{\varepsilon}^{B_H/2} \left\{ \mathcal{E}_{H,S}(x/4) + \mathcal{E}_{F,S}(x/(8L_H)) \right\} dx,$$

where $\mathcal{E}_{H,S}(u) = \sqrt{\log N_{S,F}(H, u)}$ and $\mathcal{E}_{F,S}(v) = \sqrt{\log N^{\text{ext}}(F, d_S, v)}$. Proof: FoML's Dudley integral for $H \circ F$ with the pseudometric $\|\cdot\|_S$ (`dudley_entropy_integral'`); its internal open-ball covering number at scale x is at most $N^{\text{ext}}(H \circ F, \|\cdot\|_S, y/2)$ for every $y < x$ (`lem:foml-covering-le-external-empFun`), then the composition covering lemma on the sample `lem:ent-composition-entropy-sample` at scale $y/2$ and the limit $y \uparrow x$ almost everywhere (`lem:entropy-integrand-ae-sample`).

Proof. □

Theorem 164. Deterministic entropy decomposition on the sample. Let S be a sample of size $n \geq 1$, and suppose `ass:ent-readout-sample` and `ass:ent-transition-sample` hold with $L_H > 0$, $B_H > 0$. If the entropy integrand $x \mapsto \mathcal{E}_{H,S}(x/4) + \mathcal{E}_{F,S}(x/(8L_H))$ is integrable on $[0, B_H/2]$, then

$$\hat{\mathfrak{R}}_S(H \circ F) \leq \frac{12}{\sqrt{n}} \int_0^{B_H/2} \left\{ \mathcal{E}_{H,S}(x/4) + \mathcal{E}_{F,S}(x/(8L_H)) \right\} dx,$$

where $\mathcal{E}_{H,S}(u) = \sqrt{\log N_{S,F}(H, u)}$ with $N_{S,F}(H, u) = \sup_{f \in F} N^{\text{ext}}(H, \|\cdot\|_{f \circ S}, u)$, and $\mathcal{E}_{F,S}(v) = \sqrt{\log N^{\text{ext}}(F, d_S, v)}$.

Comparison with the paper (`thm:rad.decomp.ent.ent`, Appendix D). The paper assumes H uniformly L_H -Lipschitz and bounded on $\mathcal{X}$ and covered in the sup norm, and F covered in d_∞ ; its proof passes from $\|\cdot\|_S$ to $\|\cdot\|_\infty$ at the very first step. Here (W8 of the plan) every hypothesis

is on the sample only: H is bounded and L_H -Lipschitz on the reachable points $\mathcal{R}_S(F) = \{f(x_i)\}$, F is covered in d_S , and H is covered in the empirical metrics $\|\cdot\|_{f \circ S}$ of the pushed-forward samples, uniformly in $f \in F$ (this is the honest sample analogue of the sup norm cover: in the composition estimate the output layer is evaluated at $f_b(x_i)$, not at x_i , so a cover of H for $\|\cdot\|_S$ alone would not suffice). The sup-norm assumptions imply the sample assumptions for every S (**cor:rad.decomp.ent.ent-of-sample**). The scales are $x/4$, $x/(8L_H)$ on $[0, B_H/2]$ in place of the paper's $\varepsilon/2$, $\varepsilon/(2L_H)$ on $[0, 2B_H]$: one factor 2 comes from FoML's internal open-ball covers (as in **thm:rad.decomp.ent.ent**) and, for F only, a second factor 2 from the fact that the cover of F must have its centres in F (so that the pushed-forward samples $f_b \circ S$ are admissible), see **lem:ent-composition-covering-sample**. As in the sup-norm theorem, integrability of the integrand is an additional hypothesis.

Proof: let $\varepsilon \rightarrow 0$ in **thm:rad.decomp.ent.ent-sample-foml**.

Proof. □

Theorem 165. For every sample S ($n \geq 1$) and $r \geq 0$, $N^{\text{ext}}(G, \|\cdot\|_S, r) \leq N^{\text{ext}}(G, \|\cdot\|_\infty, r)$: a sup-norm cover is an empirical cover, since $\|u - v\|_S \leq \|u - v\|_\infty$.

Proof. □

Theorem 166. For every sample S and $r \geq 0$, $N^{\text{ext}}(F, d_S, r) \leq N^{\text{ext}}(F, d_\infty, r)$, since $d_S \leq d_\infty$ (**lem:dS-le-dinf**).

Proof. □

Theorem 167. $N_{S,F}(H, r) \leq N^{\text{ext}}(H, \|\cdot\|_\infty, r)$ for every sample S ($n \geq 1$): the sup-norm covering number dominates the covering numbers on all pushed-forward samples.

Proof. □

Theorem 168. Assumption **ass:ent-readout** implies **ass:ent-readout-sample** for every sample S ($n \geq 1$) and every hidden class F .

Proof. □

Theorem 169. Assumption **ass:ent-transition** implies **ass:ent-transition-sample** for every sample S .

Proof. □

Theorem 170. If $a \leq b < \infty$ in $\mathbb{N} \cup \{\infty\}$ then $\sqrt{\log a} \leq \sqrt{\log b}$ (with $\log 0 = 0$).

Proof. □

Theorem 171. *The sup-norm theorem from the sample theorem.* Under the sup-norm Assumptions **ass:ent-readout** and **ass:ent-transition** with $L_H > 0$, $B_H > 0$, if $x \mapsto \mathcal{E}_H(x/4) + \mathcal{E}_F(x/(8L_H))$ is integrable on $[0, B_H/2]$, then for every sample S of size $n \geq 1$,

$$\hat{\mathfrak{R}}_S(H \circ F) \leq \frac{12}{\sqrt{n}} \int_0^{B_H/2} \left\{ \mathcal{E}_H(x/4) + \mathcal{E}_F(x/(8L_H)) \right\} dx.$$

This is **thm:rad.decomp.ent.ent** with the scale $x/(8L_H)$ in place of $x/(4L_H)$ for F (the price of the internal cover of F in the sample version), obtained from **thm:rad.decomp.ent.ent-sample-foml**: the sup-norm assumptions imply the sample assumptions, and $\mathcal{E}_{H,S} \leq \mathcal{E}_H$, $\mathcal{E}_{F,S} \leq \mathcal{E}_F$ pointwise at positive scales.

Proof. □

Chapter 4

Growth of the hidden class with depth

4.1 Definitions for the growth conditions

Definition 172. For a finite family $f = (f_1, \dots, f_r)$ of self-maps and a word $u = (i_1, \dots, i_k) \in [r]^k$, the associated map is $f_u = f_{i_k} \circ \dots \circ f_{i_1}$ (the first letter acts first); $f_\emptyset = \text{id}$. In Lean: ‘wordOf f [] = id’ and ‘wordOf f (i :: u) = wordOf f u ∘ f i’.

Theorem 173. $f_\emptyset = \text{id}$.

Proof.

□

Theorem 174. $f_{(i,u)} = f_u \circ f_i$.

Proof.

□

Theorem 175. $f_{(i,u)}(x) = f_u(f_i(x))$.

Proof.

□

Definition 176. For probes $P = (p_j)_{j \in J}$ (J finite) the evaluation map is $\text{ev}_P : (\mathcal{X}^{\mathcal{X}}, d_\infty) \rightarrow (\mathcal{X}^J, d_{\max})$, $\text{ev}_P(f) = (f(p_j))_{j \in J}$, where $\mathcal{X}^J$ carries the sup-metric.

Definition 177. For a monoid homomorphism $\alpha : H \rightarrow \mathcal{X}^{\mathcal{X}}$ (i.e. $\alpha(gh) = \alpha(g) \circ \alpha(h)$ and $\alpha(e) = \text{id}$), the orbit map is $g \mapsto \alpha(g) \in \mathcal{X}^{\mathcal{X}}$.

Theorem 178. $\alpha(e) = \text{id}$.

Proof.

□

Theorem 179. $\alpha(gh) = \alpha(g) \circ \alpha(h)$.

Proof.

□

Definition 180. The memory length of P1’: $m(\varepsilon) := L + \lceil \log_{1/c}(2 \text{diam}(K)/\varepsilon) \rceil$ (a natural number; the ceiling of a non-positive real is 0).

Definition 181. The memory state space $X = E^{\mathbb{N}}$ (the paper’s $\ell_\infty(E)$; we allow all sequences since the bounded metric below is finite anyway).

Theorem 182. $X = E^{\mathbb{N}}$ is nonempty (it contains 0).

Proof.

□

Theorem 183. If $a \leq b + c$ in $[0, \infty]$ then $\min\{1, a\} \leq \min\{1, b\} + \min\{1, c\}$.

Proof. If $b, c \leq 1$ this is $a \leq b + c$; otherwise the right side is at least 1.

□

Definition 184. The bounded sup metric on $X = E^{\mathbb{N}}$: $d_X(x, y) = \sup_j \min\{1, \|x_j - y_j\|_E\}$. It is a pseudo-metric (indeed a metric with values in $[0, 1]$).

Theorem 185. $d_X(x, y) = \sup_j \min\{1, \|x_j - y_j\|_E\}$.

Proof.

□

Theorem 186. $d_X(x, y) \leq 1$.

Proof.

□

Definition 187. The reset map $r(x) = 0$.

Definition 188. The writer $g_u(x_0, x_1, \dots) = (u, x_0, x_1, \dots)$.

Theorem 189. $r(x)_j = 0$.

Proof.

□

Theorem 190. $g_u(x)_0 = u$.

Proof.

□

Theorem 191. $g_u(x)_{i+1} = x_i$.

Proof.

□

Definition 192. The expander $A(x_0, x_1, \dots) = (\lambda x_0, \lambda x_1, \dots)$.

Theorem 193. $A(x)_j = \lambda x_j$.

Proof.

□

Definition 194. The hidden-layer class $F = \{r, A\} \cup \{g_u : u \in G\}$.

Definition 195. For $u = (u_0, \dots, u_{k-1}) \in E^k$ the word $w_u = A \circ g_{u_{k-1}} \circ A \circ g_{u_{k-2}} \circ \dots \circ A \circ g_{u_0} \circ r$ (so u_0 is written first and u_{k-1} last). In Lean it is defined by recursion on k : $w_{()} = r$ and $w_{(u,a)} = A \circ g_a \circ w_u$.

Theorem 196. $w_{()} = r$.

Proof.

□

Theorem 197. $w_u = A \circ g_{u_k} \circ w_{u|_k}$ for $u \in E^{k+1}$.

Proof.

□

Definition 198. $W_k = \{w_u : u \in G^k\}$.

4.2 Basic facts on covering and packing numbers

Theorem 199. (*Sub-multiplicativity.*) Let $F, G \subseteq \mathcal{X}^{\mathcal{X}}$ and suppose every $f \in F$ is λ -Lipschitz. Then for all $\varepsilon, \delta \geq 0$,

$$N^{\text{ext}}(F \circ G, d_{\infty}, \varepsilon + \lambda\delta) \leq N^{\text{ext}}(F, d_{\infty}, \varepsilon) N^{\text{ext}}(G, d_{\infty}, \delta).$$

(This is the paper's $N(FG, \varepsilon + \delta) \leq N(F, \varepsilon/\rho)N(G, \delta/\lambda)$ with $\rho = 1$, since right composition is 1-Lipschitz for d_{∞} ; we write $\lambda\delta$ in place of δ/λ to avoid dividing by $\lambda = 0$.)

Proof. Take an ε -cover C_F of F and a δ -cover C_G of G of minimal cardinality. For $f \in F$, $g \in G$ pick $f' \in C_F$, $g' \in C_G$ with $d_{\infty}(f, f') \leq \varepsilon$, $d_{\infty}(g, g') \leq \delta$; then $d_{\infty}(f \circ g, f' \circ g') \leq d_{\infty}(f \circ g, f \circ g') + d_{\infty}(f \circ g', f' \circ g') \leq \lambda\delta + \varepsilon$, so $C_F \circ C_G$ (of cardinality $\leq |C_F| |C_G|$) is a cover of $F \circ G$. $\square$

Theorem 200. The evaluation map ev_P is 1-Lipschitz from d_{∞} to the sup-metric on $\mathcal{X}^J$.

Proof. $\max_j d(f(p_j), g(p_j)) \leq \sup_x d(f(x), g(x))$. $\square$

Theorem 201. (*Probes and packing.*) Let $A \subseteq (\mathcal{X}^{\mathcal{X}}, d_{\infty})$, let P be a finite family of probes and let $T \subseteq \text{ev}_P(A)$ be δ -separated: $d_{\max}(y, z) \geq \delta$ for all distinct $y, z \in T$. Then for every ε with $2\varepsilon < \delta$, $|T| \leq N^{\text{ext}}(A, d_{\infty}, \varepsilon)$.

Proof. T is 2ε -separated (strictly), so $|T| \leq M(\text{ev}_P(A), 2\varepsilon) \leq M(A, 2\varepsilon) \leq N^{\text{ext}}(A, \varepsilon)$ by the Lipschitz embedding lemma (with $K = 1$) and the packing–covering comparison. $\square$

Theorem 202. Under the hypotheses of the previous lemma, also $|T| \leq N(A, d_{\infty}, \varepsilon)$ (internal covering number).

Proof. $\square$

Theorem 203. (*Indexed form of ‘lem:probes-packing’.*) Let $(g_i)_{i \in I}$ be a family of maps in $A \subseteq (\mathcal{X}^{\mathcal{X}}, d_{\infty})$ and P a finite family of probes such that $d_{\max}(\text{ev}_P(g_i), \text{ev}_P(g_j)) \geq \delta > 0$ for all $i \neq j$. Then $|I| \leq N^{\text{ext}}(A, d_{\infty}, \varepsilon)$ whenever $2\varepsilon < \delta$.

Proof. The map $i \mapsto \text{ev}_P(g_i)$ is injective (distinct indices have evaluations at distance $\geq \delta > 0$), and its range is a δ -separated subset of $\text{ev}_P(A)$; apply ‘lem:probes-packing’. $\square$

4.3 Compact Arzelà–Ascoli for self-maps

Theorem 204. If $\mathcal{X}$ is complete then $(\mathcal{X}^{\mathcal{X}}, d_{\infty})$ is complete (uniform limits of Cauchy sequences). This is Mathlib’s instance on ‘ $X \rightarrow X$ ’.

Proof. $\square$

Theorem 205. (*Metric modulus criterion.*) Let $(\mathcal{X}, d)$ be a totally bounded (pseudo)metric space. If $H \subseteq \mathcal{X}^{\mathcal{X}}$ has a common modulus of continuity, namely there is $\omega : \mathbb{R} \rightarrow \mathbb{R}$ with $\omega(r) \rightarrow 0$ as $r \downarrow 0$ (for every $\varepsilon > 0$ there is $r > 0$ with $\omega(t) \leq \varepsilon$ for $0 \leq t \leq r$) and

$$d(f(x), f(y)) \leq \omega(d(x, y)) \quad (f \in H, x, y \in \mathcal{X}),$$

then H is totally bounded in d_{∞} .

Proof. Fix $\eta > 0$ (below the prescribed radius). Choose $\rho > 0$ with $\omega(t) \leq \eta/4$ for $t \leq \rho$, a finite ρ -net P and a finite $\eta/4$ -net Q of $\mathcal{X}$. To $f \in H$ assign the map $P \rightarrow Q$, $p \mapsto q(f(p))$ with $d(f(p), q(f(p))) < \eta/4$. There are finitely many assignments; choose one representative $g \in H$ for each occurring assignment. If f and g have the same assignment then for every x , picking $p \in P$ with $d(x, p) < \rho$, $d(f(x), g(x)) \leq d(f(x), f(p)) + d(f(p), q(f(p))) + d(q(f(p)), g(p)) + d(g(p), g(x)) \leq \eta$. Thus the representatives form a finite η -net of H for d_∞ . $\square$

Theorem 206. (Modulus extraction on compact domains.) *Let $(\mathcal{X}, d)$ be compact and $H \subseteq \mathcal{X}^{\mathcal{X}}$ equicontinuous. Then H admits a common monotone modulus of continuity ω with $\omega(r) \rightarrow 0$ as $r \downarrow 0$ and $d(f(x), f(y)) \leq \omega(d(x, y))$ for all $f \in H$, $x, y \in \mathcal{X}$. In particular ‘thm:maa’ applies.*

Proof. Set $\omega(r) := \sup\{d(f(x), f(y)) : f \in H, d(x, y) \leq r\}$ (bounded by $\text{diam } \mathcal{X}$, and 0 for $r < 0$). It is monotone, and uniform equicontinuity (equicontinuity on a compact space) gives $\omega(r) \leq \varepsilon$ for $r \leq \delta/2$. $\square$

Theorem 207. (Forward direction of ‘thm:caa’.) *If $\mathcal{X}$ is compact and $H \subseteq \mathcal{X}^{\mathcal{X}}$ is equicontinuous, then H is totally bounded in d_∞ .*

Proof. Extract a common modulus (‘lem:aaa’) and apply the modulus criterion (‘thm:maa’); a compact space is totally bounded. $\square$

Theorem 208. (Converse direction of ‘thm:caa’.) *If $H \subseteq C(\mathcal{X}, \mathcal{X})$ is totally bounded in d_∞ then H is equicontinuous. (Continuity of the members of H is needed; compactness of $\mathcal{X}$ is not.)*

Proof. Fix x_0 and $\varepsilon > 0$, and take a finite $\varepsilon/3$ -net $t \subseteq H$. Each $g \in t$ is continuous at x_0 , so for x near x_0 we have $d(g(x_0), g(x)) < \varepsilon/3$ for all $g \in t$ simultaneously. For $f \in H$ choose $g \in t$ with $d_\infty(f, g) < \varepsilon/3$; then $d(f(x_0), f(x)) \leq d(f(x_0), g(x_0)) + d(g(x_0), g(x)) + d(g(x), f(x)) < \varepsilon$. $\square$

Theorem 209. (Compact Arzelà–Ascoli for self-maps.) *Let $(\mathcal{X}, d)$ be a compact (pseudo)metric space and let $H \subseteq C(\mathcal{X}, \mathcal{X})$. Then H is totally bounded in d_∞ if and only if H is equicontinuous.*

Proof. $\square$

Theorem 210. (Consequence of ‘thm:caa’.) *If $\mathcal{X}$ is compact and $H \subseteq \mathcal{X}^{\mathcal{X}}$ is equicontinuous, then the closure of H in $(\mathcal{X}^{\mathcal{X}}, d_\infty)$ is compact.*

Proof. $(\mathcal{X}^{\mathcal{X}}, d_\infty)$ is complete since $\mathcal{X}$ is compact (hence complete), and the closure of a totally bounded set in a complete space is compact. $\square$

Theorem 211. (Saturation for compact equicontinuous semigroups.) *Let $(\mathcal{X}, d)$ be compact and $F \subseteq \mathcal{X}^{\mathcal{X}}$. If the generated semigroup $\langle F \rangle$ is equicontinuous, then for every $\varepsilon > 0$ and every k*

$$N^{\text{ext}}(B(k, F), d_\infty, \varepsilon) \leq N^{\text{ext}}(\overline{\langle F \rangle}^{d_\infty}, d_\infty, \varepsilon) < \infty.$$

In particular the covering number of the word ball does not grow with the depth k .

Proof. $B(k, F) \subseteq \langle F \rangle \subseteq \overline{\langle F \rangle}$ and the external covering number is monotone in the set; the closure is totally bounded by ‘thm:caa’, so its covering number is finite. $\square$

Theorem 212. (Internal version of ‘cor:aa-semigroup-saturation’.) *Under the same hypotheses, for every $\varepsilon > 0$ and every k , $N(B(k, F), d_\infty, \varepsilon) \leq N(\overline{\langle F \rangle}^{d_\infty}, d_\infty, \varepsilon/2) < \infty$ (internal covering numbers are not monotone in the set, whence the loss of a factor 2 in the radius).*

Proof. Mathlib’s ‘coveringNumber_subset_le’ for the inclusion, and finiteness of the internal covering number of a totally bounded set. $\square$

4.4 Saturation: equicontinuous semigroups and contraction to an invariant set (P1, P1')

Theorem 213. $f \in B(k, F)$ if and only if $f \in F^l$ for some $l \leq k$.

Proof. Induction on k using $B(k+1, F) = B(k, F) \cup F^{k+1}$. $\square$

Theorem 214. Every word of length $n+L$ factors as $w = w_2 \circ w_1$ with $w_2 \in F^n$ and $w_1 \in F^L$.

Proof. Induction on n : for $n = 0$ take $w_2 = \text{id}$; for $w = a \circ b$ with $a \in F$, $b \in F^{n+L}$, factor $b = b_2 \circ b_1$ and take $w_2 = a \circ b_2$. $\square$

Theorem 215. If $f(A) \subseteq A$ for every $f \in F$, then $w(a) \in A$ for every word $w \in F^m$ and every $a \in A$.

Proof. Induction on m . $\square$

Theorem 216. If every $f \in F$ is c -Lipschitz then every word $w \in F^m$ is c^m -Lipschitz.

Proof. Induction on m ; Lipschitz constants multiply under composition. $\square$

Theorem 217. If the generators are non-expanding (1-Lipschitz) then every element of $B(k, F)$ is non-expanding.

Proof. Induction on k ; id is 1-Lipschitz and $1 \cdot 1 = 1$. $\square$

Theorem 218. If the generators are non-expanding then the whole semigroup $\langle F \rangle$ is non-expanding: $\text{lip } F \leq 1$ implies $\text{lip } \langle F \rangle \leq 1$.

Proof. $\langle F \rangle = \bigcup_k B(k, F)$. $\square$

Theorem 219. (P1: equicontinuous semigroup on a compact domain saturates, hypothesis 1.) Assume $\mathcal{X}$ is compact and the semigroup $\langle F \rangle$ is precompact (totally bounded) in d_∞ . Then, with $G := \overline{\langle F \rangle}^{d_\infty}$, for all $\varepsilon > 0$ and all k ,

$$N^{\text{ext}}(B(k, F), d_\infty, \varepsilon) \leq N^{\text{ext}}(G, d_\infty, \varepsilon) < \infty,$$

hence no dependence on k . (Compactness of $\mathcal{X}$ is not needed for this hypothesis.)

Proof. $B(k, F) \subseteq G$ and monotonicity of the external covering number; G is totally bounded as the closure of a totally bounded set, so its covering number is finite. $\square$

Theorem 220. (P1, hypothesis 2a.) Assume $\mathcal{X}$ is compact and the semigroup $\langle F \rangle$ is equicontinuous. Then for all $\varepsilon > 0$ and all k , $N^{\text{ext}}(B(k, F), d_\infty, \varepsilon) \leq N^{\text{ext}}(\overline{\langle F \rangle}, d_\infty, \varepsilon) < \infty$.

Proof. $\square$

Theorem 221. (P1, hypothesis 2b.) Assume $\mathcal{X}$ is compact and the semigroup $\langle F \rangle$ is uniformly Lipschitz: every $g \in \langle F \rangle$ is K -Lipschitz for a common K . Then for all $\varepsilon > 0$ and all k , $N^{\text{ext}}(B(k, F), d_\infty, \varepsilon) \leq N^{\text{ext}}(\overline{\langle F \rangle}, d_\infty, \varepsilon) < \infty$.

Proof. $\square$

Theorem 222. (P1, hypothesis 2c.) Assume $\mathcal{X}$ is compact and the generators are non-expanding: $\text{lip } F \leq 1$. Then for all $\varepsilon > 0$ and all k , $N^{\text{ext}}(B(k, F), d_\infty, \varepsilon) \leq N^{\text{ext}}(\overline{\langle F \rangle}, d_\infty, \varepsilon) < \infty$.

Proof. □

Theorem 223. *If $0 < c < 1$, $\varepsilon > 0$ and $m(\varepsilon) \leq n + L$ then $c^n \text{diam}(K) \leq \varepsilon/2$.*

Proof. If $\text{diam}(K) = 0$ this is trivial. Otherwise $\log_{1/c}(2 \text{diam}(K)/\varepsilon) \leq \lceil \cdot \rceil \leq n$, i.e. $2 \text{diam}(K)/\varepsilon \leq (1/c)^n$, which rearranges to the claim. □

Theorem 224. *(Long words are almost constant.) Under the hypotheses of ‘cond:p1-ucont’, for every word $w \in F^l$ with $l \geq m(\varepsilon)$ and all $x, y \in \mathcal{X}$,*

$$d(w(x), w(y)) \leq c^{l-L} \text{diam}(K) \leq \varepsilon/2.$$

In particular $d_\infty(w, \text{const}_{w(a_0)}) \leq \varepsilon/2$ for every a_0 .

Proof. Write $l = n + L$ and $w = w_2 \circ w_1$ with $w_2 \in F^n$ (c^n -Lipschitz) and $w_1 \in F^L$ (with values in K). Then $d(w(x), w(y)) \leq c^n d(w_1(x), w_1(y)) \leq c^n \text{diam}(K) \leq \varepsilon/2$. □

Theorem 225. *(Long words.) Under the hypotheses of ‘cond:p1-ucont’, the set of all words of length $\geq m(\varepsilon)$ satisfies*

$$N^{\text{ext}}\left(\bigcup_{l \geq m(\varepsilon)} F^l, d_\infty, \varepsilon\right) \leq N^{\text{ext}}(A, d, \varepsilon/2).$$

Indeed each such word w is $\varepsilon/2$ -close to the constant map $\text{const}_{w(a_0)}$ with $w(a_0) \in A$, so the constant maps at the points of an $\varepsilon/2$ -net of A form an ε -cover.

Proof. Take a minimal $\varepsilon/2$ -cover C of A ; for a long word w pick $q \in C$ with $d(w(a_0), q) \leq \varepsilon/2$; then $d(w(x), q) \leq d(w(x), w(a_0)) + d(w(a_0), q) \leq \varepsilon$ for every x , so $\{\text{const}_q : q \in C\}$ is an ε -cover. □

Theorem 226. *(Short words; subadditivity over finitely many sets.) For a finite family $(A_i)_{i \in s}$, $N^{\text{ext}}(\bigcup_{i \in s} A_i, \varepsilon) \leq \sum_{i \in s} N^{\text{ext}}(A_i, \varepsilon)$. Applied to $A_l = F^l$, $l < m(\varepsilon)$, this covers the short words.*

Proof. Induction on the finite index set using ‘lem:subadditivity’. □

Theorem 227. *(P1’: contraction to a compact invariant set.) Assume*

1. (uniform contraction) every $f \in F$ is c -Lipschitz with $0 < c < 1$;
2. (invariant set) there is a nonempty F -invariant set $A \subseteq \mathcal{X}$ ($f(A) \subseteq A$ for all $f \in F$);
3. (bounded absorbing set) there are $L \in \mathbb{N}$ and a bounded set $K \subseteq \mathcal{X}$ with $f(\mathcal{X}) \subseteq K$ for every word $f \in F^L$ of length L .

Then for every $\varepsilon > 0$, with $m(\varepsilon) := L + \lceil \log_{1/c}(2 \text{diam}(K)/\varepsilon) \rceil$, for every $k \geq 0$,

$$N^{\text{ext}}(B(k, F), d_\infty, \varepsilon) \leq N^{\text{ext}}(A, d, \varepsilon/2) + \sum_{l=0}^{m(\varepsilon)-1} N^{\text{ext}}(F^l, d_\infty, \varepsilon).$$

In particular the right-hand side is independent of k . (This generalizes the paper, which assumes A compact, K compact and $k \geq m(\varepsilon)$: the proof uses of A only that it is nonempty and invariant, of K only its boundedness, and the bound holds for every k . The right-hand side is finite whenever each $N^{\text{ext}}(F^l, d_\infty, \varepsilon)$, $l < m(\varepsilon)$, is finite, which the paper does not assume.)

Proof. $B(k, F) \subseteq \bigcup_{l \geq m(\varepsilon)} F^l \cup \bigcup_{l < m(\varepsilon)} F^l$; apply monotonicity, subadditivity, ‘lem:p1ucont-long-words’ and ‘lem:p1ucont-short-words’. □

4.5 Polynomial growth under nilpotent control (P2)

Theorem 228. *Let $\alpha : H \rightarrow \mathcal{X}^{\mathcal{X}}$ be a homomorphism, assume the length $g \mapsto d_H(e, g)$ is subadditive, $d_H(e, gh) \leq d_H(e, g) + d_H(e, h)$, and let $S \subseteq H$ satisfy $d_H(e, s) \leq R_S$ for all $s \in S$ and $F \subseteq \alpha(S)$. Then for every k ,*

$$B(k, F) \subseteq \alpha(\overline{B}_H(e, kR_S)).$$

Proof. Induction on k . For $k = 0$, $\text{id} = \alpha(e)$ and $e \in \overline{B}_H(e, 0)$. For $k + 1$: an element of $B(k, F)$ lies in $\alpha(\overline{B}_H(e, kR_S)) \subseteq \alpha(\overline{B}_H(e, (k+1)R_S))$; an element $\alpha(s) \circ \alpha(h)$ with $s \in S$, $d_H(e, h) \leq kR_S$ equals $\alpha(sh)$ and $d_H(e, sh) \leq d_H(e, s) + d_H(e, h) \leq R_S + kR_S$. $\square$

Theorem 229. *If the orbit map is L_α -Lipschitz, $d_\infty(\alpha(g), \alpha(h)) \leq L_\alpha d_H(g, h)$, then for every $B \subseteq H$ and $\delta \geq 0$,*

$$N^{\text{ext}}(\alpha(B), d_\infty, L_\alpha \delta) \leq N^{\text{ext}}(B, d_H, \delta).$$

Proof. This is the Lipschitz embedding lemma ‘lem:lipschitz-embedding’. $\square$

Theorem 230. *Under the hypotheses of ‘lem:p2-wordball-subset-ball’ and ‘lem:p2-transfer’, for every k and $\delta \geq 0$,*

$$N^{\text{ext}}(B(k, F), d_\infty, L_\alpha \delta) \leq N^{\text{ext}}(\overline{B}_H(e, kR_S), d_H, \delta).$$

Proof. Monotonicity of the external covering number in the set, then ‘lem:p2-transfer’. $\square$

Theorem 231. *For $\varepsilon, L_\alpha > 0$, $R_S \geq 0$, $k \geq 0$ and $D \geq 0$,*

$$\left(1 + \frac{kR_S}{\varepsilon/L_\alpha}\right)^D \leq \max(1, R_S L_\alpha)^D \left(1 + \frac{k}{\varepsilon}\right)^D.$$

Proof. With $M = \max(1, R_S L_\alpha)$ we have $1 \leq M$ and $R_S L_\alpha \leq M$, so $1 + (R_S L_\alpha)(k/\varepsilon) \leq M + M(k/\varepsilon) = M(1 + k/\varepsilon)$; raise to the power $D \geq 0$ and use $(ab)^D = a^D b^D$. $\square$

Theorem 232. *(P2: nilpotent control grows polynomially.) Let (H, d_H) be a group with a pseudo-metric, identity e , and assume the length $g \mapsto d_H(e, g)$ is subadditive: $d_H(e, gh) \leq d_H(e, g) + d_H(e, h)$. Assume its balls have polynomial entropy of degree $D \geq 0$: there is $1 \leq C_H < \infty$ such that for all $R \geq 0$ and $\delta > 0$,*

$$N^{\text{ext}}(\overline{B}_H(e, R), d_H, \delta) \leq C_H \left(1 + \frac{R}{\delta}\right)^D.$$

Suppose H acts on $\mathcal{X}$ through a homomorphism $\alpha : H \rightarrow \mathcal{X}^{\mathcal{X}}$, that there is a bounded set $S \subseteq H$ ($d_H(e, s) \leq R_S$ for $s \in S$) with $F \subseteq \alpha(S)$, and that the orbit map is Lipschitz in the uniform metric: $d_\infty(\alpha(g), \alpha(h)) \leq L_\alpha d_H(g, h)$ with $0 < L_\alpha < \infty$. Then with

$$C := C_H \max(1, R_S L_\alpha)^D$$

one has, for every $\varepsilon > 0$ and every $k \geq 0$,

$$N^{\text{ext}}(B(k, F), d_\infty, \varepsilon) \leq C \left(1 + \frac{k}{\varepsilon}\right)^D.$$

(In Lean the ball-entropy hypothesis and the conclusion are stated in $[0, \infty]$ as $N \leq \text{ofReal}(\dots)$, which in particular asserts finiteness; the bound holds for $k = 0$ as well.)

Proof. Put $\delta := \varepsilon/L_\alpha$, so $L_\alpha\delta = \varepsilon$. By ‘lem:p2-transfer-wordball’, $N^{\text{ext}}(B(k, F), d_\infty, \varepsilon) \leq N^{\text{ext}}(\bar{B}_H(e, kR_S), d_H, \delta) \leq C_H(1 + kR_S/\delta)^D$, and ‘lem:p2-poly-factor’ bounds the last factor by $\max(1, R_S L_\alpha)^D(1 + k/\varepsilon)^D$. $\square$

Theorem 233. (*Real-valued form of P2.*) Under the hypotheses of ‘cond:p2-nilp’, for every $\varepsilon > 0$ and $k \geq 0$, $N^{\text{ext}}(B(k, F), d_\infty, \varepsilon) \leq C_H \max(1, R_S L_\alpha)^D(1 + k/\varepsilon)^D$ as real numbers (the covering number being finite).

Proof. Take $C = C_H \max(1, R_S L_\alpha)^D$ and apply ‘cond:p2-nilp’; a quantity bounded by $\text{offReal}(b)$ is finite and has real part $\leq b$. $\square$

Theorem 234. (*Diameter envelope.*) Under the hypotheses of ‘lem:p2-wordball-subset-ball’ and with the orbit map L_α -Lipschitz, for all $f, g \in B(k, F)$,

$$d_\infty(f, g) \leq 2L_\alpha R_S k,$$

i.e. $\text{diam}_\infty B(k, F) \leq 2L_\alpha R_S k$ (and a fortiori $D_k(S) \leq 2L_\alpha R_S k$ for every sample S).

Proof. Write $f = \alpha(a)$, $g = \alpha(b)$ with $d_H(e, a), d_H(e, b) \leq kR_S$ (‘lem:p2-wordball-subset-ball’). Then $d_\infty(\alpha(a), \alpha(b)) \leq L_\alpha d_H(a, b) \leq L_\alpha(d_H(a, e) + d_H(e, b)) \leq 2L_\alpha R_S k$. $\square$

Theorem 235. On a bounded state space, $d(x, y) \leq D_X$ for all $x, y \in \mathcal{X}$, every pair of self-maps satisfies $d_\infty(f, g) \leq D_X$; in particular $\text{diam}_\infty B(k, F) \leq \text{diam}(\mathcal{X})$ for every k .

Proof. $\sup_x d(f(x), g(x)) \leq D_X$ termwise. $\square$

4.6 Exponential growth: free semigroups and ping-pong coding (E1, E1’, E2)

Theorem 236. If $w \in F^m$ and $g \in F$ then $w \circ g \in F^{m+1}$: the words of length m are also closed under right composition with generators.

Proof. Induction on m : $\text{id} \circ g = g \circ \text{id}$, and $(a \circ b) \circ g = a \circ (b \circ g)$. $\square$

Theorem 237. $f_u \in F^{|u|}$ where $F = \{f_1, \dots, f_r\}$.

Proof. Induction on u using the previous lemma. $\square$

Theorem 238. $F^m \subseteq B(m, F)$.

Proof. $\square$

Theorem 239. If $|u| \leq k$ then $f_u \in B(k, F)$ where $F = \{f_1, \dots, f_r\}$.

Proof. $\square$

Theorem 240. Let $f = (f_1, \dots, f_r)$ and let P be a finite family of probes such that the evaluations $\text{ev}_P(f_u)$, $u \in [r]^k$, are pairwise at sup-distance $\geq \delta > 0$. Then $N^{\text{ext}}(B(k, F), d_\infty, \varepsilon) \geq r^k$ for every ε with $2\varepsilon < \delta$.

Proof. Index the words of length k by $[r]^{[k]}$ (‘List.ofFn’), which has r^k elements, and apply ‘lem:probes-packing-family’. $\square$

Theorem 241. (*E1: free semigroup with one-point uniform separation.*) Let $F = \{f_1, \dots, f_r\}$, $r \geq 2$, and suppose there are a base point $x_* \in \mathcal{X}$ and $\delta > 0$ such that for every k and all distinct words $u \neq v \in [r]^k$,

$$d(f_u(x_*), f_v(x_*)) \geq \delta.$$

Then for every k and every $\varepsilon < \delta/2$, $N^{\text{ext}}(B(k, F), d_\infty, \varepsilon) \geq r^k$. (The paper's hypothesis (1), that $u \mapsto f_u$ is injective on $[r]^k$, follows from the separation hypothesis since $\delta > 0$, so it is omitted; the hypothesis $r \geq 2$ is not used in the proof.)

Proof. Apply ‘lem:pow-le-covering-of-probes’ with the single probe x_* . $\square$

Theorem 242. (*E1': equal-length coding.*) Let $F = \{f_1, \dots, f_r\}$, $r \geq 2$, and suppose there are $x_* \in \mathcal{X}$ and $\delta > 0$ with $d(f_u(x_*), f_v(x_*)) \geq \delta$ for all distinct $u, v \in [r]^k$ and all k . Then $N^{\text{ext}}(B(k, F), d_\infty, \varepsilon) \geq r^k$ for all k and all $\varepsilon < \delta/2$. (This generalizes the paper, which assumes in addition that the generators are isometries, that d_∞ is finite on $\langle F \rangle$ and that $u \mapsto f_u$ is injective on $[r]^k$: none of these is needed for the lower bound, which follows from ‘cond:e1-free-iso’ in one line; the isometry hypothesis only serves, in the paper, to make the covering numbers meaningful.)

Proof. $\square$

Theorem 243. If $A \subseteq \mathcal{X}$ satisfies $f_i(A) \subseteq A$ for all i , then $f_u(A) \subseteq A$ for every word u .

Proof. Induction on u . $\square$

Theorem 244. (Probe construction for E2.) Under the hypotheses of ‘cond:e2-pingpong’, for every word u there is a point $x_u \in Q = \{q\} \cup \bigcup_j V_j$ with $f_u(x_u) = q$ and $f_v(x_u) \in A = \{a_1, \dots, a_r\}$ for every word $v \neq u$ of the same length.

Proof. Induction on u , building the probe backwards. For $u = \emptyset$ take $x = q$. For $u = (i, u')$ with probe x' of u' , pick $x \in V_i$ with $f_i(x) = x'$ (coding core). Then $f_u(x) = f_{u'}(x') = q$. If $v = (j, v') \neq u$: when $j = i$, $v' \neq u'$ and $f_v(x) = f_{v'}(x') \in A$ by induction; when $j \neq i$, $x \in U_i$ lies outside U_j (chambers are disjoint), so $f_j(x) = a_j \in A$ (reset) and $f_{v'}(a_j) \in A$ (invariance of A). $\square$

Theorem 245. (*E2: ping-pong coding.*) Let $F = \{f_1, \dots, f_r\}$, $r \geq 2$, and suppose there are sets $V_i \subseteq U_i \subseteq \mathcal{X}$, anchors $a_1, \dots, a_r \in \mathcal{X}$, a marker $q \in \mathcal{X}$ and a constant $\alpha > 0$ such that, with $A = \{a_1, \dots, a_r\}$ and $Q = \{q\} \cup \bigcup_j V_j$:

1. (disjoint chambers) $U_i \cap U_j = \emptyset$ for $i \neq j$;
2. (coding cores) $Q \subseteq f_i(V_i)$ for every i ;
3. (reset) $f_i(x) = a_i$ for $x \notin U_i$, and $f_i(A) \subseteq A$;
4. (marker separation) $d(q, a_i) \geq \alpha$ for every i .

Then (a) $N^{\text{ext}}(B(k, F), d_\infty, \varepsilon) \geq r^k$ for every k and every $\varepsilon < \alpha/2$, and (b) $u \mapsto f_u$ is injective on $[r]^k$ for every k . (This generalizes the paper, which assumes in (1) a uniform separation $d(U_i, U_j) \geq \Delta > 0$ and additionally $V_i \neq \emptyset$: the proof uses of (1) only the disjointness of the chambers, and $V_i \neq \emptyset$ follows from (2) since $q \in f_i(V_i)$; the hypothesis $r \geq 2$ is not used.)

Proof. Choose the probes x_u of ‘lem:e2-probe’. For (a), use the r^k probes $P = (x_v)_{v \in [r]^k}$: for $u \neq w$ in $[r]^k$ the evaluations differ at the coordinate w , where $f_u(x_w) \in A$ and $f_w(x_w) = q$, so they are at distance $\geq \alpha$; conclude by ‘lem:pow-le-covering-of-probes’. For (b), if $f_u = f_v$ with $u \neq v$ of the same length then $q = f_u(x_u) = f_v(x_u) \in A$, contradicting $\alpha > 0$. $\square$

4.7 Memory-preserving expansion: super- and double-exponential growth (E3)

Theorem 246. $w_u \in B(2k+1, F)$ for every $u \in G^k$.

Proof. Induction on k : $r \in F \subseteq B(1, F)$, and $A, g_{u_k} \in F$ add two letters. $\square$

Theorem 247. $W_k \subseteq B(2k+1, F)$.

Proof. $\square$

Theorem 248. (**E3 (i).**) For every $x \in X$, $w_u(x) = (\lambda u_{k-1}, \lambda^2 u_{k-2}, \dots, \lambda^k u_0, 0, 0, \dots)$: coordinate $j < k$ of $w_u(x)$ is $\lambda^{j+1} u_{k-1-j}$ and the coordinates $j \geq k$ vanish. In particular w_u is a constant map.

Proof. Induction on k . For $k+1$: coordinate 0 of $A(g_a(y))$ is λa with $a = u_k$, and coordinate $j+1$ is λy_j where $y = w_{u_k}(x)$; apply the induction hypothesis to y . $\square$

Theorem 249. w_u is constant: $w_u(x) = w_u(y)$ for all x, y .

Proof. $\square$

Theorem 250. For $\lambda \neq 0$ the map $u \mapsto w_u$ is injective on E^k .

Proof. Coordinate $k-1-i$ of $w_u(0)$ is $\lambda^{k-i} u_i$, and $\lambda^{k-i} \neq 0$. $\square$

Theorem 251. If $f : \mathbb{N} \rightarrow [0, \infty]$ vanishes on $j \geq k$ then $\sup_{j \in \mathbb{N}} f(j) = \max_{j < k} f(j)$.

Proof. $\square$

Theorem 252. (**E3 (ii).**) For $u, v \in E^k$, $d_\infty(w_u, w_v) = \max_{0 \leq j < k} \min\{1, \lambda^{j+1} \|u_{k-1-j} - v_{k-1-j}\|_E\}$ (the paper's $\max_{1 \leq j \leq k} \min\{1, \lambda^{k-j+1} \|u_j - v_j\|\}$ after reindexing).

Proof. Both maps are constant, so $d_\infty(w_u, w_v) = d_X(w_u(0), w_v(0))$; the coordinates $j \geq k$ of both vanish and coordinate $j < k$ contributes $\min\{1, \lambda^{j+1} \|u_{k-1-j} - v_{k-1-j}\|\}$. $\square$

Theorem 253. The same identity for the extended distance of $(\mathcal{X}^X, d_\infty)$.

Proof. $\square$

Theorem 254. (**E3 (iii), upper bound.**) For every $\varepsilon \geq 0$ (and $\lambda \neq 0$),

$$N^{\text{ext}}(W_k, d_\infty, \varepsilon) \leq \prod_{j=0}^{k-1} N^{\text{ext}}(G, \|\cdot\|_E, \varepsilon / \lambda^{j+1}).$$

Proof. Take minimal $\varepsilon / \lambda^{j+1}$ -covers C_j of G . For $u \in G^k$ choose $c_i \in C_{k-1-i}$ with $\|u_i - c_i\| \leq \varepsilon / \lambda^{k-i}$; then by (ii) $d_\infty(w_u, w_c) \leq \max_j \lambda^{j+1} \|u_{k-1-j} - c_{k-1-j}\| \leq \varepsilon$. So $w(\prod_i C_{k-1-i})$ is an ε -cover of W_k of cardinality $\leq \prod_j |C_j|$. $\square$

Theorem 255. (*Product packing.*) Let $2\varepsilon < 1$, $\lambda \neq 0$, and let $P_j \subseteq G$ be $2\varepsilon / \lambda^{j+1}$ -separated ($0 \leq j < k$). Then $\prod_j |P_j| \leq M(W_k, d_\infty, 2\varepsilon)$.

Proof. The set $w(\prod_i P_{k-1-i}) \subseteq W_k$ is 2ε -separated: if $u \neq v$ then $u_i \neq v_i$ for some i , and with $j = k-1-i$ the j -th term of (ii) is $\min\{1, \lambda^{j+1} \|u_i - v_i\|\} > \min\{1, 2\varepsilon\} = 2\varepsilon$. Since w is injective its cardinality is $\prod_i |P_{k-1-i}| = \prod_j |P_j|$. $\square$

Theorem 256. In $\mathbb{N} \cup \{\infty\}$, a finite product of suprema over nonempty index types is the supremum of the products: $\prod_{i \in s} \sup_{a \in A_i} g_i(a) = \sup_{x \in \prod_i A_i} \prod_{i \in s} g_i(x_i)$.

Proof. Induction on s , distributing multiplication over suprema ('ENat.mul_iSup', 'ENat.iSup_mul') and identifying pairs (a, x) with the update $x[i := a]$. $\square$

Theorem 257. (E3 (iii), lower bound.) For every $0 \leq \varepsilon < 1/2$ (and $\lambda \neq 0$),

$$\prod_{j=0}^{k-1} M(G, \|\cdot\|_E, 2\varepsilon/\lambda^{j+1}) \leq N^{\text{ext}}(W_k, d_\infty, \varepsilon).$$

Proof. Each packing number is a supremum over separated subsets of G ; by 'lem:enat-prod-isup' the product is the supremum over families $(P_j)_j$ of $\prod_j |P_j|$, which is bounded by $M(W_k, 2\varepsilon) \leq N^{\text{ext}}(W_k, \varepsilon)$ by 'lem:e3-prod-packing-family' and 'lem:packing-covering'. $\square$

Theorem 258. $N^{\text{ext}}(W_k, d_\infty, \varepsilon) \leq N^{\text{ext}}(B(2k+1, F), d_\infty, \varepsilon)$ (monotonicity of the external covering number under $W_k \subseteq B(2k+1, F)$).

Proof. $\square$

Theorem 259. (E3: memory-preserving expansion grows super- or double-exponentially.)

Let E be a normed space, $G \subseteq E$, $\lambda > 1$, $X = E^{\mathbb{N}}$ with the bounded sup metric, $F = \{r, A\} \cup \{g_u : u \in G\}$ and $W_k = \{w_u : u \in G^k\} \subseteq B(2k+1, F)$. Then (i) each w_u is the constant map with value $(\lambda u_{k-1}, \dots, \lambda^k u_0, 0, \dots)$ ('cond:e3-i'), (ii) $d_\infty(w_u, w_v) = \max_{j < k} \min\{1, \lambda^{j+1} \|u_{k-1-j} - v_{k-1-j}\|\}$ ('cond:e3-ii'), and (iii) for every $0 < \varepsilon < 1/2$,

$$\prod_{j < k} M\left(G, \frac{2\varepsilon}{\lambda^{j+1}}\right) \leq N^{\text{ext}}(W_k, d_\infty, \varepsilon) \leq \prod_{j < k} N^{\text{ext}}\left(G, \frac{\varepsilon}{\lambda^{j+1}}\right).$$

Consequently $N^{\text{ext}}(B(2k+1, F), d_\infty, \varepsilon) \geq N^{\text{ext}}(W_k, d_\infty, \varepsilon)$.

Proof. $\square$

Theorem 260. For finite nonzero $n_i \in \mathbb{N} \cup \{\infty\}$, $\log \prod_i n_i = \sum_i \log n_i$ (with the coercion to $[0, \infty]$ and 'toReal').

Proof. $\square$

Theorem 261. If $G \neq \emptyset$ and all $N^{\text{ext}}(G, \varepsilon/\lambda^{j+1})$ are finite, then $\log N^{\text{ext}}(W_k, \varepsilon) \leq \sum_{j < k} \log N^{\text{ext}}(G, \varepsilon/\lambda^{j+1})$.

Proof. Take logarithms in 'cond:e3-iii-upper'; all quantities are finite and positive since G (hence W_k) is nonempty. $\square$

Theorem 262. If $G \neq \emptyset$, $\varepsilon < 1/2$, all $M(G, 2\varepsilon/\lambda^{j+1})$ and $N^{\text{ext}}(W_k, \varepsilon)$ are finite, then $\sum_{j < k} \log M(G, 2\varepsilon/\lambda^{j+1}) \leq \log N^{\text{ext}}(W_k, \varepsilon)$.

Proof. Take logarithms in 'cond:e3-iii-lower'. $\square$

Theorem 263. $\sum_{j < k} \log(\lambda^{j+1}/a) = \frac{k(k+1)}{2} \log \lambda - k \log a$ for $a, \lambda > 0$.

Proof. Induction on k using $\log(\lambda^{j+1}/a) = (j+1) \log \lambda - \log a$. $\square$

Theorem 264. If $0 \leq a < \lambda \delta_0$ and $\lambda > 1$ then $a/\lambda^{j+1} < \delta_0$ for every j .

Proof. □

Theorem 265. *If $c_- \log(1/\delta) \leq \log M(G, \delta)$ for all small $\delta > 0$ with $c_- > 0$, then $G \neq \emptyset$.*

Proof. Otherwise $M(G, \delta) = 0$ and $\log 0 = 0 < c_- \log(1/\delta)$ for $\delta = \min\{\delta_0/2, 1/2\}$. □

Theorem 266. *(Upper bound in sum form.) Suppose $N^{\text{ext}}(G, \delta) < \infty$ for all $\delta > 0$, $G \neq \emptyset$, and $\log N^{\text{ext}}(G, \delta) \leq \varphi(\delta)$ for $0 < \delta < \delta_0$. If $0 < \varepsilon < \lambda\delta_0$ then $\log N^{\text{ext}}(W_k, \varepsilon) \leq \sum_{j < k} \varphi(\varepsilon/\lambda^{j+1})$ for every k .*

Proof. All radii $\varepsilon/\lambda^{j+1} \leq \varepsilon/\lambda < \delta_0$, so ‘lem:e3-log-covering-le-sum’ and the hypothesis apply termwise. □

Theorem 267. *If $N^{\text{ext}}(G, \delta) < \infty$ for all $\delta > 0$ then $N^{\text{ext}}(W_k, \varepsilon) < \infty$ for all $\varepsilon > 0$.*

Proof. □

Theorem 268. *(Lower bound in sum form.) Suppose $N^{\text{ext}}(G, \delta) < \infty$ for all $\delta > 0$, $G \neq \emptyset$, and $\varphi(\delta) \leq \log M(G, \delta)$ for $0 < \delta < \delta_0$. If $0 < \varepsilon < 1/2$ and $2\varepsilon < \lambda\delta_0$ then $\sum_{j < k} \varphi(2\varepsilon/\lambda^{j+1}) \leq \log N^{\text{ext}}(W_k, \varepsilon)$ for every k .*

Proof. All radii $2\varepsilon/\lambda^{j+1} \leq 2\varepsilon/\lambda < \delta_0$; the packing numbers are finite since $M(G, 2\delta) \leq N^{\text{ext}}(G, \delta)$; apply ‘lem:e3-sum-log-packing-le’ and the hypothesis termwise. □

Theorem 269. *(Super-exponential regime.) Assume there are $c_-, c_+, \delta_0 > 0$ with $c_- \log(1/\delta) \leq \log M(G, \delta)$ and $\log N^{\text{ext}}(G, \delta) \leq c_+ \log(1/\delta)$ for all $0 < \delta < \delta_0$, and that $N^{\text{ext}}(G, \delta) < \infty$ for all $\delta > 0$ (automatic for compact G). Then for every fixed ε with $0 < \varepsilon < 1/2$ and $\varepsilon < \lambda\delta_0/2$ there are $C_1, C_2 > 0$ and k_0 such that $C_1 k^2 \leq \log N^{\text{ext}}(W_k, d_\infty, \varepsilon) \leq C_2 k^2$ for all $k \geq k_0$. (The smallness condition $\varepsilon < \lambda\delta_0/2$ makes every radius $2\varepsilon/\lambda^{j+1} \leq 2\varepsilon/\lambda$ fall below δ_0 ; it cannot be replaced by “large k ” since the $j = 0$ radius does not shrink with k .)*

Proof. By the sum forms of (iii) and the entropy hypotheses,

$$c_- \sum_{j < k} \log \frac{\lambda^{j+1}}{2\varepsilon} \leq \log N^{\text{ext}}(W_k, \varepsilon) \leq c_+ \sum_{j < k} \log \frac{\lambda^{j+1}}{\varepsilon},$$

and $\sum_{j < k} \log(\lambda^{j+1}/a) = \frac{k(k+1)}{2} \log \lambda + k \log(1/a)$. For $k \geq 1$ and $a \leq 1$ this is between $\frac{\log \lambda}{2} k^2$ and $(\log \lambda + \log(1/a))k^2$; take $C_1 = c_- \log \lambda/2$, $C_2 = c_+(\log \lambda - \log \varepsilon)$, $k_0 = 1$. □

Theorem 270. $\sum_{j < k} (a/\lambda^{j+1})^{-p} = a^{-p} \sum_{j < k} (\lambda^p)^{j+1}$ for $a, \lambda > 0$.

Proof. Termwise: $(a/\lambda^{j+1})^{-p} = (\lambda^{j+1})^p/a^p = a^{-p}(\lambda^p)^{j+1}$. □

Theorem 271. For $q > 1$ and $k \geq 1$: $q^k \leq \sum_{j < k} q^{j+1} \leq \frac{q}{q-1} q^k$.

Proof. The lower bound is the last term; the upper bound is the geometric sum formula $q(q^k - 1)/(q - 1)$. □

Theorem 272. *If $c_- \delta^{-p} \leq \log M(G, \delta)$ for all small $\delta > 0$ with $c_- > 0$, then $G \neq \emptyset$.*

Proof. Otherwise $M(G, \delta) = 0$ and $\log 0 = 0 < c_- \delta^{-p}$ for $\delta = \delta_0/2$. □

Theorem 273. (*Double-exponential regime.*) Assume there are $p, c_-, c_+, \delta_0 > 0$ with $c_- \delta^{-p} \leq \log M(G, \delta)$ and $\log N^{\text{ext}}(G, \delta) \leq c_+ \delta^{-p}$ for all $0 < \delta < \delta_0$, and that $N^{\text{ext}}(G, \delta) < \infty$ for all $\delta > 0$. Then for every fixed ε with $0 < \varepsilon < 1/2$ and $\varepsilon < \lambda \delta_0/2$ there are $C_1, C_2 > 0$ and k_0 such that $C_1 \lambda^{pk} \leq \log N^{\text{ext}}(W_k, d_\infty, \varepsilon) \leq C_2 \lambda^{pk}$ for all $k \geq k_0$.

Proof. By the sum forms of (iii), with $q = \lambda^p > 1$,

$$c_-(2\varepsilon)^{-p} \sum_{j < k} q^{j+1} \leq \log N^{\text{ext}}(W_k, \varepsilon) \leq c_+ \varepsilon^{-p} \sum_{j < k} q^{j+1},$$

and $q^k \leq \sum_{j < k} q^{j+1} \leq \frac{q}{q-1} q^k$ for $k \geq 1$; take $C_1 = c_-(2\varepsilon)^{-p}$, $C_2 = c_+ \varepsilon^{-p} q/(q-1)$, $k_0 = 1$. $\square$

Chapter 5

From growth mechanisms to variance profiles

5.1 An elementary logarithmic splitting lemma

Theorem 274. For $\overline{D} > 0$, $k \geq 0$ and $0 < \varepsilon \leq \overline{D}$, $\log(1 + \frac{k}{\varepsilon}) \leq \log(1 + \frac{k}{\overline{D}}) + \log \frac{\overline{D}}{\varepsilon}$.

Proof. $(1 + k/\overline{D})(\overline{D}/\varepsilon) = \overline{D}/\varepsilon + k/\varepsilon \geq 1 + k/\varepsilon$ since $\overline{D}/\varepsilon \geq 1$; take logarithms. $\square$

5.2 Variance profiles from growth and diameter

Theorem 275. $\log N(A, \varepsilon) \geq 0$ for every A and ε .

Proof. $\square$

Theorem 276. The metric entropy is antitone in the scale: if $\varepsilon \leq \delta$ and $N(A, \varepsilon) < \infty$ then $\log N(A, \delta) \leq \log N(A, \varepsilon)$.

Proof. $\square$

Theorem 277. $\varepsilon \mapsto \sqrt{\log N(A, \varepsilon^+)}$ is a measurable function on $\mathbb{R}$ (it is the composition of the antitone map $\varepsilon \mapsto N(A, \varepsilon^+)$ with measurable maps).

Proof. $\varepsilon \mapsto N(A, \varepsilon^+) \in [0, \infty]$ is antitone, hence measurable; compose with $x \mapsto x.\text{toReal}$, $\log$ and $\sqrt{\cdot}$. $\square$

Theorem 278. $V(D, A) \geq 0$ for $D \geq 0$.

Proof. $\square$

Theorem 279. If $\sqrt{\log N(A, \varepsilon)} \leq g(\varepsilon)$ on $(0, D]$ for an interval-integrable g on $[0, D]$, then $\varepsilon \mapsto \sqrt{\log N(A, \varepsilon)}$ is interval-integrable on $[0, D]$.

Proof. The integrand is measurable and nonnegative, and dominated by $|g|$ on $(0, D]$. $\square$

Theorem 280. For $0 < a \leq D$ with $N(A, a) < \infty$, $\varepsilon \mapsto \sqrt{\log N(A, \varepsilon)}$ is interval-integrable on $[a, D]$ (it is antitone and bounded there).

Proof. Antitone functions on a compact interval are interval-integrable. $\square$

Theorem 281. $V(D, A) \leq V(D', A)$ for $0 \leq D \leq D'$, provided the integrand is interval-integrable on $[0, D']$ (the integrand is nonnegative).

Proof. $\square$

Theorem 282. If $\sqrt{\log N(A, \varepsilon)} \leq g(\varepsilon)$ on $(0, D]$ for an interval-integrable g on $[0, D]$, then $V(D, A) \leq \int_0^D g$.

Proof. Monotonicity of the integral; the integrand is integrable by domination. $\square$

Theorem 283. Combined comparison: if $0 \leq D \leq \bar{D}$ and $\sqrt{\log N(A, \varepsilon)} \leq g(\varepsilon)$ on $(0, \bar{D}]$ for an interval-integrable g on $[0, \bar{D}]$, then $V(D, A) \leq \int_0^{\bar{D}} g$.

Proof. $V(D, A) \leq V(\bar{D}, A) \leq \int_0^{\bar{D}} g$. $\square$

Theorem 284. (Saturation.) Let A_k be sets with diameters $0 \leq D_k \leq \bar{D}$ and $N(A_k, \varepsilon) \leq N_\infty(\varepsilon)$ for all k and ε , with $N_\infty(\varepsilon) < \infty$ for $\varepsilon > 0$ and $V_\infty := \int_0^{\bar{D}} \sqrt{\log N_\infty(\varepsilon)} d\varepsilon < \infty$ (the integrand is interval-integrable). Then $V(D_k, A_k) \leq V_\infty$ for all k .

Proof. Both the integrand and the upper limit are dominated. $\square$

Theorem 285. If $N(A, \varepsilon) \leq C_0(1 + k/\varepsilon)^D$ for $\varepsilon > 0$, with $C_0 \geq 1$, $D \geq 0$, then for $0 < \varepsilon \leq \bar{D}$, $\sqrt{\log N(A, \varepsilon)} \leq \sqrt{\log C_0} + \sqrt{D} \sqrt{\log(1 + k/\bar{D})} + \sqrt{D} \sqrt{\log(\bar{D}/\varepsilon)}$.

Proof. $\log N \leq \log C_0 + D \log(1 + k/\varepsilon) \leq \log C_0 + D \log(1 + k/\bar{D}) + D \log(\bar{D}/\varepsilon)$ by the log-splitting inequality, then $\sqrt{a + b + c} \leq \sqrt{a} + \sqrt{b} + \sqrt{c}$. $\square$

Theorem 286. (Polynomial growth, bounded diameter; single-set form.) If $0 \leq D \leq \bar{D}$, $\bar{D} > 0$, $C_0 \geq 1$, $D \geq 0$ (the exponent) and $N(A, \varepsilon) \leq C_0(1 + k/\varepsilon)^D$ for all $\varepsilon > 0$, then $V(D, A) \leq \bar{D} \sqrt{D} \left(\sqrt{\log(1 + k/\bar{D})} + \frac{\sqrt{\pi}}{2} \right) + \bar{D} \sqrt{\log C_0}$.

Proof. Integrate the pointwise bound over $(0, \bar{D}]$ and use $\int_0^{\bar{D}} \sqrt{\log(\bar{D}/\varepsilon)} d\varepsilon = \frac{\sqrt{\pi}}{2} \bar{D}$. $\square$

Theorem 287. (Polynomial growth, bounded diameter.) If $0 \leq D_k \leq \bar{D}$, $\bar{D} > 0$, and $N(A_k, \varepsilon) \leq C_0(1 + k/\varepsilon)^D$ for all $k \geq 1$ and $\varepsilon > 0$ (with $C_0 \geq 1$, $D \geq 0$), then for $k \geq 1$

$$V(D_k, A_k) \leq \bar{D} \sqrt{D} \left(\sqrt{\log(1 + k/\bar{D})} + \frac{\sqrt{\pi}}{2} \right) + \bar{D} \sqrt{\log C_0} = O(\sqrt{D \log k}).$$

Proof. $\square$

Theorem 288. (Polynomial growth, linearly growing diameter.) If $0 \leq D_k \leq D_1 k$ with $D_1 > 0$, and $N(A_k, \varepsilon) \leq C_0(1 + k/\varepsilon)^D$ for all $k \geq 1$ and $\varepsilon > 0$ (with $C_0 \geq 1$, $D \geq 0$), then for $k \geq 1$

$$V(D_k, A_k) \leq D_1 k \left(\sqrt{D} \left(\sqrt{\log(1 + 1/D_1)} + \frac{\sqrt{\pi}}{2} \right) + \sqrt{\log C_0} \right) = O(k \sqrt{D}).$$

Proof. Apply the bounded-diameter estimate with $\bar{D} = D_1 k$ and simplify $k/(D_1 k) = 1/D_1$. $\square$

Theorem 289. (*Exponential growth, bounded diameter.*) If $0 \leq D_k \leq \bar{D}$ and $\log N(A_k, \varepsilon) \leq \alpha k + \psi(\varepsilon)$ for all k and $\varepsilon > 0$, with $\Psi := \int_0^{\bar{D}} \sqrt{\psi(\varepsilon)} d\varepsilon < \infty$ (i.e. $\sqrt{\psi}$ is interval-integrable on $[0, \bar{D}]$), then $V(D_k, A_k) \leq \bar{D}\sqrt{\alpha k} + \Psi = O(\sqrt{k})$.

Proof. $\sqrt{\log N_k(\varepsilon)} \leq \sqrt{\alpha k} + \sqrt{\psi(\varepsilon)}$; integrate. $\square$

Theorem 290. (*Finite classes.*) If $|A_k| \leq r^{k+1}$ and $0 \leq D_k \leq \bar{D}$, then $N(A_k, \varepsilon) \leq r^{k+1}$ at every scale and $V(D_k, A_k) \leq \bar{D}\sqrt{(k+1)\log r}$.

Proof. $N(A_k, \varepsilon) \leq |A_k| \leq r^{k+1}$, so $\log N(A_k, \varepsilon) \leq (k+1)\log r$; integrate the constant bound. $\square$

5.3 The variance term and the table of profiles

Theorem 291. Since $d_S \leq d_\infty$, the identity map $(\mathcal{X}^X, d_\infty) \rightarrow (\mathcal{X}^X, d_S)$ is 1-Lipschitz.

Proof. This is ‘lem:dS-le-dinf’. $\square$

Theorem 292. For every $A \subseteq \mathcal{X}^X$ and $\varepsilon \geq 0$, $N^{\text{ext}}(A, d_S, \varepsilon) \leq N^{\text{ext}}(A, d_\infty, \varepsilon)$: covering numbers in the empirical metric are dominated by those in the uniform metric, so all hypotheses may be verified in d_∞ .

Proof. ‘lem:lipschitz-embedding’ with $K = 1$ and $\varphi = \text{id}$. $\square$

Theorem 293. $N(A, d_S, \varepsilon) \leq N(A, d_\infty, \varepsilon)$ (internal covering numbers).

Proof. ‘lem:lipschitz-embedding-internal’ with $K = 1$ and $\varphi = \text{id}$. $\square$

Theorem 294. $M(A, d_S, \varepsilon) \leq M(A, d_\infty, \varepsilon)$ (packing numbers).

Proof. ‘lem:lipschitz-embedding-packing’ with $K = 1$ and $\varphi = \text{id}$. $\square$

Theorem 295. The bridge used by all profiles: for every $A \subseteq \mathcal{X}^X$ and $\varepsilon \geq 0$, $N(A, d_S, \varepsilon) \leq N^{\text{ext}}(A, d_\infty, \varepsilon/2)$ (internal covering number on the left, external on the right; the factor 2 is the price of comparing internal with external covers).

Proof. $N(A, d_S, \varepsilon) = N(A, d_S, 2 \cdot \varepsilon/2) \leq N^{\text{ext}}(A, d_S, \varepsilon/2) \leq N^{\text{ext}}(A, d_\infty, \varepsilon/2)$. $\square$

Theorem 296. If $N^{\text{ext}}(A, d_\infty, \varepsilon/2) \leq b$ with $b \geq 0$, then $N(A, d_S, \varepsilon) \leq b$ as real numbers (the covering number being finite).

Proof. $\square$

Theorem 297. $\text{diam}_S(A) \geq 0$.

Proof. $\square$

Theorem 298. (Diameter envelopes transfer from d_∞ to d_S .) If $d_\infty(f, g) \leq D$ for all $f, g \in A$, with $D \geq 0$, then $\text{diam}_S(A) \leq D$ for every sample S .

Proof. $d_S(f, g) \leq d_\infty(f, g) \leq D$ for all $f, g \in A$; take the supremum (which is $0 \leq D$ if $A = \emptyset$). $\square$

Theorem 299. If $d_\infty(f, g) \leq \bar{D} < \infty$ for all $f, g \in A$ then $\text{diam}_S(A) \leq \bar{D}$.

Proof. $\square$

Theorem 300. *On a bounded state space, $d(x, y) \leq D_{\mathcal{X}}$ for all x, y ($D_{\mathcal{X}} \geq 0$), every class satisfies $\text{diam}_S(A) \leq D_{\mathcal{X}}$; in particular $D_k(S) \leq \text{diam}(\mathcal{X})$ for all k .*

Proof. □

Theorem 301. *On a compact state space, $D_k(S) \leq \text{diam}(\mathcal{X})$ for every class A and every sample S .*

Proof. □

Theorem 302. *(Diameter envelope under P2, non-compact case.) Under the hypotheses of ‘lem:p2-diameter’, $D_k(S) = \text{diam}_S B(k, F) \leq 2L_{\alpha} R_S k$ for every sample S .*

Proof. ‘lem:p2-diameter’ gives $d_{\infty}(f, g) \leq 2L_{\alpha} R_S k$ on $B(k, F)$; transfer to d_S . □

Theorem 303. *(Restatement of ‘thm:hidden-decomp-depth’.) Under the hypotheses of ‘thm:hidden-decomp-depth’ (including the integrability of the entropy integrand of $B(k, F_0)$ on $[0, D_k(S)]$), $\hat{\mathfrak{R}}_S(\mathcal{H}_k) \leq \hat{\mathfrak{R}}_S(H) + \text{var}(k, n)$.*

Proof. □

Theorem 304. *(Plugging a profile into the decomposition.) Under the hypotheses of ‘thm:hidden-decomp-depth’ (including the integrability of the entropy integrand), if $\mathbf{V}_k(S) \leq B$ then $\hat{\mathfrak{R}}_S(\mathcal{H}_k) \leq \hat{\mathfrak{R}}_S(H) + \frac{12A_H L}{\sqrt{n}} B$. The statements about $\text{var}(k, n)$ in ‘prop:profiles’ follow by multiplying the profile bounds with $12A_H L/\sqrt{n}$.*

Proof. ‘lem:hidden-decomp-var’ and $12A_H L/\sqrt{n} \geq 0$. □

Theorem 305. *(P1 $\Rightarrow$ saturation; general form.) Let the state metric be bounded, $d(x, y) \leq D_{\mathcal{X}}$ ($D_{\mathcal{X}} \geq 0$), and let the semigroup $\langle F \rangle$ be totally bounded in d_{∞} (‘cond:p1’). Put $N_{\infty}(\varepsilon) := N^{\text{ext}}(\langle F \rangle, d_{\infty}, \varepsilon/2)$ (finite for $\varepsilon > 0$; the factor $\frac{1}{2}$ comes from comparing internal with external covers) and assume $V_{\infty} := \int_0^{D_{\mathcal{X}}} \sqrt{\log N_{\infty}(\varepsilon)} d\varepsilon < \infty$ (interval-integrability of the majorant). Then for every k and every sample S , $\mathbf{V}_k(S) \leq V_{\infty}$: profile (i), $\text{var}(k, n) = O(n^{-1/2})$.*

Proof. ‘prop:profiles-i’ with $A_k = B(k, F)$, $D_k = D_k(S) \leq D_{\mathcal{X}}$ (‘lem:empDiam-le-of-bounded’) and the majorant N_{∞} : $N(B(k, F), d_S, \varepsilon) \leq N^{\text{ext}}(B(k, F), d_{\infty}, \varepsilon/2) \leq N^{\text{ext}}(\langle F \rangle, d_{\infty}, \varepsilon/2)$ (‘lem:covering-empSpace-le-external-unifMaps’ and monotonicity), finite by ‘cond:p1’. □

Theorem 306. *(P1 $\Rightarrow$ saturation; table row “P1 (compact, equicontinuous)”.) Let $\mathcal{X}$ be compact and the semigroup $\langle F \rangle$ equicontinuous. With $N_{\infty}(\varepsilon) := N^{\text{ext}}(\langle F \rangle, d_{\infty}, \varepsilon/2)$ and $V_{\infty} := \int_0^{\text{diam}(\mathcal{X})} \sqrt{\log N_{\infty}(\varepsilon)} d\varepsilon < \infty$ (assumed interval-integrable), $D_k(S) \leq \text{diam}(\mathcal{X})$ and $\mathbf{V}_k(S) \leq V_{\infty}$ for all k : profile (i), $\mathbf{V}_k = O(1)$, $\text{var}(k, n) = O(n^{-1/2})$.*

Proof. □

Theorem 307. *(P1 with non-expanding generators.) Let $\mathcal{X}$ be compact and every $f \in F$ be 1-Lipschitz. Then the conclusion of ‘cor:profile-p1’ holds.*

Proof. □

Theorem 308. (*P2 in the empirical metric.*) Under the hypotheses of ‘cond:p2-nilp’, for every k and $\varepsilon > 0$,

$$N(B(k, F), d_S, \varepsilon) \leq C_0 \left(1 + \frac{k}{\varepsilon}\right)^D, \quad C_0 := 2^D C_H \max(1, R_S L_\alpha)^D,$$

as real numbers. (The factor 2^D comes from $N(\cdot, d_S, \varepsilon) \leq N^{\text{ext}}(\cdot, d_\infty, \varepsilon/2)$ and $1 + 2k/\varepsilon \leq 2(1 + k/\varepsilon)$.)

Proof. Apply ‘cond:p2-nilp’ at scale $\varepsilon/2$ and ‘lem:covering-empSpace-toReal-le’. $\square$

Theorem 309. $C_0 = 2^D C_H \max(1, R_S L_\alpha)^D \geq 1$.

Proof. $\square$

Theorem 310. (*P2 on a bounded state space; table row “P2, compact $\mathcal{X}$ ”.*) Under the hypotheses of ‘cond:p2-nilp’, if moreover $d(x, y) \leq D_{\mathcal{X}}$ for all x, y ($D_{\mathcal{X}} > 0$), then $D_k(S) \leq D_{\mathcal{X}}$ and for every $k \geq 1$

$$V_k(S) \leq D_{\mathcal{X}} \sqrt{D} \left(\sqrt{\log(1 + k/D_{\mathcal{X}})} + \frac{\sqrt{\pi}}{2} \right) + D_{\mathcal{X}} \sqrt{\log C_0} = O(\sqrt{D \log k}),$$

with $C_0 = 2^D C_H \max(1, R_S L_\alpha)^D$: profile (ii), $\text{var}(k, n) = O(\sqrt{D \log k/n})$.

Proof. $\square$

Theorem 311. (*P2 on a general state space; table row “P2, non-compact $\mathcal{X}$ ”.*) Under the hypotheses of ‘cond:p2-nilp’ with $R_S > 0$, $D_k(S) \leq 2L_\alpha R_S k$ and for every $k \geq 1$

$$V_k(S) \leq 2L_\alpha R_S k \left(\sqrt{D} (\sqrt{\log(1 + 1/(2L_\alpha R_S))} + \frac{\sqrt{\pi}}{2}) + \sqrt{\log C_0} \right) = O(k\sqrt{D}),$$

with $C_0 = 2^D C_H \max(1, R_S L_\alpha)^D$: profile (iv), $\text{var}(k, n) = O(k\sqrt{D/n})$.

Proof. $\square$

Theorem 312. (*Finite hidden-layer classes; table row “E1’, E2 with $|F| = r$ ”.*) If F is finite with $|F| = r \geq 2$ and the state metric is bounded, $d(x, y) \leq D_{\mathcal{X}}$ ($D_{\mathcal{X}} \geq 0$), then $|B(k, F)| \leq r^{k+1}$, hence $N(B(k, F), d_S, \varepsilon) \leq r^{k+1}$ for every $\varepsilon > 0$, and

$$V_k(S) \leq D_{\mathcal{X}} \sqrt{(k+1) \log r} = O(\sqrt{k \log r})$$

for every k : profile (iii), $\text{var}(k, n) = O(\sqrt{k/n})$.

Proof. $\square$

Theorem 313. If $0 \leq D \leq \bar{D}$ and $\sqrt{\log N(A, \varepsilon)} \leq g(\varepsilon)$ on $(0, \bar{D}]$ for an interval-integrable g on $[0, \bar{D}]$, then $\varepsilon \mapsto \sqrt{\log N(A, \varepsilon)}$ is interval-integrable on $[0, D]$.

Proof. $\square$

Theorem 314. (*Integrability under P1.*) Under the hypotheses of ‘cor:profile-p1-totallyBounded’ (in particular the interval-integrability of the majorant $\sqrt{\log N_\infty}$ on $[0, D_{\mathcal{X}}]$), the entropy integrand $\varepsilon \mapsto \sqrt{\log N(B(k, F), d_S, \varepsilon)}$ is interval-integrable on $[0, D_k(S)]$ for every k : the hypothesis of ‘thm:hidden-decomp-depth’ is automatic.

Proof. Domination by the majorant N_∞ , as in ‘cor:profile-p1-totallyBounded’. $\square$

Theorem 315. (Integrability under P2.) Under the hypotheses of ‘cond:p2-nilp’, if $D_k(S) \leq \bar{D}$ with $\bar{D} > 0$, then the entropy integrand of $B(k, F)$ is interval-integrable on $[0, D_k(S)]$ (it is dominated by $\sqrt{\log C_0} + \sqrt{\bar{D}}\sqrt{\log(1 + k/\bar{D})} + \sqrt{\bar{D}}\sqrt{\log(\bar{D}/\varepsilon)}$, ‘lem:sqrt-metric-entropy-le-of-poly’).

Proof. The polynomial majorant is interval-integrable on $[0, \bar{D}]$ (‘intervalIntegrable_sqrt_log_div’). $\square$

Theorem 316. (Integrability for finite hidden-layer classes.) If F is finite with $|F| = r \geq 2$ and $d(x, y) \leq D_x$, then the entropy integrand of $B(k, F)$ is bounded by the constant $\sqrt{(k+1)\log r}$, hence interval-integrable on $[0, D_k(S)]$.

Proof. $N(B(k, F), d_S, \varepsilon) \leq |B(k, F)| \leq r^{k+1}$ (‘lem:wordball-card’). $\square$

Theorem 317. (Estimation term under P1.) Let $\mathcal{X}$ be compact, the semigroup $\langle F \rangle$ equicontinuous, and V_∞ as in ‘cor:profile-p1’. Under the hypotheses of ‘thm:hidden-decomp-depth’, $\hat{\mathfrak{R}}_S(\mathcal{H}_k) \leq \hat{\mathfrak{R}}_S(H) + \frac{12A_H L}{\sqrt{n}} V_\infty$ for every k : $\text{var}(k, n) = O(n^{-1/2})$ uniformly in the depth.

Proof. $\square$

Theorem 318. (Estimation term under P2, bounded state space.) Under the hypotheses of ‘cor:profile-p2-compact’ and ‘thm:hidden-decomp-depth’, for $k \geq 1$, $\hat{\mathfrak{R}}_S(\mathcal{H}_k) \leq \hat{\mathfrak{R}}_S(H) + \frac{12A_H L}{\sqrt{n}} \left(D_x \sqrt{\bar{D}} (\sqrt{\log(1 + k/D_x)} + \frac{\sqrt{\pi}}{2}) + D_x \sqrt{\log C_0} \right)$: $\text{var}(k, n) = O(\sqrt{\bar{D} \log k/n})$.

Proof. $\square$

Theorem 319. (Estimation term under P2, general state space.) Under the hypotheses of ‘cor:profile-p2-noncompact’ and ‘thm:hidden-decomp-depth’, for $k \geq 1$, $\hat{\mathfrak{R}}_S(\mathcal{H}_k) \leq \hat{\mathfrak{R}}_S(H) + \frac{12A_H L}{\sqrt{n}} 2L_\alpha R_S k \left(\sqrt{\bar{D}} (\sqrt{\log(1 + 1/(2L_\alpha R_S))} + \frac{\sqrt{\pi}}{2}) + \sqrt{\log C_0} \right)$: $\text{var}(k, n) = O(k\sqrt{\bar{D}/n})$.

Proof. $\square$

Theorem 320. (Estimation term for finite hidden-layer classes.) Under the hypotheses of ‘cor:profile-finite’ and ‘thm:hidden-decomp-depth’, $\hat{\mathfrak{R}}_S(\mathcal{H}_k) \leq \hat{\mathfrak{R}}_S(H) + \frac{12A_H L}{\sqrt{n}} D_x \sqrt{(k+1)\log r}$ for every k : $\text{var}(k, n) = O(\sqrt{k/n})$.

Proof. $\square$

Chapter 6

Depth bias–variance trade-offs

6.1 Bias laws, variance profiles, balancing depths (definitions)

Definition 321. Exponentially decaying approximation error: $\text{bias}(k) = e^{-\alpha k}$, $\alpha > 0$.

Definition 322. Polynomially decaying approximation error: $\text{bias}(k) = k^{-\beta}$, $\beta > 0$.

Definition 323. Root-logarithmic estimation profile: $\text{var}(k, n) = \sqrt{\log k/n}$ (??(i)–(ii)).

Definition 324. Root-polynomial estimation profile: $\text{var}(k, n) = \sqrt{k^\gamma/n}$, $\gamma > 0$ (??(iii)–(iv)).

Definition 325. The depth-dependent part of the excess-risk bound of Theorem 98: $\text{gen}(k, n) = \text{bias}(k) + \text{var}(k, n)$.

Definition 326. PP balancing depth $k^* = n^{1/(2\beta+\gamma)}$.

Definition 327. EP balancing depth (leading terms): $k^* = \frac{1}{2\alpha}(\log n - \gamma \log \log n)$.

Definition 328. EP balanced rate $n^{-1/2}(\log n)^{\gamma/2}$.

Definition 329. EL balancing depth (leading terms): $k^* = \frac{1}{2\alpha}(\log n - \log \log \log n)$.

Definition 330. EL balanced rate $\sqrt{\log \log n/n}$.

Definition 331. PL balancing depth $k^* = (2\beta n / \log(2\beta n))^{1/(2\beta)}$, the leading term of $\exp(W(2\beta n)/(2\beta))$ (Lambert W , $W(x) \sim \log x$).

Definition 332. PL balanced rate $\sqrt{\log n/n}$.

6.2 Balancing depths and balanced rates (EL, EP, PL, PP)

Theorem 333. For $\alpha > 0$, $k \mapsto e^{-\alpha k}$ is nonincreasing.

Proof. $\exp$ is monotone and $k \mapsto -\alpha k$ is nonincreasing. □

Theorem 334. For $\beta > 0$, $k \mapsto k^{-\beta}$ is nonincreasing on $(0, \infty)$.

Proof. Power with a nonpositive exponent is antitone on the positive reals. $\square$

Theorem 335. For $n \geq 0$, $k \mapsto \sqrt{\log k/n}$ is nondecreasing on $(0, \infty)$.

Proof. $\log$ is monotone on $(0, \infty)$; divide by $n \geq 0$ and take square roots. $\square$

Theorem 336. For $\gamma \geq 0$ and $n \geq 0$, $k \mapsto \sqrt{k^\gamma/n}$ is nondecreasing on $(0, \infty)$.

Proof. $k \mapsto k^\gamma$ is monotone for $\gamma \geq 0$; divide by n and take roots. $\square$

Theorem 337. If $\text{bias} \geq 0$ and $\text{var}(\cdot, n) \geq 0$ on a set S of depths, then $\text{gen}(k, n) \geq \max\{\text{bias}(k), \text{var}(k, n)\}$ for every $k \in S$.

Proof. Each summand is bounded by the sum because the other one is nonnegative. $\square$

Theorem 338. *Balancing principle.* Let $S \subseteq \mathbb{R}$ be a set of depths on which $\text{bias} \geq 0$ is nonincreasing and $\text{var}(\cdot, n) \geq 0$ is nondecreasing. If $k_0 \in S$ balances the two terms, $\text{bias}(k_0) = \text{var}(k_0, n)$, then $\text{gen}(k, n) \geq \text{bias}(k_0)$ for every $k \in S$, while $\text{gen}(k_0, n) = 2\text{bias}(k_0)$; hence k_0 minimizes the bound over S up to the factor 2.

Proof. Case split on $k \leq k_0$ or $k \geq k_0$: in the first case $\text{bias}(k) \geq \text{bias}(k_0)$ and $\text{var}(k, n) \geq 0$; in the second $\text{var}(k, n) \geq \text{var}(k_0, n) = \text{bias}(k_0)$ and $\text{bias}(k) \geq 0$. The identity at k_0 is the balance equation. $\square$

Theorem 339. For $\beta, \gamma > 0$ and $n > 0$, $\text{bias}(k^*) = k^{*\beta} = n^{-\beta/(2\beta+\gamma)}$.

Proof. $(n^{1/(2\beta+\gamma)})^{-\beta} = n^{-\beta/(2\beta+\gamma)}$ by the power rule. $\square$

Theorem 340. For $\beta, \gamma > 0$ and $n > 0$, $\text{var}(k^*, n) = \sqrt{k^{*\gamma}/n} = n^{-\beta/(2\beta+\gamma)}$.

Proof. $k^{*\gamma}/n = n^{\gamma/(2\beta+\gamma)-1} = n^{-2\beta/(2\beta+\gamma)}$, and the square root halves the exponent. $\square$

Theorem 341. *PP regime (??).* Let $\beta, \gamma > 0$ and $n > 0$. At the balancing depth $k^* = n^{1/(2\beta+\gamma)}$, $\text{gen}(k^*, n) = k^{*\beta} + \sqrt{k^{*\gamma}/n} = 2n^{-\beta/(2\beta+\gamma)}$, and for every depth $k > 0$, $\text{gen}(k, n) = k^{-\beta} + \sqrt{k^\gamma/n} \geq n^{-\beta/(2\beta+\gamma)}$. Thus k^* is optimal up to the factor 2 and $\text{gen}(k^*, n) \asymp n^{-\beta/(2\beta+\gamma)}$.

Proof. Both terms equal $n^{-\beta/(2\beta+\gamma)}$ at k^* (Theorem 339, Theorem 340), so the balancing principle Theorem 338 on $S = (0, \infty)$ gives both claims. $\square$

Theorem 342. For every $c > 0$, $\log x \leq cx$ for all sufficiently large x .

Proof. $\log x = o(x)$ (Mathlib: ‘Real.isLittleO_log_id_atTop’). $\square$

Theorem 343. For every $c > 0$, $\log \log n \leq c \log n$ for all sufficiently large n .

Proof. Compose Theorem 342 with $\log n \rightarrow \infty$. $\square$

Theorem 344. $\log x < x$ for $x > 0$.

Proof. $\log x \leq x - 1 < x$. $\square$

Theorem 345. For $n > 0$ and $L \geq 0$, $n^{-1/2}L^{\gamma/2} = \sqrt{L^\gamma/n}$.

Proof. $\sqrt{L^\gamma/n} = \sqrt{L^\gamma}/\sqrt{n} = L^{\gamma/2}n^{-1/2}$. $\square$

Theorem 346. For $c \geq 0$, $\sqrt{cx/n} = \sqrt{c} \sqrt{x/n}$.

Proof. Multiplicativity of the square root. $\square$

Theorem 347. $n^{-1/2}(\log n)^{\gamma/2} \geq 0$ for $n \geq 1$.

Proof. Product of two nonnegative powers. $\square$

Theorem 348. For $\alpha > 0$ and $n > 1$, $e^{-\alpha k^*} = n^{-1/2}(\log n)^{\gamma/2}$ exactly.

Proof. $-\alpha k^* = -\frac{1}{2} \log n + \frac{\gamma}{2} \log \log n$; exponentiate. $\square$

Theorem 349. For $\alpha, \gamma > 0$ and $\log n \geq 1$, $k^* \leq \log n / (2\alpha)$.

Proof. $\gamma \log \log n \geq 0$ when $\log n \geq 1$. $\square$

Theorem 350. For $\alpha, \gamma > 0$, $n > 1$, $\log n \geq 1$ and $\gamma \log \log n \leq \log n$: $\sqrt{k^*/n} \leq \sqrt{(2\alpha)^{-\gamma}} n^{-1/2}(\log n)^{\gamma/2}$.

Proof. $0 \leq k^* \leq \log n / (2\alpha)$, so $k^{*\gamma} \leq (2\alpha)^{-\gamma}(\log n)^{\gamma}$; divide by n and take roots. $\square$

Theorem 351. For $\alpha, \gamma > 0$, eventually in n : $n^{-1/2}(\log n)^{\gamma/2} \leq \text{gen}(k^*, n) \leq (1 + \sqrt{(2\alpha)^{-\gamma}}) n^{-1/2}(\log n)^{\gamma/2}$.

Proof. The bias term equals the rate exactly (Theorem 348) and the variance term is at most $\sqrt{(2\alpha)^{-\gamma}}$ times the rate (Theorem 350); the side conditions hold for large n by Theorem 343. $\square$

Theorem 352. If $0 \leq g$ eventually, $c > 0$, and $g \leq f \leq c g$ eventually, then $f = \Theta(g)$.

Proof. Both O -directions, with constants c and 1. $\square$

Theorem 353. *EP regime (??).* Let $\alpha, \gamma > 0$ and $k^*(n) = \frac{1}{2\alpha}(\log n - \gamma \log \log n)$. Then $\text{gen}(k^*, n) = e^{-\alpha k^*} + \sqrt{k^*/n} \asymp n^{-1/2}(\log n)^{\gamma/2}$ as $n \rightarrow \infty$ (more precisely, between 1 and $1 + (2\alpha)^{-\gamma/2}$ times the rate).

Proof. Apply Theorem 352 to the two-sided bound Theorem 351. $\square$

Theorem 354. EP regime, upper bound: $\text{gen}(k^*, n) = O(n^{-1/2}(\log n)^{\gamma/2})$.

Proof. The O -half of Theorem 353. $\square$

Theorem 355. For $\alpha > 0$ and $n > e$, $e^{-\alpha k^*} = \sqrt{\log \log n / n}$ exactly.

Proof. $-\alpha k^* = -\frac{1}{2} \log n + \frac{1}{2} \log \log \log n$; exponentiate and rewrite $n^{-1/2}(\log \log n)^{1/2} = \sqrt{\log \log n / n}$. $\square$

Theorem 356. For $\alpha > 0$ and $\log n \geq e$: $0 < k^* \leq \log n / (2\alpha)$ and $\log k^* \leq \log \log n + |\log(2\alpha)|$.

Proof. $\log \log \log n \geq 0$ gives the upper bound; $\log \log \log n < \log \log n < \log n$ gives positivity; then $\log k^* \leq \log(\log n / (2\alpha)) = \log \log n - \log(2\alpha)$. $\square$

Theorem 357. For $\alpha > 0$ and $\log n \geq e$: $\sqrt{\log k^* / n} \leq \sqrt{1 + |\log(2\alpha)|} \sqrt{\log \log n / n}$.

Proof. $\log k^* \leq \log \log n + |\log 2\alpha| \leq (1 + |\log 2\alpha|) \log \log n$ because $\log \log n \geq 1$ (Theorem 356). $\square$

Theorem 358. For $\alpha > 0$, eventually in n : $\sqrt{\log \log n / n} \leq \text{gen}(k^*, n) \leq (1 + \sqrt{1 + |\log(2\alpha)|}) \sqrt{\log \log n / n}$.

Proof. Bias equals the rate (Theorem 355); variance is at most $\sqrt{1 + |\log 2\alpha|}$ times the rate (Theorem 357). $\square$

Theorem 359. *EL regime (??).* Let $\alpha > 0$ and $k^*(n) = \frac{1}{2\alpha}(\log n - \log \log \log n)$. Then $\text{gen}(k^*, n) = e^{-\alpha k^*} + \sqrt{\log k^*/n} \asymp \sqrt{\log \log n/n}$ as $n \rightarrow \infty$.

Proof. Apply Theorem 352 to Theorem 358. □

Theorem 360. *EL regime, upper bound:* $\text{gen}(k^*, n) = O(\sqrt{\log \log n/n})$.

Proof. The O -half of Theorem 359. □

Theorem 361. For $\beta > 0$, $n > 0$ and $L = \log(2\beta n) > 0$: $k^{*- \beta} = \sqrt{L/(2\beta n)}$.

Proof. $k^{*- \beta} = (2\beta n/L)^{-1/2} = (L/(2\beta n))^{1/2}$. □

Theorem 362. For $\beta > 0$, $n > 0$ and $L = \log(2\beta n) > 0$: $\log k^* = (L - \log L)/(2\beta)$.

Proof. $\log$ of a power and of a quotient. □

Theorem 363. The PL balancing equation $k^{2\beta} \log k \asymp n$ holds exactly up to the factor $1 - \log L/L$: for $\beta > 0$, $n > 0$ and $L = \log(2\beta n) > 0$, $k^{*2\beta} \log k^* = n(1 - \log L/L)$.

Proof. $k^{*2\beta} = 2\beta n/L$ and $\log k^* = (L - \log L)/(2\beta)$ (Theorem 362); multiply out. □

Theorem 364. $k^{*2\beta} \log k^*/n \rightarrow 1$ as $n \rightarrow \infty$, i.e. $k^{*2\beta} \log k^* \sim n$.

Proof. By Theorem 363 the ratio is $1 - \log L/L$ with $L = \log(2\beta n) \rightarrow \infty$, and $\log L/L \rightarrow 0$. □

Theorem 365. For $\beta > 0$, $n > 0$ and $L = \log(2\beta n) \geq 1$: $\sqrt{\log k^*/n} \leq \sqrt{L/(2\beta n)}$.

Proof. $\log k^* = (L - \log L)/(2\beta) \leq L/(2\beta)$ since $\log L \geq 0$. □

Theorem 366. For $\beta > 0$ and $\log n \geq \max\{1, 2|\log 2\beta|\}$: $\frac{1}{2} \log n \leq \log(2\beta n) \leq (1 + |\log 2\beta|) \log n$.

Proof. $\log(2\beta n) = \log 2\beta + \log n$ and $-|c| \leq c \leq |c| \leq |c| \log n$. □

Theorem 367. For $\beta > 0$, eventually in n : $\sqrt{1/(4\beta)} \sqrt{\log n/n} \leq \text{gen}(k^*, n) \leq 2\sqrt{(1 + |\log 2\beta|)/(2\beta)} \sqrt{\log n/n}$.

Proof. Both terms are at most $\sqrt{L/(2\beta n)}$ with $L = \log(2\beta n)$ (Theorem 361, Theorem 365), the bias term equals it, and $\frac{1}{2} \log n \leq L \leq (1 + |\log 2\beta|) \log n$ (Theorem 366). □

Theorem 368. *PL regime (??).* Let $\beta > 0$ and $k^*(n) = (2\beta n / \log(2\beta n))^{1/(2\beta)}$. Then $\text{gen}(k^*, n) = k^{*- \beta} + \sqrt{\log k^*/n} \asymp \sqrt{\log n/n}$ as $n \rightarrow \infty$.

Proof. Rescale the lower bound of Theorem 367 by $\sqrt{1/(4\beta)}^{-1}$ and apply Theorem 352 to the function $\sqrt{1/(4\beta)} \sqrt{\log n/n}$, which is $\Theta(\sqrt{\log n/n})$. □

Theorem 369. *PL regime, upper bound:* $\text{gen}(k^*, n) = O(\sqrt{\log n/n})$.

Proof. The O -half of Theorem 368. □

Theorem 370. For $n > 1$, $\sqrt{\log n/n} = n^{-1/2}(\log n)^{1/2}$.

Proof. Theorem 345 with $\gamma = 1$. □

Theorem 371. *Ordering note, first half.* If $\gamma \leq 1$ then the EP rate is no worse than the PL rate: $n^{-1/2}(\log n)^{\gamma/2} = O(\sqrt{\log n/n})$.

Proof. For $\log n \geq 1$, $(\log n)^{\gamma/2} \leq (\log n)^{1/2}$. $\square$

Theorem 372. *Ordering note, second half.* If $\gamma \geq 1$ then the PL rate is no worse than the EP rate: $\sqrt{\log n/n} = O(n^{-1/2}(\log n)^{\gamma/2})$.

Proof. For $\log n \geq 1$, $(\log n)^{1/2} \leq (\log n)^{\gamma/2}$. $\square$

Theorem 373. *Ordering note (??).* The EP and PL balanced rates are not uniformly ordered: $EP \lesssim PL$ when $\gamma \leq 1$, and $PL \lesssim EP$ when $\gamma \geq 1$; for $\gamma = 1$ they coincide up to constants.

Proof. Theorem 371 and Theorem 372. $\square$

Chapter 7

Examples

7.1 Shared glue for the worked examples (regimes)

Theorem 374. $L[f] \geq 0$ since the loss is nonnegative.

Proof. □

Theorem 375. (*Approximation transfer, ‘eq:approx-transfer’.*) Let ℓ be β_ℓ -Lipschitz in its first argument, $\mathcal{H}$ and $\mathcal{C} \neq \emptyset$ classes of measurable functions, and $B \in \mathbb{R}$. If every $c \in \mathcal{C}$ is uniformly approximable from $\mathcal{H}$ within $B + \varepsilon$ for every $\varepsilon > 0$ (in particular if $\sup_{c \in \mathcal{C}} \inf_{f \in \mathcal{H}} \|f - c\|_\infty \leq B$), then

$$\varepsilon_{\text{model}} = \inf_{\mathcal{H}} L - \inf_{\mathcal{C}} L \leq \beta_\ell B.$$

Proof. For $c \in \mathcal{C}$ and $\varepsilon > 0$ pick $f \in \mathcal{H}$ with $\|f - c\|_\infty \leq B + \varepsilon$; then $\inf_{\mathcal{H}} L \leq L[f] \leq L[c] + \beta_\ell(B + \varepsilon)$ by ‘lem:risk-lipschitz’. Let $\varepsilon \rightarrow 0$ and take the infimum over c . □

Theorem 376. For $h = \langle w, \Phi(\cdot) \rangle \in H_R(\Phi)$, $|h(x)| \leq R \|\Phi(x)\|$.

Proof. □

Theorem 377. If Φ is L_Φ -Lipschitz and $R \geq 0$ then every $h \in H_R(\Phi)$ is RL_Φ -Lipschitz.

Proof. □

Theorem 378. For $R > 0$, $H_R(\Phi) = H_1(R\Phi)$: the radius can be absorbed into the feature map.

Proof. □

Theorem 379. (‘prop:hilbert-sg’ for the radius- R class.) If Φ is L_Φ -Lipschitz and $R > 0$, then $H_R(\Phi)$ satisfies the sub-Gaussian increment condition ‘ass:sg-increment-main’ with $A_H = 1$ and $L = RL_\Phi$, for every hidden-layer class $\mathfrak{F}$.

Proof. $H_R(\Phi) = H_1(R\Phi)$ and $R\Phi$ is RL_Φ -Lipschitz. □

Theorem 380. If $|g(x)| \leq C$ for all $g \in G$ and x ($C \geq 0$), then the Rademacher averages $\{\frac{1}{n} \sum_i \sigma_i g(x_i) : g \in G\}$ are bounded above by C , for every sample and sign pattern.

Proof. □

Theorem 381. (*Rademacher complexity of the linear readout class.*) If $\|\Phi(x)\| \leq M$ for all x ($R, M \geq 0$), then for every sample S of size n ,

$$\hat{\mathfrak{R}}_S(H_R(\Phi)) \leq \frac{RM}{\sqrt{n}}.$$

(Proof: $\hat{\mathfrak{R}}_S(H_R(\Phi)) \leq R \mathbb{E}_\sigma \|n^{-1} \sum_i \sigma_i \Phi(x_i)\| \leq R \sqrt{\sum_i \|\Phi(x_i)\|^2 / n} \leq RM / \sqrt{n}$, via FoML's `hilbertPredictor_empiricalRademacherComplexity_le.`)

Proof. □

Theorem 382. If F is finite then every word ball $B(k, F)$ is finite.

Proof. □

Theorem 383. If Φ is continuous and every $f \in \mathfrak{F}$ is continuous, then every element of $H_R(\Phi) \circ \mathfrak{F}$ is (Borel) measurable.

Proof. □

Theorem 384. If $\|\Phi\| \leq M$ and $R \geq 0$ then $|g(x)| \leq RM$ for every $g \in H_R(\Phi) \circ \mathfrak{F}$ and every x .

Proof. □

Definition 385. A hidden-layer class $\mathfrak{F}$ has a countable uniformly dense subset if there is a countable $D \subseteq \mathfrak{F}$ such that every $f \in \mathfrak{F}$ is within uniform distance ε of some $g \in D$, for every $\varepsilon > 0$.

Theorem 386. A finite hidden-layer class has a countable uniformly dense subset (itself).

Proof. □

Theorem 387. A hidden-layer class which is totally bounded in d_∞ has a countable uniformly dense subset (it is separable in the pseudo-metrizable space $(\mathcal{X}^X, d_\infty)$).

Proof. □

Theorem 388. (*Sup-norm separability of $H_R(\Phi) \circ \mathfrak{F}$.*) Let $\mathcal{H}$ be a separable real inner product space, $\Phi : \mathcal{X} \rightarrow \mathcal{H}$ be L_Φ -Lipschitz with $\|\Phi\| \leq M$, $R \geq 0$, and let $\mathfrak{F}$ have a countable uniformly dense subset. Then $H_R(\Phi) \circ \mathfrak{F}$ is sup-norm separable: the countable set $\{\langle w, \Phi(g(\cdot)) \rangle : w \in W, g \in D\}$, with W a countable dense subset of the ball of radius R , is uniformly dense, since $|\langle w, \Phi(fx) \rangle - \langle w', \Phi(gx) \rangle| \leq \|w - w'\|M + RL_\Phi d(fx, gx)$.

Proof. □

Theorem 389. A class which is totally bounded in d_∞ is totally bounded in d_S for every sample S (the identity is 1-Lipschitz, 'lem:to-emp-space-lipschitz').

Proof. □

Theorem 390. Every element of the uniform closure of a class of measurable functions is measurable (a uniform limit of measurable functions is a pointwise limit).

Proof. □

Definition 391. The estimation-plus-deviation term of ‘thm:bv’ (with the constants of ‘thm:bv-general’), as a function of an upper bound B for $\hat{\mathfrak{R}}_S(\mathcal{H})$:

$$\text{dev}_{\ell,n,\delta}(B) := 4\beta_\ell B + 6b\sqrt{\frac{2\log(4/\delta)}{n}}.$$

(All regime propositions below are stated in terms of this quantity, so that the constants of ‘thm:bv’ enter in one place only.)

Theorem 392. $B \mapsto \text{dev}_{\ell,n,\delta}(B)$ is nondecreasing when $\beta_\ell \geq 0$.

Proof. □

Theorem 393. (*‘thm:bv’ with $\iota = \text{id}$ and explicit bounds.*) Let $\mathcal{H}$ be a sup-norm separable, pointwise bounded class of measurable functions, $\mathcal{C} \neq \emptyset$ a class of measurable benchmarks, ℓ measurable, bounded by $b > 0$ and β_ℓ -Lipschitz, $n \geq 1$, $\eta \geq 0$, $\delta \in (0, 1)$. Suppose $\hat{\mathfrak{R}}_S(\mathcal{H}) \leq B$ for every sample S of size n and $\varepsilon_{\text{model}} \leq \text{bias}$. Then with probability at least $1 - \delta$ over $\mathcal{D} \sim P^{\otimes n}$, every η -empirical minimizer $\hat{f} \in \mathcal{H}$ satisfies

$$L[\hat{f}] - \inf_{\mathcal{C}} L \leq \text{bias} + \eta + \text{dev}_{\ell,n,\delta}(B) = \text{bias} + \eta + 4\beta_\ell B + 6b\sqrt{\frac{2\log(4/\delta)}{n}}.$$

Proof. ‘thm:bv-general’ with $\iota = \text{id}$, $d_T = 0$, $\varepsilon_{\text{imp}} = 0$, then monotonicity of the bad event. □

Theorem 394. (*Truncation, Sec. ‘sec:examples-regime’.*) Let every $h \in H$ be L_H -Lipschitz, every $f \in F$ be c -Lipschitz, and $d(x, y) \leq D_{\mathcal{X}}$ on $\mathcal{X}$. Then every $g = h \circ w \in H \circ \langle F \rangle$ is within uniform distance $L_H c^k D_{\mathcal{X}}$ of $\mathcal{H}_k = H \circ B(k, F)$: if $w = w_2 \circ w_1$ with $|w_2| = k$ then $|h(w_2(w_1 x)) - h(w_2 x)| \leq L_H c^k d(w_1 x, x)$.

Proof. □

Theorem 395. (*Truncation for the uniform closure.*) Under the hypotheses of ‘lem:truncation-bias-exact’, every g in the uniform closure $\overline{H \circ \langle F \rangle}^{d_\infty}$ satisfies, for every $\varepsilon > 0$, $\inf_{f \in \mathcal{H}_k} \|f - g\|_\infty \leq L_H c^k D_{\mathcal{X}} + \varepsilon$.

Proof. □

Definition 396. The depth-independent entropy integral of ‘cor:profile-p1’: $V_\infty(F) := \int_0^{\text{diam}(\mathcal{X})} \sqrt{\log N^{\text{ext}}(\langle F \rangle, d_\infty, \varepsilon/2)}$,

Definition 397. The paper’s condition $V_\infty(F) < \infty$: the majorant $\varepsilon \mapsto \sqrt{\log N^{\text{ext}}(\langle F \rangle, d_\infty, \varepsilon/2)}$ is interval-integrable on $[0, \text{diam}(\mathcal{X})]$.

Theorem 398. (*EL balance with saturated variance.*) For $0 < \theta < 1$ and $n \geq 1$, the depth $k := \lceil \log n / (2 \log(1/\theta)) \rceil$ satisfies $\theta^k \leq n^{-1/2}$: the bias θ^k is balanced against the saturated variance $n^{-1/2}$, and $k = \frac{\log n}{2 \log(1/\theta)} + O(1)$.

Proof. $k \log(1/\theta) \geq \frac{1}{2} \log n = \log \sqrt{n}$, so $(1/\theta)^k \geq \sqrt{n}$. □

Theorem 399. (*PL balance with saturated variance, $p = 1$.*) For $n \geq 1$ the depth $k := \lceil \sqrt{n} \rceil \geq 1$ satisfies $1/k \leq n^{-1/2}$: the bias k^{-1} is balanced against the saturated variance $n^{-1/2}$, and $k \asymp n^{1/2}$.

Proof. □

7.2 Implementation with controlled uniform error

Definition 400. For a list of layers $u = [f_m, \dots, f_1]$ (an explicit representation of a word) the composed map is $f_u = f_m \circ \dots \circ f_1$, with $f_{[]} = \text{id}$. In Lean: ‘compList [] = id’ and ‘compList (f :: u) = f ∘ compList u’ (the head of the list acts last, matching the recursion $B(k+1, F) = B(k, F) \cup F \circ B(k, F)$).

Theorem 401. $f_{[]} = \text{id}$.

Proof. □

Theorem 402. $f_{f::u} = f \circ f_u$.

Proof. □

Definition 403. Given an implementation $f \mapsto \tilde{f}$ of the layers, the implemented word of a representation $u = [f_m, \dots, f_1]$ is $\tilde{f}_u = \tilde{f}_m \circ \dots \circ \tilde{f}_1$.

Theorem 404. $\tilde{f}_{[]} = \text{id}$.

Proof. □

Theorem 405. $\tilde{f}_{f::u} = \tilde{f} \circ \tilde{f}_u$.

Proof. □

Theorem 406. If all layers of u lie in F and $|u| \leq k$ then $f_u \in B(k, F)$.

Proof. Induction on u : $\text{id} \in B(k, F)$, and $f \circ f_u \in B(k'+1, F)$ when $f \in F$ and $f_u \in B(k', F)$. □

Theorem 407. Every $g \in B(k, F)$ has a representation $g = f_m \circ \dots \circ f_1$ with $f_i \in F$ and $m \leq k$, i.e. $g = f_u$ for a list u of elements of F with $|u| \leq k$.

Proof. Induction on k along the recursion $B(k+1, F) = B(k, F) \cup F \circ B(k, F)$. □

Theorem 408. (Telescoping error propagation.) Let every $f \in F$ be Λ -Lipschitz and let $\tilde{f}$ satisfy $d_\infty(f, \tilde{f}) \leq \delta$ for $f \in F$, where $\delta \geq 0$. Then for every representation $u = [f_m, \dots, f_1]$ of layers in F and every x ,

$$d(f_m \circ \dots \circ f_1(x), \tilde{f}_m \circ \dots \circ \tilde{f}_1(x)) \leq \delta \sum_{i=0}^{m-1} \Lambda^i.$$

Proof. Induction on u . For $u = f :: u'$ write $y = f_{u'}(x)$, $\tilde{y} = \tilde{f}_{u'}(x)$; then $d(f(y), \tilde{f}(\tilde{y})) \leq d(f(y), f(\tilde{y})) + d(f(\tilde{y}), \tilde{f}(\tilde{y})) \leq \Lambda d(y, \tilde{y}) + \delta \leq \Lambda \delta \sum_{i < m} \Lambda^i + \delta = \delta \sum_{i < m+1} \Lambda^i$. □

Theorem 409. (Existence of the layerwise implementation map.) Given implementations $f \mapsto \tilde{f}$ of the transitions and $h \mapsto \tilde{h}$ of the output layers, there is a map $\iota : \mathbb{R}^{\mathcal{X}} \rightarrow \mathbb{R}^{\mathcal{X}}$ such that every $g \in \mathcal{H}_k$ has a representation $g = h \circ f_m \circ \dots \circ f_1$ with $h \in H$, $f_i \in F$, $m \leq k$ and $\iota(g) = \tilde{h} \circ \tilde{f}_m \circ \dots \circ \tilde{f}_1$.

Proof. For each $g \in \mathcal{H}_k$ choose (by the axiom of choice) a representation, using ‘lem:exists-comp-list-of-mem-wordball’; outside $\mathcal{H}_k$ the value of ι is irrelevant. □

Theorem 410. (Layerwise implementation.) Let $F \subseteq \mathcal{X}^{\mathcal{X}}$ with $\text{lip}(f) \leq \Lambda$ for all $f \in F$, and $H \subseteq \mathbb{R}^{\mathcal{X}}$ with $\text{lip}(h) \leq L_H$ for all $h \in H$. Suppose every $f \in F$ is assigned an implemented map $\tilde{f}$ with $d_{\infty}(f, \tilde{f}) \leq \delta$ ($\delta \geq 0$) and every $h \in H$ an implemented $\tilde{h}$ with $\|h - \tilde{h}\|_{\infty} \leq \delta_H$. Let ι be an implementation map that sends each $g \in \mathcal{H}_k$, for one representation $g = h \circ f_m \circ \dots \circ f_1$ with $m \leq k$, to $\tilde{h} \circ \tilde{f}_m \circ \dots \circ \tilde{f}_1$. Then

$$\varepsilon_{\text{imp}}(k) \leq \delta_H + L_H \delta \sum_{i=0}^{k-1} \Lambda^i.$$

(Stated for an arbitrary ι with this property; such an ι exists by ‘lem:impl-map-exists’.)

Proof. Fix $g = h \circ f_u \in \mathcal{H}_k$ and x ; put $T_0 = f_u(x)$, $T_m = \tilde{f}_u(x)$. By ‘lem:impl-word-error’, $d(T_0, T_m) \leq \delta \sum_{i < m} \Lambda^i \leq \delta \sum_{i < k} \Lambda^i$ (as $\Lambda \geq 0$ and $m \leq k$). Then $|h(T_0) - \tilde{h}(T_m)| \leq |h(T_0) - h(T_m)| + |h(T_m) - \tilde{h}(T_m)| \leq L_H d(T_0, T_m) + \delta_H$; take suprema over x and g . $\square$

Theorem 411. Under the hypotheses of ‘prop:implementation-b’ there exists an implementation map ι with $\varepsilon_{\text{imp}}(k) \leq \delta_H + L_H \delta \sum_{i < k} \Lambda^i$.

Proof. Combine ‘lem:impl-map-exists’ and ‘prop:implementation-b’. $\square$

Theorem 412. If $0 \leq \Lambda \leq 1$ then $\sum_{i < k} \Lambda^i \leq k$.

Proof. $\square$

Theorem 413. If $0 \leq \Lambda < 1$ then $\sum_{i < k} \Lambda^i \leq 1/(1 - \Lambda)$.

Proof. $\sum_{i < k} \Lambda^i = (1 - \Lambda^k)/(1 - \Lambda) \leq 1/(1 - \Lambda)$. $\square$

Theorem 414. (Non-expanding transitions.) If moreover $\Lambda \leq 1$, then $\varepsilon_{\text{imp}}(k) \leq \delta_H + L_H k \delta$.

Proof. ‘prop:implementation-b’ and $\sum_{i < k} \Lambda^i \leq k$. $\square$

Theorem 415. (Contractive transitions.) If moreover $\Lambda < 1$, then $\varepsilon_{\text{imp}}(k) \leq \delta_H + L_H \delta/(1 - \Lambda)$.

Proof. ‘prop:implementation-b’ and $\sum_{i < k} \Lambda^i \leq 1/(1 - \Lambda)$. $\square$

Definition 416. $\text{relu}(t) = \max\{0, t\}$.

Definition 417. A one-hidden-layer ReLU layer on $\mathbb{R}^d$ with hidden index set m (width $|m|$) is $x \mapsto W_2 \text{relu}(W_1 x) + b$, with $W_1 \in \mathbb{R}^{m \times d}$, $W_2 \in \mathbb{R}^{d \times m}$, $b \in \mathbb{R}^d$ and relu applied coordinatewise.

Theorem 418. $t = \text{relu}(t) - \text{relu}(-t)$.

Proof. $\square$

Theorem 419. (Exact realization of an affine map.) For $A \in \mathbb{R}^{d \times d}$ and $b \in \mathbb{R}^d$, with $W_1 = [I_d; -I_d] \in \mathbb{R}^{2d \times d}$ and $W_2 = [A, -A] \in \mathbb{R}^{d \times 2d}$,

$$W_2 \text{relu}(W_1 x) + b = Ax + b \quad (x \in \mathbb{R}^d),$$

i.e. $x \mapsto Ax + b$ is one ReLU layer of width $2d$.

Proof. $\text{relu}(W_1 x) = (\text{relu}(x), \text{relu}(-x))$ and $[A, -A](v_1, v_2) = A(v_1 - v_2)$; conclude with $\text{relu}(t) - \text{relu}(-t) = t$. $\square$

Definition 420. The parameters (W_2, W_1, b) of a ReLU layer of width $2d$ on $\mathbb{R}^d$.

Definition 421. The ReLU layer of width $2d$ with parameters (W_2, W_1, b) .

Definition 422. A map $g : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a ReLU network of depth $\leq k$ and width $2d$ if it is a composition of at most k ReLU layers of width $2d$.

Theorem 423. If every layer of the representation u is affine, then f_u is a composition of $|u|$ ReLU layers of width $2d$.

Proof. Induction on u , replacing each affine layer by the ReLU layer of ‘lem:affine-eq-relu’. $\square$

Theorem 424. The identity implementation map has zero implementation error: $\varepsilon_{\text{imp}} = 0$ for $\iota = \text{id}$ on any class.

Proof. $\square$

Theorem 425. (Exact implementation of affine transitions.) Let $\mathcal{X} = \mathbb{R}^d$ and let every $f \in F$ be affine, $f(x) = Ax + b$. Then every element of $B(k, F)$ is a ReLU network of depth $\leq k$ and width $2d$, and with the identity implementation map $\iota = \text{id}$ (the abstract class $\mathcal{H}_k$ is itself the implemented class) one has $\varepsilon_{\text{imp}}(k) = 0$.

Proof. Take a representation of $g \in B(k, F)$ (‘lem:exists-comp-list-of-mem-wordball’) and apply ‘lem:comp-list-affine-eq-relu-net’; the second claim is ‘lem:impl-error-id’. $\square$

Theorem 426. (Net-based implementation.) Let $\mathcal{H} \subseteq \mathbb{R}^{\mathcal{X}}$ be totally bounded in the uniform norm and let $\mathcal{A} \subseteq \mathbb{R}^{\mathcal{X}}$ be uniformly dense on $\mathcal{H}$ (for every $g \in \mathcal{H}$ and $\eta > 0$ there is $a \in \mathcal{A}$ with $\|g - a\|_{\infty} \leq \eta$). Then for every $\varepsilon > 0$ there exist a finite class $\mathcal{H}_{\text{imp}}^{\varepsilon} \subseteq \mathcal{A}$ with $|\mathcal{H}_{\text{imp}}^{\varepsilon}| \leq N(\mathcal{H}, \|\cdot\|_{\infty}, \varepsilon/2)$ and an implementation map $\iota : \mathcal{H} \rightarrow \mathcal{H}_{\text{imp}}^{\varepsilon}$ with $\sup_{g \in \mathcal{H}} \|g - \iota(g)\|_{\infty} \leq \varepsilon$. (The paper’s compact domain K and the continuity of the functions are only used to guarantee total boundedness and density, which are the hypotheses here.)

Proof. Let $C \subseteq \mathcal{H}$ be a minimal internal $\varepsilon/2$ -net (finite by total boundedness). For each $c \in C$ choose $a_c \in \mathcal{A}$ with $\|c - a_c\|_{\infty} \leq \varepsilon/2$, set $\mathcal{H}_{\text{imp}} = \{a_c : c \in C\}$ and $\iota(g) = a_{c(g)}$ where $c(g) \in C$ satisfies $\|g - c(g)\|_{\infty} \leq \varepsilon/2$. Then $\|g - \iota(g)\|_{\infty} \leq \varepsilon/2 + \varepsilon/2$. $\square$

7.3 Output-layer realization of hidden geometry

Definition 427. For a real Hilbert space $\mathcal{H}$, a feature map $\Phi : \mathcal{X} \rightarrow \mathcal{H}$ and a radius $R_H \geq 0$, the norm-bounded linear output-layer class is

$$H_{R_H}(\Phi) = \{h_w(x) = \langle w, \Phi(x) \rangle_{\mathcal{H}} : \|w\|_{\mathcal{H}} \leq R_H\}.$$

Definition 428. The reachable sample set of a hidden class B_k on the sample $S = (x_1, \dots, x_n)$ is $U_{k,S} = \{f(x_i) : f \in B_k, i \in [n]\}$.

Theorem 429. (Global scalar observable.) Fix a sample $S = (x_i)_{i=1}^n$, a hidden class B_k and $U_{k,S} = \{f(x_i) : f \in B_k, i \in [n]\}$. Let $\Phi : \mathcal{X} \rightarrow \mathcal{H}$ and assume there are $u \in \mathcal{H}$ with $\|u\| \leq R_H$ and constants $\kappa, R_{\Phi} > 0$ such that

$$|\langle u, \Phi(x) - \Phi(y) \rangle| \geq \kappa d(x, y) \quad (x, y \in U_{k,S}), \quad \sup_{x \in \mathcal{X}} \|\Phi(x)\| \leq R_{\Phi}.$$

Then the readout-realization assumption holds for $H_{R_H}(\Phi)$ on B_k with constants κ and $R_{\text{out}} = R_H R_{\Phi}$, with the single choice $h_f = h_u = \langle u, \Phi(\cdot) \rangle$ for all $f \in B_k$: $\|h_u \circ f - h_u \circ g\|_S \geq \kappa d_S(f, g)$ and $\|h_u \circ f\|_{S, \infty} \leq R_H R_{\Phi}$ for $f, g \in B_k$.

Proof. Termwise, $\kappa^2 d(f(x_i), g(x_i))^2 \leq |\langle u, \Phi(f(x_i)) - \Phi(g(x_i)) \rangle|^2$ by co-Lipschitzness on $U_{k,S}$; average and take square roots. The bound $|\langle u, \Phi(f(x)) \rangle| \leq \|u\| \|\Phi(f(x))\| \leq R_H R_\Phi$ is Cauchy–Schwarz. $\square$

Theorem 430. (*Map-dependent finite-set interpolation criterion for linear heads.*) Fix $f_1, \dots, f_M \in B_k$ and a sample $S = (x_i)_{i=1}^n$, and let $\Phi : \mathcal{X} \rightarrow \mathcal{H}$. Assume that for each j the evaluation operator $T_j : w \mapsto (\langle w, \Phi(f_j(x_i)) \rangle)_{i \in [n]}$ admits a right inverse of norm $\leq \Lambda_j$: for every $c \in \mathbb{R}^n$ there is $w \in \mathcal{H}$ with $\|w\| \leq \Lambda_j \|c\|_2$ and $\langle w, \Phi(f_j(x_i)) \rangle = c_i$ for all i . Then for every family of code vectors $u^{(1)}, \dots, u^{(M)} \in \mathbb{R}^n$ with $\Lambda_j \|u^{(j)}\|_2 \leq R_H$ there are output layers $h_j \in H_{R_H}(\Phi)$ with $h_j(f_j(x_i)) = u_i^{(j)}$. Consequently, if $(\frac{1}{n} \sum_i |u_i^{(j)} - u_i^{(\ell)}|^2)^{1/2} \geq \rho$ for $j \neq \ell$ then $\|h_j \circ f_j - h_\ell \circ f_\ell\|_S \geq \rho$ for $j \neq \ell$, and if $\max_{j,i} |u_i^{(j)}| \leq R_{\text{out}}$ (with $R_{\text{out}} \geq 0$) then $\|h_j \circ f_j\|_{S,\infty} \leq R_{\text{out}}$.

Proof. Set $w_j := R_j u^{(j)}$ and $h_j := \langle w_j, \Phi(\cdot) \rangle$; then $h_j(f_j(x_i)) = u_i^{(j)}$ and $\|w_j\| \leq \Lambda_j \|u^{(j)}\|_2 \leq R_H$. The separation and boundedness claims follow by evaluating on the sample. $\square$

Theorem 431. If $|u_i| \leq R_{\text{out}}$ for all $i \in [n]$ then $\|u\|_2 \leq \sqrt{n} R_{\text{out}}$.

Proof. $\square$

Theorem 432. (*Bounded codes.*) In ‘prop:linear-interpolation’, for R_{out} -bounded codes it suffices to take $R_H \geq \sqrt{n} R_{\text{out}} \max_j \Lambda_j$ (with $\Lambda_j \geq 0$), which is independent of the number M of maps.

Proof. ‘lem:code-norm-le’ gives $\Lambda_j \|u^{(j)}\|_2 \leq \Lambda_j \sqrt{n} R_{\text{out}} \leq R_H$; apply ‘prop:linear-interpolation’. $\square$

Theorem 433. (*Readout realization from interpolation.*) Under the hypotheses of ‘prop:linear-interpolation’, if moreover $\kappa d_S(f_j, f_\ell) \leq \rho$ for all $j \neq \ell$, then the readout-realization assumption holds for $H_{R_H}(\Phi)$ on $B_k = \{f_1, \dots, f_M\}$ with constants κ and R_{out} .

Proof. Take the h_j of ‘prop:linear-interpolation’ and set $h_{f_j} := h_j$ (choosing an index j for each element of the range). For $f_j \neq f_\ell$ one has $j \neq \ell$, so $\kappa d_S(f_j, f_\ell) \leq \rho \leq \|h_j \circ f_j - h_\ell \circ f_\ell\|_S$; for $f_j = f_\ell$ the left-hand side vanishes. $\square$

Theorem 434. (*Right inverse from a well-conditioned Gram matrix.*) Let $\varphi_1, \dots, \varphi_n \in \mathcal{H}$ and suppose the Gram matrix $G = (\langle \varphi_i, \varphi_{i'} \rangle)_{i,i'}$ satisfies $c^\top G c \geq \lambda_{\min} \|c\|_2^2$ for all $c \in \mathbb{R}^n$, with $\lambda_{\min} > 0$. Then for every $c \in \mathbb{R}^n$ there is $w \in \mathcal{H}$ with $\langle w, \varphi_i \rangle = c_i$ for all i and $\|w\| \leq \lambda_{\min}^{-1/2} \|c\|_2$.

Proof. G is injective (if $Ga = 0$ then $\lambda_{\min} \|a\|^2 \leq a^\top G a = 0$), hence surjective: pick a with $Ga = c$ and set $w = \sum_i a_i \varphi_i$. Then $\langle w, \varphi_i \rangle = (Ga)_i = c_i$ and $p := \|w\|^2 = a^\top G a = a^\top c$. From $\lambda_{\min} \|a\|^2 \leq p$ and $p^2 \leq \|a\|^2 \|c\|^2$ (Cauchy–Schwarz) we get $\lambda_{\min} p \leq \|c\|^2$, i.e. $\|w\| \leq \lambda_{\min}^{-1/2} \|c\|_2$. $\square$

Theorem 435. (*RKHS / kernel output layer.*) Let $\Phi : \mathcal{X} \rightarrow \mathcal{H}$ be a feature map (e.g. the canonical feature map of a positive definite kernel K , so that $\langle \Phi(a), \Phi(a') \rangle = K(a, a')$) and fix $f_1, \dots, f_M$ and a sample S . Suppose that for each j the Gram matrix $G_j = (\langle \Phi(f_j(x_i)), \Phi(f_j(x_{i'})) \rangle)_{i,i'}$ satisfies $c^\top G_j c \geq \lambda_{\min,j} \|c\|_2^2$ with $\lambda_{\min,j} > 0$. Then ‘prop:linear-interpolation’ applies with $\Lambda_j = \lambda_{\min,j}^{-1/2}$: for all codes $u^{(j)}$ with $\lambda_{\min,j}^{-1/2} \|u^{(j)}\|_2 \leq R_H$, ρ -separated and R_{out} -bounded, there are $h_j \in H_{R_H}(\Phi)$ with $h_j(f_j(x_i)) = u_i^{(j)}$, $\|h_j \circ f_j - h_\ell \circ f_\ell\|_S \geq \rho$ ($j \neq \ell$) and $\|h_j \circ f_j\|_{S,\infty} \leq R_{\text{out}}$.

Proof. ‘lem:gram-right-inverse’ applied to $\varphi_i = \Phi(f_j(x_i))$ gives the right-inverse hypothesis of ‘prop:linear-interpolation’ with $\Lambda_j = \lambda_{\min,j}^{-1/2}$. $\square$

Theorem 436. (*Finite-dimensional feature map.*) Let $\mathcal{H} = \mathbb{R}^m$ and $\Phi : \mathcal{X} \rightarrow \mathbb{R}^m$. If for each j the feature matrix $\Phi_j = (\Phi(f_j(x_i)))_{i \in [n]} \in \mathbb{R}^{m \times n}$ has full column rank, quantified as $\sigma_{\min}(\Phi_j)^2 = \lambda_{\min}(\Phi_j^\top \Phi_j) \geq \lambda_{\min,j} > 0$, i.e. $c^\top \Phi_j^\top \Phi_j c \geq \lambda_{\min,j} \|c\|_2^2$, then ‘prop:linear-interpolation’ applies with $\Lambda_j = \lambda_{\min,j}^{-1/2} = \sigma_{\min}(\Phi_j)^{-1}$. (This is ‘cor:rkhs-readout’ for $\mathcal{H} = \mathbb{R}^m$, since $\Phi_j^\top \Phi_j$ is the Gram matrix.)

Proof. Specialization of ‘cor:rkhs-readout’ to $\mathcal{H} = \mathbb{R}^m$. □

7.4 Chain-of-thought style symbolic computation

Definition 437. The scratchpad space is $\mathcal{X} = \mathcal{A}^{\mathbb{N}} = \{x = (x_0, x_1, \dots) : x_j \in \mathcal{A}\}$, the space of infinite symbol sequences over the alphabet $\mathcal{A}$ (a type synonym of $\mathbb{N} \rightarrow \mathcal{A}$ carrying the parameter θ of the metric below).

Definition 438. For $x \neq y$ the first mismatch index is $n(x, y) := \min\{j \geq 0 : x_j \neq y_j\}$.

Theorem 439. $x_{n(x,y)} \neq y_{n(x,y)}$.

Proof. □

Theorem 440. $L \leq n(x, y)$ if and only if $x_j = y_j$ for all $j < L$.

Proof. □

Theorem 441. $x_j = y_j$ for all $j < n(x, y)$.

Proof. □

Theorem 442. $n(x, y) = n(y, x)$.

Proof. Both indices are characterized by the same prefix-agreement property. □

Definition 443. The ultrametric on $\mathcal{A}^{\mathbb{N}}$: $d_\theta(x, y) = \theta^{n(x,y)}$ for $x \neq y$ and $d_\theta(x, x) = 0$, for a fixed $\theta \in (0, 1)$.

Theorem 444. $0 < \theta$ (as a real number).

Proof. □

Theorem 445. $\theta < 1$ (as a real number).

Proof. □

Theorem 446. For $x \neq y$, $d_\theta(x, y) = \theta^{n(x,y)}$.

Proof. □

Theorem 447. $d_\theta(x, y) = d_\theta(y, x)$.

Proof. □

Theorem 448. (Prefix characterization.) $d_\theta(x, y) \leq \theta^L$ if and only if $x_j = y_j$ for all $j < L$.

Proof. If $x = y$ both sides hold. Otherwise $\theta^{n(x,y)} \leq \theta^L \iff L \leq n(x, y)$ since $0 < \theta < 1$, and $L \leq n(x, y)$ is the prefix-agreement property. □

Theorem 449. $d_\theta(x, y) \geq 0$.

Proof. □

Theorem 450. $d_\theta(x, y) \leq 1$: the scratchpad space has diameter at most 1.

Proof. The case $L = 0$ of the prefix characterization. □

Theorem 451. (Ultrametric inequality.) $d_\theta(x, z) \leq \max\{d_\theta(x, y), d_\theta(y, z)\}$.

Proof. If $x = y$ or $y = z$ this is trivial. Otherwise let $L = \min\{n(x, y), n(y, z)\}$; then x, y and y, z agree on the first L symbols, hence so do x, z , so $d_\theta(x, z) \leq \theta^L = \max\{\theta^{n(x, y)}, \theta^{n(y, z)}\}$. □

Theorem 452. $d_\theta(x, y) = 0$ implies $x = y$.

Proof. □

Definition 453. $(\mathcal{A}^\mathbb{N}, d_\theta)$ is a metric space (indeed an ultrametric space).

Theorem 454. On $\mathcal{A}^\mathbb{N}$ the distance is d_θ .

Proof. □

Theorem 455. Two sequences are within distance θ^L if and only if they agree in the first L symbols: $d_\theta(x, y) \leq \theta^L \iff \forall j < L, x_j = y_j$.

Proof. □

Theorem 456. $d_\theta(x, y) \leq 1$, i.e. $\text{diam}(\mathcal{A}^\mathbb{N}) \leq 1$.

Proof. □

Theorem 457. $d_\theta(x, y) \leq 1$ as an extended distance.

Proof. □

Theorem 458. If $x_0 \neq y_0$ then $d_\theta(x, y) = 1$.

Proof. $x \neq y$ and $n(x, y) = 0$. □

Theorem 459. If x and y do not agree on the first L symbols then $d_\theta(x, y) \geq \theta^{L-1}$.

Proof. $x \neq y$ and $n(x, y) < L$, so $n(x, y) \leq L - 1$ and $\theta^{L-1} \leq \theta^{n(x, y)}$. □

Definition 460. The identity map $\mathcal{A}^\mathbb{N} \rightarrow (\mathcal{A}^\mathbb{N}, d_\theta)$ from the product of the discrete spaces.

Theorem 461. For a finite alphabet, $(\mathcal{A}^\mathbb{N}, d_\theta)$ is compact. (The identity from the product of discrete spaces is continuous, since a d_θ -ball is a cylinder set.)

Proof. Equip $\mathcal{A}$ with the discrete topology; $\mathcal{A}^\mathbb{N}$ with the product topology is compact (Tychonoff). The identity map to $(\mathcal{A}^\mathbb{N}, d_\theta)$ is continuous: given $\varepsilon > 0$ pick L with $\theta^L < \varepsilon$; the cylinder $\{y : y_j = x_j, j < L\}$ is a product neighbourhood of x contained in the ε -ball. The continuous image of a compact space is compact. □

Theorem 462. Since $\text{diam}(\mathcal{A}^\mathbb{N}) \leq 1$ and $d_S \leq d_\infty$, the empirical diameter of any class of self-maps of $\mathcal{A}^\mathbb{N}$ is at most 1: $D_k(S) \leq 1$.

Proof. □

Definition 463. A write step prepends a symbol: $g_a(x) := (a, x_0, x_1, \dots)$ for $a \in \mathcal{A}$.

Definition 464. The append-only step family $F_w := \{g_a : a \in \mathcal{A}\}$.

Theorem 465. $g_a(x)_0 = a$.

Proof. □

Theorem 466. $g_a(x)_{i+1} = x_i$.

Proof. □

Theorem 467. $a \mapsto g_a$ is injective, so $|F_w| = |\mathcal{A}| = m$.

Proof. □

Theorem 468. F_w is finite.

Proof. □

Theorem 469. $|F_w| = m = |\mathcal{A}|$.

Proof. □

Theorem 470. Each write step is θ -Lipschitz: $d_\theta(g_a(x), g_a(y)) \leq \theta d_\theta(x, y)$ (in fact with equality). Hence F_w is a contractive hidden-layer class with $\text{lip}(g_a) = \theta < 1$.

Proof. If $x = y$ there is nothing to prove. Otherwise $g_a(x)$ and $g_a(y)$ agree on the first $n(x, y) + 1$ symbols, so $d_\theta(g_a(x), g_a(y)) \leq \theta^{n(x, y)+1} = \theta d_\theta(x, y)$. □

Theorem 471. Each write step is non-expanding (1-Lipschitz).

Proof. □

Theorem 472. (Saturation via P1.) For a finite alphabet and every $\varepsilon > 0$, $N^{\text{ext}}(B(k, F_w), d_\infty, \varepsilon) \leq N^{\text{ext}}(\langle F_w \rangle, d_\infty, \varepsilon) < \infty$ uniformly in k .

Proof. □

Theorem 473. A word of ℓ write steps determines the first ℓ symbols of its output independently of the input: for $w \in F_w^\ell$ and all x, y , $w(x)_j = w(y)_j$ for $j < \ell$.

Proof. Induction on ℓ : for $w = g_a \circ w'$, the symbol 0 is a for both inputs and the symbols $1, \dots, \ell$ are the first $\ell - 1$ symbols of $w'(x)$, $w'(y)$. □

Definition 474. The resolution length $\ell(\varepsilon) := \lceil \log(1/\varepsilon) / \log(1/\theta) \rceil$ (a natural number; 0 for $\varepsilon \geq 1$).

Theorem 475. $\log(1/\theta) > 0$.

Proof. □

Theorem 476. $\theta^{\ell(\varepsilon)} \leq \varepsilon$ for every $\varepsilon > 0$.

Proof. $\ell(\varepsilon) \geq \log(1/\varepsilon) / \log(1/\theta)$, so $\log(1/\varepsilon) \leq \log((1/\theta)^{\ell(\varepsilon)})$ and $1/\varepsilon \leq 1/\theta^{\ell(\varepsilon)}$. □

Theorem 477. For $0 < \varepsilon \leq 1$, $\ell(\varepsilon) \leq \log(1/\varepsilon) / \log(1/\theta) + 1$.

Proof. $\lceil a \rceil < a + 1$ for $a \geq 0$; here $a \geq 0$ since $\varepsilon \leq 1$. □

Theorem 478. *(The short words cover the word ball.) For every $\varepsilon > 0$ and k , the word ball $B(\min\{k, \ell(\varepsilon)\}, F_w) \subseteq B(k, F_w)$ is an ε -cover of $B(k, F_w)$ in d_∞ : a word g_u with $|u| > \ell$ is within $\theta^\ell \leq \varepsilon$ of the word $g_{u'}$ formed by its last ℓ letters.*

Proof. Let $f \in F_w^l$ with $l \leq k$. If $l \leq \ell(\varepsilon)$ then f itself lies in the cover. Otherwise factor $f = w_2 \circ w_1$ with $w_2 \in F_w^{\ell(\varepsilon)}$ (the last $\ell(\varepsilon)$ letters); $w_2(w_1(x))$ and $w_2(x)$ agree on the first $\ell(\varepsilon)$ symbols, so $d_\infty(f, w_2) \leq \theta^{\ell(\varepsilon)} \leq \varepsilon$. $\square$

Theorem 479. *(Saturation for append-only steps.) Let $|\mathcal{A}| = m \geq 2$. For $\varepsilon > 0$ let $\ell(\varepsilon) := \lceil \log(1/\varepsilon) / \log(1/\theta) \rceil$. Then for all $k \geq 0$,*

$$N^{\text{ext}}(B(k, F_w), d_\infty, \varepsilon) \leq m^{\min\{k, \ell(\varepsilon)\}+1}.$$

Proof. The cover of ‘lem:cot-append-cover’ has at most $\sum_{j \leq \min\{k, \ell(\varepsilon)\}} m^j \leq m^{\min\{k, \ell(\varepsilon)\}+1}$ elements (‘lem:wordball-card’). $\square$

Theorem 480. *The same bound for the internal covering number in d_∞ and in d_S : $N(B(k, F_w), d_S, \varepsilon) \leq N(B(k, F_w), d_\infty, \varepsilon) \leq m^{\min\{k, \ell(\varepsilon)\}+1}$ (the cover consists of elements of $B(k, F_w)$).*

Proof. ‘lem:covering-empSpace-le-unifMaps-internal’, then the internal cover $B(\min\{k, \ell(\varepsilon)\}, F_w) \subseteq B(k, F_w)$. $\square$

Theorem 481. *For $0 < \varepsilon \leq 1$ and $m \geq 2$, $\sqrt{\log N(B(k, F_w), d_S, \varepsilon)} \leq \sqrt{\log m} \left(\frac{\sqrt{\log(1/\varepsilon)}}{\sqrt{\log(1/\theta)}} + \sqrt{2} \right)$, using $\ell(\varepsilon) + 1 \leq \log(1/\varepsilon) / \log(1/\theta) + 2$ and $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$.*

Proof. $\log N \leq (\ell(\varepsilon) + 1) \log m \leq (\log(1/\varepsilon) / \log(1/\theta) + 2) \log m$; take square roots. $\square$

Theorem 482. *(Saturated variance profile for append-only steps.) Let $|\mathcal{A}| = m \geq 2$. For every sample S and every $k \geq 0$,*

$$\mathbb{V}_k(S) = \int_0^{D_k(S)} \sqrt{\log N(B(k, F_w), d_S, \varepsilon)} d\varepsilon \leq \sqrt{\log m} \left(\frac{\sqrt{\pi}}{2\sqrt{\log(1/\theta)}} + \sqrt{2} \right) =: \mathbb{V}_\infty.$$

(Case (i) of ‘prop:profiles’: the entropy integral does not depend on the depth k .)

Proof. Since $D_k(S) \leq 1$, dominate the integrand on $(0, 1]$ by ‘lem:cot-append-entropy-pointwise’ and use $\int_0^1 \sqrt{\log(1/\varepsilon)} d\varepsilon = \sqrt{\pi}/2$ (‘lem:log-split’). $\square$

Definition 483. The alphabet $\mathcal{A} = [r] \cup \{\bullet\}$ with r active symbols and one padding symbol $\bullet$.

Definition 484. The shift $\sigma(x) = (x_1, x_2, \dots)$.

Definition 485. The constant sequence $\bar{a} = (a, a, \dots)$.

Definition 486. A guarded step f_a , $a \in [r]$, reads the current symbol: if it equals a the step consumes it, $f_a(x) = \sigma(x)$; otherwise it resets to the constant error state $\bar{a}$.

Definition 487. The branching step family $F_b := \{f_a : a \in [r]\}$.

Theorem 488. $\sigma(g_a(y)) = y$.

Proof. $\square$

Theorem 489. $\sigma(\bar{a}) = \bar{a}$.

Proof. □

Theorem 490. *If $x_0 \neq a$ then $f_a(x) = \bar{a}$.*

Proof. □

Theorem 491. *If $x_0 = a$ then $f_a(x) = \sigma(x)$.*

Proof. □

Theorem 492. *$f_a(\bar{a}) = \bar{a}$ and $f_a(\bar{b}) = \bar{a}$ for $b \neq a$: the anchors are mapped to anchors.*

Proof. □

Theorem 493. *$a \mapsto f_a$ is injective, so $|F_b| = r$.*

Proof. $f_a(\bar{a}) = \bar{a}$ while $f_b(\bar{a}) = \bar{b}$. □

Theorem 494. *$|F_b| = r$.*

Proof. □

Theorem 495. *(The branching family is a ping-pong family, ‘ex:e2-pingpong-subshift’.) With chambers $U_a = V_a = \{x : x_0 = a\}$, anchors $\bar{a}$, marker $\bar{\bullet}$ and $\alpha = 1$, F_b satisfies the hypotheses of ‘cond:e2-pingpong’ (the chambers are disjoint; they are in fact 1-separated, which the paper’s form of E2 requires). Consequently, for $r \geq 2$, $N^{\text{ext}}(B(k, F_b), d_\infty, \varepsilon) \geq r^k$ for every k and every $\varepsilon < 1/2$, and $u \mapsto f_u$ is injective on $[r]^k$.*

Proof. Verify the four ping-pong conditions: (1) sequences in different chambers differ at index 0, so the chambers are disjoint; (2) $y = f_a(g_a(y))$ with $g_a(y) \in V_a$; (3) the reset is the definition of f_a off U_a , and anchors go to anchors (‘lem:cot-guarded-const’); (4) $\bar{\bullet}_0 = \bullet \neq a = \bar{a}_0$, so $d(\bar{\bullet}, \bar{a}) = 1$. □

Theorem 496. *(Exponential growth for branching steps.) For $r \geq 2$, every $k \geq 0$ and every $\varepsilon < 1/2$,*

$$r^k \leq N^{\text{ext}}(B(k, F_b), d_\infty, \varepsilon) \leq r^{k+1}.$$

The lower bound is the ping-pong bound ‘cond:e2-pingpong’; the upper bound holds because F_b is finite, so $|B(k, F_b)| \leq \sum_{j \leq k} r^j \leq r^{k+1}$.

Proof. □

Theorem 497. *For $r \geq 2$, every sample S and every k , $V_k(S) \leq \sqrt{(k+1) \log r}$ (case (iii) of ‘prop:profiles’, with $\bar{D} = \text{diam}(\mathcal{A}^{\mathbb{N}}) = 1$).*

Proof. ‘prop:profiles-finite’ with $|B(k, F_b)| \leq r^{k+1}$ and $D_k(S) \leq 1$. □

Definition 498. For $L \geq 1$ the window feature map $\Phi_L : \mathcal{A}^{\mathbb{N}} \rightarrow \mathbb{R}^{\mathcal{A}^L}$ is the one-hot encoding of the window $(x_0, \dots, x_{L-1})$.

Theorem 499. *$\|\Phi_L(x)\| = 1$ (in particular $\|\Phi_L\| \leq 1$).*

Proof. □

Theorem 500. *If x, y agree on the first L symbols then $\Phi_L(x) = \Phi_L(y)$.*

Proof. □

Theorem 501. *Distinct one-hot vectors are at Euclidean distance $\sqrt{2}$: $\|e_u - e_v\| = \sqrt{2}$ for $u \neq v$.*

Proof. $\|e_u - e_v\|^2 = \|e_u\|^2 - 2\langle e_u, e_v \rangle + \|e_v\|^2 = 1 - 0 + 1.$ $\square$

Theorem 502. *(Window output features.) Φ_L is $\sqrt{2}\theta^{1-L}$ -Lipschitz with respect to d_θ , and $\|\Phi_L\| \leq 1$. (If $d_\theta(x, y) \leq \theta^L$ the first L symbols agree and $\Phi_L(x) = \Phi_L(y)$; otherwise $d_\theta(x, y) \geq \theta^{L-1}$ and $\|\Phi_L(x) - \Phi_L(y)\| = \sqrt{2} \leq \sqrt{2}\theta^{1-L}d_\theta(x, y)$.)*

Proof. Case analysis on whether the windows agree. $\square$

Theorem 503. *Consequently (via ‘prop:hilbert-sg’) the sub-Gaussian increment condition holds for the linear readout class $H_L = \{x \mapsto \langle w, \Phi_L(x) \rangle : \|w\| \leq 1\}$ and every hidden-layer class on $\mathcal{A}^\mathbb{N}$, with $A_H = 1$ and $L = \sqrt{2}\theta^{1-L}$.*

Proof. $\square$

Definition 504. The depth-independent entropy integral of ‘lem:cot-append-profile’: $V_\infty(m, \theta) := \sqrt{\log m} \left(\frac{\sqrt{\pi}}{2\sqrt{\log(1/\theta)}} + \sqrt{2} \right).$

Theorem 505. $V_\infty(m, \theta) \geq 0.$

Proof. $\square$

Definition 506. The teacher–student target class of arbitrarily long programs read out linearly: $\mathcal{C} := \overline{H_R(\Phi) \circ \langle F_w \rangle}^{d_\infty}$, the uniform closure of the readouts of all finite compositions of write steps.

Theorem 507. *Every element of $B(k, F_w)$ and of $\langle F_w \rangle$ is 1-Lipschitz, hence continuous.*

Proof. $\square$

Theorem 508. *(Measurability.) For a continuous feature map Φ (with the Borel σ -algebra on $\mathcal{A}^\mathbb{N}$), every element of $\mathcal{H}_k = H_R(\Phi) \circ B(k, F_w)$ is measurable.*

Proof. $\square$

Theorem 509. *Every element of the target class $\mathcal{C}$ is measurable (a uniform limit of continuous functions).*

Proof. $\square$

Theorem 510. $\mathcal{C} \neq \emptyset$ (it contains the zero readout) when $R \geq 0.$

Proof. $\square$

Theorem 511. *(Pointwise boundedness.) If $\|\Phi\| \leq M_\Phi$ and $R \geq 0$ then $|g(x)| \leq RM_\Phi$ for every $g \in \mathcal{H}_k$.*

Proof. $\square$

Theorem 512. *(Sup-norm separability.) For a separable feature space, a Lipschitz feature map Φ with $\|\Phi\| \leq M_\Phi$ and $R \geq 0$, the class $\mathcal{H}_k = H_R(\Phi) \circ B(k, F_w)$ is sup-norm separable ($B(k, F_w)$ is finite and the ball of radius R has a countable dense subset).*

Proof. $\square$

Theorem 513. *(Rademacher complexity of the readout class, ‘lem:cot-output’.) If $\|\Phi\| \leq M_\Phi$ then $\hat{\mathfrak{R}}_S(H_R(\Phi)) \leq RM_\Phi/\sqrt{n}$ for every sample S (for the window features, $M_\Phi = 1$).*

Proof. □

Theorem 514. (*Estimation term for append-only steps.*) Let $|\mathcal{A}| = m \geq 2$, Φ be L_Φ -Lipschitz with $\|\Phi\| \leq M_\Phi$ and $R > 0$. Then for every sample S of size $n \geq 1$ and every depth k ,

$$\hat{\mathcal{R}}_S(\mathcal{H}_k) \leq \frac{RM_\Phi}{\sqrt{n}} + \frac{12 \cdot 1 \cdot RL_\Phi}{\sqrt{n}} V_\infty(m, \theta)$$

(‘thm:hidden-decomp-depth’ with $A_H = 1$, ‘lem:cot-append-profile’ and ‘lem:cot-readout-rademacher’).

Proof. □

Theorem 515. (*Bias for append-only steps.*) Let Φ be L_Φ -Lipschitz and $R \geq 0$. Against the teacher–student class $\mathcal{C} = \overline{H_R(\Phi) \circ \langle F_w \rangle}^{d_\infty}$,

$$\text{bias}(k) = \varepsilon_{\text{model}}(k) \leq \beta_\ell RL_\Phi \theta^k :$$

for a teacher $c = h_w \circ g_u \circ g_v$ with $|u| = k$, $|c(x) - h_w(g_u(x))| \leq RL_\Phi d_\theta(g_u(g_v x), g_u x) \leq RL_\Phi \theta^k$ since $\text{lip}(g_u) = \theta^k$ and $\text{diam} = 1$, and the same bound holds on the uniform closure; conclude with ‘lem:approx-transfer’.

Proof. □

Theorem 516. (*Append-only scratchpad, rigorous form.*) Let $|\mathcal{A}| = m \geq 2$, let $\Phi : \mathcal{A}^{\mathbb{N}} \rightarrow \mathcal{H}$ be an L_Φ -Lipschitz feature map into a separable Hilbert space with $\|\Phi\| \leq M_\Phi$ (e.g. the window features Φ_L), $R > 0$, $H = H_R(\Phi)$, $F = F_w$, and let the target class be $\mathcal{C} = \overline{H \circ \langle F_w \rangle}^{d_\infty}$. For a measurable loss $\ell : \mathbb{R} \times \mathcal{Y} \rightarrow [0, b]$, β_ℓ -Lipschitz in its first argument, $n \geq 1$, $\eta \geq 0$ and $\delta \in (0, 1)$: with probability at least $1 - \delta$ over $\mathcal{D} \sim P^{\otimes n}$, every η -empirical minimizer $\hat{h} \in \mathcal{H}_k = H \circ B(k, F_w)$ satisfies

$$L[\hat{h}] - \inf_{\mathcal{C}} L \leq \beta_\ell RL_\Phi \theta^k + \eta + \text{dev}_{\ell, n, \delta} \left(\frac{RM_\Phi}{\sqrt{n}} + \frac{12RL_\Phi}{\sqrt{n}} V_\infty(m, \theta) \right),$$

where $\text{dev}_{\ell, n, \delta}(B) = 4\beta_\ell B + 6b\sqrt{2\log(4/\delta)/n}$ is the estimation-plus-deviation term of ‘thm:bv’ (‘def:bv-dev’). This is the EL regime with $\alpha = \log(1/\theta)$ and saturated variance. (The implementation map is the identity, $\varepsilon_{\text{imp}} = 0$.)

Proof. □

Theorem 517. (*EL balancing with saturated variance.*) For $n \geq 1$ the depth $k^* := \lceil \log n / (2\log(1/\theta)) \rceil = \frac{\log n}{2\log(1/\theta)} + O(1)$ satisfies $\theta^{k^*} \leq n^{-1/2}$, so the bias term matches the saturated variance $n^{-1/2}$.

Proof. □

Theorem 518. (*Balanced value $O(n^{-1/2})$.*) Under the hypotheses of ‘prop:cot-append’, at the depth $k^* = \lceil \log n / (2\log(1/\theta)) \rceil$ the bound reads

$$L[\hat{h}] - \inf_{\mathcal{C}} L \leq \frac{\beta_\ell RL_\Phi}{\sqrt{n}} + \eta + \text{dev}_{\ell, n, \delta} \left(\frac{RM_\Phi + 12RL_\Phi V_\infty(m, \theta)}{\sqrt{n}} \right)$$

with probability at least $1 - \delta$: every term is of order $n^{-1/2}$ (up to the $\sqrt{\log(1/\delta)}$ factor of the deviation term), so the balanced value is $O(n^{-1/2})$.

Proof. □

7.5 Unrolled fixed-point iterations and ODE solvers

Theorem 519. (*Saturated profile from total boundedness.*) Let $\mathcal{X}$ be compact, let $G \subseteq \mathcal{X}^{\mathcal{X}}$ be totally bounded in d_{∞} and $A_k \subseteq G$ for all k . With $N_{\infty}(\varepsilon) := N(\overline{G}, d_{\infty}, \varepsilon/2) < \infty$ and $\overline{D} := \text{diam}(\mathcal{X})$, if $\int_0^{\overline{D}} \sqrt{\log N_{\infty}(\varepsilon)} d\varepsilon < \infty$ then $V(\text{diam}_S(A_k), A_k) \leq \int_0^{\overline{D}} \sqrt{\log N_{\infty}(\varepsilon)} d\varepsilon$ for every sample S and every k (case (i) of ‘prop:profiles’).

Proof. ‘prop:profiles-i’ with $N(A_k, d_S, \varepsilon) \leq N(A_k, d_{\infty}, \varepsilon) \leq N(\overline{G}, d_{\infty}, \varepsilon/2)$ (‘lem:covering-empSpace-le-unifMaps-internal’ and Mathlib’s ‘coveringNumber_subset_le’), which is finite since $\overline{G}$ is totally bounded, and $D_k(S) \leq \text{diam}(\mathcal{X})$. $\square$

Theorem 520. (*Integrability of the entropy integrand from total boundedness.*) Under the hypotheses of ‘lem:ode-profile-saturation-generic’, the entropy integrand $\varepsilon \mapsto \sqrt{\log N(A_k, d_S, \varepsilon)}$ is interval-integrable on $[0, \text{diam}_S(A_k)]$ (it is dominated by the integrable majorant).

Proof. $\square$

Definition 521. A map $\Pi_K : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a (Euclidean) projection onto K if $\Pi_K(x) \in K$ for all x , $\Pi_K(x) = x$ for $x \in K$, and Π_K is 1-Lipschitz. (For a nonempty closed convex K the nearest-point map has these properties; Mathlib provides only the existence of nearest points, so we take the projection and its properties as hypotheses.)

Definition 522. The projected step of size h along a vector field $s : K \rightarrow \mathbb{R}^d$: $T_s(x) := \Pi_K(x + h s(x))$, a self-map of K .

Theorem 523. Since Π_K is 1-Lipschitz, $\|T_s(x) - T_s(y)\| \leq \|(x - y) + h(s(x) - s(y))\|$.

Proof. $\square$

Theorem 524. A step of size 0 is the identity: $T_{s,0} = \text{id}$.

Proof. $\square$

Definition 525. A vector field $s : K \rightarrow \mathbb{R}^d$ is μ -strongly monotone and Λ -co-coercive (the properties of $s = \nabla \phi$ for ϕ μ -strongly concave and Λ -smooth) if for all $x, y \in K$

$$\langle x - y, s(x) - s(y) \rangle \leq -\mu \|x - y\|^2, \quad \langle x - y, s(x) - s(y) \rangle \leq -\frac{\mu \Lambda}{\mu + \Lambda} \|x - y\|^2 - \frac{1}{\mu + \Lambda} \|s(x) - s(y)\|^2.$$

(The first is the gradient characterization of strong concavity; the second is the standard consequence of strong concavity and smoothness. We take both as the definition.)

Theorem 526. (*Contraction of projected gradient steps.*) Let $0 < \mu \leq \Lambda$, let s be μ -strongly monotone and Λ -co-coercive, and let $0 < h \leq 2/(\mu + \Lambda)$. Then $T_s = \Pi_K(\cdot + h s(\cdot))$ is λ -Lipschitz on K with $\lambda := 1 - h\mu \in [0, 1)$.

Proof. Write $u = x - y$, $v = s(x) - s(y)$, $a = \|u\|$, $b = \|v\|$. Strong monotonicity and Cauchy-Schwarz give $\mu a \leq b$. Then $\|u + hv\|^2 = a^2 + 2h\langle u, v \rangle + h^2 b^2 \leq a^2(1 - \frac{2h\mu\Lambda}{\mu + \Lambda}) + (h^2 - \frac{2h}{\mu + \Lambda})b^2$ and, since $h^2 - 2h/(\mu + \Lambda) \leq 0$ and $b^2 \geq \mu^2 a^2$, this is at most $a^2(1 - \frac{2h\mu\Lambda}{\mu + \Lambda}) + (h^2 - \frac{2h}{\mu + \Lambda})\mu^2 a^2 = (1 - h\mu)^2 a^2$. Finally Π_K is 1-Lipschitz. $\square$

Theorem 527. (*Stability of one step.*) If s is Λ_s -Lipschitz and $h \geq 0$ then $x \mapsto \Pi_K(x + h s(x))$ is $(1 + h\Lambda_s)$ -Lipschitz.

Proof. $\|(x - y) + h(s(x) - s(y))\| \leq \|x - y\| + h\Lambda_s\|x - y\|.$ $\square$

Definition 528. The fixed-point refinement class $F_{\text{fp}} := \{T_s : x \mapsto \Pi_K(x + hs(x)) : s \in \mathcal{S}\}$ for a class $\mathcal{S}$ of vector fields on K and a fixed step size h .

Theorem 529. Every $T \in F_{\text{fp}}$ is $(1 - h\mu)$ -Lipschitz, under the hypotheses of ‘lem:fp-contraction’ for every $s \in \mathcal{S}$.

Proof. $\square$

Theorem 530. Every $T \in F_{\text{fp}}$ is non-expanding: $\text{lip } F_{\text{fp}} \leq 1 - h\mu \leq 1$.

Proof. $\square$

Theorem 531. (*Saturation for fixed-point refinement, ‘cond:p1.’*) If K is compact then for every $\varepsilon > 0$ and every k , $N^{\text{ext}}(B(k, F_{\text{fp}}), d_\infty, \varepsilon) \leq N^{\text{ext}}(\overline{\langle F_{\text{fp}} \rangle}, d_\infty, \varepsilon) < \infty$; in particular $\sup_k N^{\text{ext}}(B(k, F_{\text{fp}}), d_\infty, \varepsilon) < \infty$.

Proof. ‘cond:p1-2c’ on the compact state space K with non-expanding generators. $\square$

Theorem 532. (*Explicit entropy bound via ‘cond:p1-ucont.’*) Assume moreover $h\mu < 1$ and $K \neq \emptyset$, and let $\lambda = 1 - h\mu$, $m(\varepsilon) = \lceil \log_{1/\lambda}(2D_K/\varepsilon) \rceil$ with $D_K = \text{diam}(K)$. Then for every $\varepsilon > 0$ and every k (generalizing the paper, which assumes $k \geq m(\varepsilon)$; see ‘cond:p1-ucont’),

$$N^{\text{ext}}(B(k, F_{\text{fp}}), d_\infty, \varepsilon) \leq N^{\text{ext}}(K, \varepsilon/2) + \sum_{j < m(\varepsilon)} N^{\text{ext}}(F_{\text{fp}}^j, d_\infty, \varepsilon).$$

(In Lean the absorbing set is K itself, with $L = 0$.)

Proof. ‘cond:p1-ucont’ with $c = 1 - h\mu \in (0, 1)$, invariant set $A = K$ and bounded absorbing set K (words of length $L = 0$). $\square$

Theorem 533. (*Saturated variance profile for fixed-point refinement.*) Under the hypotheses of ‘lem:fp-saturation’, with $N_\infty(\varepsilon) := N(\overline{\langle F_{\text{fp}} \rangle}, d_\infty, \varepsilon/2) < \infty$ and $D_K = \text{diam}(K)$: if $\int_0^{D_K} \sqrt{\log N_\infty(\varepsilon)} d\varepsilon < \infty$ then $\mathbb{V}_k(S) \leq \mathbb{V}_\infty := \int_0^{D_K} \sqrt{\log N_\infty(\varepsilon)} d\varepsilon$ for all k and all samples S (case (i) of ‘prop:profiles’).

Proof. The non-expanding semigroup is equicontinuous, hence totally bounded in d_∞ (‘thm:caa’); apply ‘lem:ode-profile-saturation-generic’. $\square$

Definition 534. An explicit Euler layer of step size h at time stamp τ for a time-dependent vector field $s : K \times [0, T] \rightarrow \mathbb{R}^d$: $T_{s,h,\tau}(x) := \Pi_K(x + hs(x, \tau))$.

Definition 535. The Euler layer class $F_T := \{T_{s,h,\tau} : s \in \mathcal{S}_T, h \in [0, h_0], \tau \in [0, T]\}$.

Theorem 536. Each Euler layer with $h \geq 0$ and Λ_s -Lipschitz drift $s(\cdot, \tau)$ is $(1 + h\Lambda_s)$ -Lipschitz.

Proof. $\square$

Definition 537. A scheme of n steps with a single drift s , step sizes $h = (h_0, \dots, h_{n-1})$ and time stamps $\tau = (\tau_0, \dots, \tau_{n-1})$ is the composition $T_{s,h_{n-1},\tau_{n-1}} \circ \dots \circ T_{s,h_0,\tau_0}$ (the empty scheme is id).

Theorem 538. A scheme of n steps with drift $s \in \mathcal{S}_T$, step sizes in $[0, h_0]$ and time stamps in $[0, T]$ belongs to $B(n, F_T)$.

Proof. □

Theorem 539. (*Stability of schemes.*) If $s(\cdot, \tau)$ is Λ_s -Lipschitz for every τ and $h_i \geq 0$, then the scheme with steps $h_0, \dots, h_{n-1}$ is $\prod_i (1 + h_i \Lambda_s) \leq \exp(\Lambda_s \sum_i h_i)$ -Lipschitz.

Proof. □

Definition 540. The class $B_T(k)$ of schemes of at most k steps with a single drift $s \in \mathcal{S}_T$, step sizes $h_i \in [0, h_0]$, time stamps $\tau_i \in [0, T]$ and total time $\sum_i h_i \leq T$ (it contains id , the empty scheme). The consistency condition $\tau_i = \sum_{j < i} h_j$ of the paper is dropped; the bounds below hold for this larger class.

Definition 541. All schemes: $\bigcup_k B_T(k)$.

Theorem 542. $B_T(k) \subseteq B_T(k+1)$: the scheme classes are nested.

Proof. □

Theorem 543. $B_T(k) \subseteq B(k, F_T)$.

Proof. □

Theorem 544. Every scheme in $\bigcup_k B_T(k)$ is $e^{\Lambda_s T}$ -Lipschitz on K .

Proof. ‘lem:ode-scheme-lipschitz’ and $\sum_i h_i \leq T$. □

Theorem 545. (*Stability and saturation.*) Let K be compact and let every drift $s \in \mathcal{S}_T$ be Λ_s -Lipschitz in x . Every scheme in $\bigcup_k B_T(k)$ is $e^{\Lambda_s T}$ -Lipschitz on K . Consequently $\bigcup_k B_T(k)$ is equicontinuous on the compact set K , hence totally bounded in d_∞ (‘thm:caa’), and

$$\sup_k N^{\text{ext}}(B_T(k), d_\infty, \varepsilon) \leq N^{\text{ext}}\left(\bigcup_k B_T(k), d_\infty, \varepsilon\right) =: N_\infty(\varepsilon) < \infty \quad (\varepsilon > 0).$$

Proof. A uniformly Lipschitz family is (uniformly) equicontinuous; Arzelà–Ascoli (‘thm:caa-of-equicontinuous’) gives total boundedness in d_∞ , monotonicity of the external covering number and ‘lem:aa-external-covering-ne-top’ the rest. □

Theorem 546. (*Saturated variance profile for Euler schemes.*) Under the hypotheses of ‘lem:ode-saturation’, with $N_\infty(\varepsilon) := N(\bigcup_k B_T(k), d_\infty, \varepsilon/2) < \infty$ and $D_K = \text{diam}(K)$: if $\int_0^{D_K} \sqrt{\log N_\infty(\varepsilon)} d\varepsilon < \infty$ then $V_k(S) \leq V_\infty := \int_0^{D_K} \sqrt{\log N_\infty(\varepsilon)} d\varepsilon$ for all k and all samples S (case (i) of ‘prop:profiles’).

Proof. ‘lem:ode-profile-saturation-generic’ with $G = \bigcup_k B_T(k)$. □

Definition 547. The identity feature map $K \hookrightarrow \mathbb{R}^d$, so that $H_R = \{x \mapsto \langle w, x \rangle : \|w\| \leq R\}$ is the linear readout class $H_R(\Phi)$ with $\Phi = \text{id}_K$.

Theorem 548. id_K is 1-Lipschitz and continuous.

Proof. □

Definition 549. The teacher–student target class of the fixed-point example: $\mathcal{C} := \overline{H_R \circ \langle F_{\text{fp}} \rangle}^{d_\infty}$, which contains the readouts $x \mapsto \langle w, x_s^* \rangle$ of the fixed points (limits of the refinement).

Theorem 550. Under the step-size condition $0 < h \leq 2/(\mu + \Lambda)$ with $0 < \mu \leq \Lambda$, $0 \leq 1 - h\mu$.

Proof. □

Theorem 551. *Every element of $\langle F_{\text{fp}} \rangle$ is 1-Lipschitz, hence continuous.*

Proof. □

Theorem 552. *For compact K , $\langle F_{\text{fp}} \rangle$ is totally bounded in d_∞ (Arzelà–Ascoli for the non-expanding semigroup).*

Proof. □

Theorem 553. (*Estimation term for fixed-point refinement.*) *Let K be compact with $\|x\| \leq M_K$ on K , $R > 0$, and assume $V_\infty(F_{\text{fp}}) < \infty$ (‘def:sat-integrable’). Then for every sample S of size $n \geq 1$ and every depth k ,*

$$\hat{\mathfrak{R}}_S(H_R \circ B(k, F_{\text{fp}})) \leq \frac{RM_K}{\sqrt{n}} + \frac{12 \cdot 1 \cdot R}{\sqrt{n}} V_\infty(F_{\text{fp}})$$

(‘cor:var-profiles-p1’ with $A_H = 1$, $L = R$, and ‘lem:linear-readouts-rademacher’).

Proof. □

Theorem 554. *Every element of $H_R \circ B(k, F_{\text{fp}})$ is Borel measurable.*

Proof. □

Theorem 555. *Every element of the target class $\mathcal{C}$ is measurable.*

Proof. □

Theorem 556. $\mathcal{C} \neq \emptyset$ for $R \geq 0$.

Proof. □

Theorem 557. (*Bias for fixed-point refinement.*) *Let K be compact with $D_K = \text{diam}(K)$ and $R \geq 0$. Against the teacher–student class $\mathcal{C} = \overline{H_R \circ \langle F_{\text{fp}} \rangle}^{d_\infty}$,*

$$\text{bias}(k) = \varepsilon_{\text{model}}(k) \leq \beta_\ell R D_K (1 - h\mu)^k$$

by the truncation argument ‘lem:truncation-bias’ ($\text{lip}(h) \leq R$, $\text{lip}(u) \leq (1 - h\mu)^k$ for $u \in F_{\text{fp}}^k$) and ‘lem:approx-transfer’.

Proof. □

Theorem 558. (*Fixed-point refinement, rigorous form.*) *Let $K \subseteq \mathbb{R}^d$ (a separable Hilbert space) be compact with $\|x\| \leq M_K$ on K and $D_K = \text{diam}(K)$, let F_{fp} be the projected gradient steps of ‘lem:fp-contraction’ with $0 < h \leq 2/(\mu + \Lambda)$, $H = H_R$ ($R > 0$), and let the target class be $\mathcal{C} = \overline{H_R \circ \langle F_{\text{fp}} \rangle}^{d_\infty}$. Assume $V_\infty(F_{\text{fp}}) < \infty$. For a measurable loss $\ell : \mathbb{R} \times \mathcal{Y} \rightarrow [0, b]$, β_ℓ -Lipschitz in its first argument, $n \geq 1$, $\eta \geq 0$, $\delta \in (0, 1)$: with probability at least $1 - \delta$ over $\mathcal{D} \sim P^{\otimes n}$, every η -empirical minimizer $\hat{h} \in H_R \circ B(k, F_{\text{fp}})$ satisfies*

$$L[\hat{h}] - \inf_{\mathcal{C}} L \leq \beta_\ell R D_K (1 - h\mu)^k + \eta + \text{dev}_{\ell, n, \delta} \left(\frac{RM_K}{\sqrt{n}} + \frac{12R}{\sqrt{n}} V_\infty(F_{\text{fp}}) \right),$$

with $\text{dev}_{\ell, n, \delta}(B) = 4\beta_\ell B + 6b\sqrt{2\log(4/\delta)/n}$ (‘def:bv-dev’): the EL regime with $\alpha = \log(1/(1 - h\mu)) \geq h\mu$ and saturated variance.

Proof. □

Theorem 559. (*EL balancing for fixed-point refinement.*) If $h\mu < 1$ and $n \geq 1$, the depth $k^* := \lceil \log n / (2 \log(1/(1-h\mu))) \rceil = \frac{\log n}{2 \log(1/(1-h\mu))} + O(1) \leq \frac{\log n}{2h\mu} + O(1)$ satisfies $(1-h\mu)^{k^*} \leq n^{-1/2}$.

Proof. □

Theorem 560. (*Balanced value* $O(n^{-1/2})$.) Under the hypotheses of ‘prop:ode-fixedpoint’ with $h\mu < 1$, at the depth $k^* = \lceil \log n / (2 \log(1/(1-h\mu))) \rceil$,

$$L[\hat{h}] - \inf_{\mathcal{C}} L \leq \frac{\beta_{\ell} R D_K}{\sqrt{n}} + \eta + \text{dev}_{\ell, n, \delta} \left(\frac{R M_K + 12 R V_{\infty}(F_{\text{fp}})}{\sqrt{n}} \right)$$

with probability at least $1 - \delta$: the balanced value is of order $n^{-1/2}$.

Proof. □

Definition 561. The explicit Euler iterates on the ambient space with constant step h from time 0: $y_0 = x$, $y_{i+1} = y_i + h s(y_i, ih)$.

Theorem 562. (*One-step consistency of the Euler scheme.*) Let s be Λ_s -Lipschitz in x , Λ_{τ} -Lipschitz in τ and bounded by M_s , and let x solve $\dot{x}(t) = s(x(t), t)$ on $[t_0, t_0 + h]$ ($h \geq 0$). Then

$$\|x(t_0 + h) - x(t_0) - h s(x(t_0), t_0)\| \leq \frac{(\Lambda_s M_s + \Lambda_{\tau}) h^2}{2}.$$

(The function $g(t) = x(t) - x(t_0) - (t - t_0)s(x(t_0), t_0)$ has $\|g'(t)\| \leq \Lambda_s \|x(t) - x(t_0)\| + \Lambda_{\tau}(t - t_0) \leq (\Lambda_s M_s + \Lambda_{\tau})(t - t_0)$, and the mean value inequality with the quadratic boundary $B(t) = (\Lambda_s M_s + \Lambda_{\tau})(t - t_0)^2/2$ gives the claim.)

Proof. □

Theorem 563. (*Global error of the explicit Euler scheme.*) Let s be Λ_s -Lipschitz in x , Λ_{τ} -Lipschitz in τ and bounded by M_s , let x solve $\dot{x}(t) = s(x(t), t)$ on $[0, T]$ ($T \geq 0$), and let $y_0, \dots, y_k$ be the Euler iterates with $k \geq 1$ equal steps $h = T/k$ started at $y_0 = x(0)$. Then

$$\|x(T) - y_k\| \leq C_E \frac{T^2}{k}, \quad C_E := \frac{(\Lambda_s M_s + \Lambda_{\tau}) e^{\Lambda_s T}}{2}.$$

(Discrete Grönwall: $e_{i+1} \leq (1 + h\Lambda_s)e_i + ch^2/2$ with $c = \Lambda_s M_s + \Lambda_{\tau}$, hence $e_k \leq \frac{ch^2}{2} \sum_{j < k} (1 + h\Lambda_s)^j \leq \frac{ch^2}{2} k e^{\Lambda_s T}$.)

Proof. □

Definition 564. The equal-step scheme with k Euler steps of size $h = T/k$ and time stamps $\tau_i = ih$: $T_{s, h, \tau_{k-1}} \circ \dots \circ T_{s, h, \tau_0}$.

Theorem 565. For $s \in \mathcal{S}_T$, $T \geq 0$, $k \geq 1$ and $T/k \leq h_0$, the equal-step scheme with k steps belongs to $B_T(k)$.

Proof. □

Theorem 566. (*Inactive projection.*) Let $\tilde{s} : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}^d$ extend the drift s from K , and suppose K is invariant under the Euler steps, $x + h s(x, \tau) \in K$ for $x \in K$. Then the projection is inactive and the scheme with k equal steps of size h coincides with the Euler iterates: $T_{s, h, (k-1)h} \circ \dots \circ T_{s, h, 0}(x_0) = y_k(x_0)$.

Proof. □

Definition 567. The depth-independent entropy integral of ‘lem:ode-saturation-profile’: $V_\infty := \int_0^{D_K} \sqrt{\log N(\bigcup_k B_T(k), d_\infty, \varepsilon/2)} d\varepsilon$.

Definition 568. The paper’s condition $\int_0^{D_K} \sqrt{\log N_\infty(\varepsilon)} d\varepsilon < \infty$ for the scheme classes: the majorant is interval-integrable.

Theorem 569. (*Estimation term for fixed-horizon schemes.*) Let K be compact with $\|x\| \leq M_K$ on K , every drift in $\mathcal{S}_T \neq \emptyset$ be Λ_s -Lipschitz in x , $T \geq 0$, $R > 0$, and assume the entropy condition of ‘lem:ode-saturation-profile’. Then for every sample S of size $n \geq 1$ and every k ,

$$\hat{\mathfrak{R}}_S(H_R \circ B_T(k)) \leq \frac{RM_K}{\sqrt{n}} + \frac{12 \cdot 1 \cdot R}{\sqrt{n}} V_\infty$$

(‘thm:hidden-decomp’ on the scheme class, ‘lem:ode-saturation-profile’ and ‘lem:linear-readouts-rademacher’).

Proof. □

Theorem 570. Every scheme in $\bigcup_k B_T(k)$ is Lipschitz, hence continuous.

Proof. □

Theorem 571. Every element of $H_R \circ B_T(k)$ is Borel measurable.

Proof. □

Theorem 572. (*Bias for fixed-horizon integration.*) Let the target class be the readouts of the endpoint map of the exact flow of $s^* \in \mathcal{S}_T$: $\mathcal{C} = \{x_0 \mapsto \langle w, \Phi_T(x_0) \rangle : \|w\| \leq R\}$, where for every $x_0 \in K$ the flow $\Phi_T(x_0) = x(T)$ is given by a solution of $\dot{x} = \tilde{s}(x, t)$, $x(0) = x_0$, for an extension $\tilde{s}$ of s^* which is Λ_s -Lipschitz in x , Λ_τ -Lipschitz in τ and bounded by M_s . Assume the projection is inactive along the Euler steps of size T/k (K is invariant), $k \geq 1$ and $T/k \leq h_0$. Then

$$\text{bias}(k) = \varepsilon_{\text{model}}(k) \leq \beta_\ell R C_E \frac{T^2}{k}, \quad C_E = \frac{(\Lambda_s M_s + \Lambda_\tau) e^{\Lambda_s T}}{2},$$

by ‘lem:ode-euler-error’, ‘lem:ode-scheme-eq-euler’ and ‘lem:approx-transfer’.

Proof. □

Theorem 573. (*Fixed-horizon integration, rigorous form, $p = 1$.*) Let $K \subseteq \mathbb{R}^d$ (a separable Hilbert space) be compact with $\|x\| \leq M_K$ on K , let every drift in $\mathcal{S}_T$ be Λ_s -Lipschitz in x , $H = H_R$ ($R > 0$), and let the target class be the flow readouts $\mathcal{C} = \{\langle w, \Phi_T(\cdot) \rangle : \|w\| \leq R\}$ of a teacher $s^* \in \mathcal{S}_T$ as in ‘lem:ode-horizon-bias’ (projection inactive, $k \geq 1$, $T/k \leq h_0$). Assume the entropy condition $V_\infty < \infty$ of ‘lem:ode-saturation-profile’. For a measurable loss $\ell : \mathbb{R} \times \mathcal{Y} \rightarrow [0, b]$, β_ℓ -Lipschitz in its first argument, $n \geq 1$, $\eta \geq 0$, $\delta \in (0, 1)$: with probability at least $1 - \delta$ over $\mathcal{D} \sim P^{\otimes n}$, every η -empirical minimizer $\hat{h} \in H_R \circ B_T(k)$ satisfies

$$L[\hat{h}] - \inf_{\mathcal{C}} L \leq \beta_\ell R C_E \frac{T^2}{k} + \eta + \text{dev}_{\ell, n, \delta} \left(\frac{RM_K}{\sqrt{n}} + \frac{12R}{\sqrt{n}} V_\infty \right),$$

with $C_E = (\Lambda_s M_s + \Lambda_\tau) e^{\Lambda_s T} / 2$ and $\text{dev}_{\ell, n, \delta}(B) = 4\beta_\ell B + 6b\sqrt{2 \log(4/\delta)/n}$ (‘def:bv-dev’): the PL regime with $\beta = p = 1$ and saturated variance.

Proof.

□

Theorem 574. (*PL balancing with saturated variance, $p = 1$.*) For $n \geq 1$ the depth $k^* := \lceil \sqrt{n} \rceil \geq 1$ satisfies $1/k^* \leq n^{-1/2}$, so $k^* \asymp n^{1/2} = n^{1/(2p)}$ balances the bias k^{-1} against the saturated variance $n^{-1/2}$.

Proof.

□

Theorem 575. (*Balanced value $O(n^{-1/2})$.*) Under the hypotheses of ‘prop:ode-horizon’ with $k^* = \lceil \sqrt{n} \rceil$ (and $T/k^* \leq h_0$, projection inactive at step size T/k^*),

$$L[\hat{h}] - \inf_{\mathcal{C}} L \leq \frac{\beta_{\ell} RC_E T^2}{\sqrt{n}} + \eta + \text{dev}_{\ell, n, \delta} \left(\frac{RM_K + 12R\mathbf{V}_{\infty}}{\sqrt{n}} \right)$$

with probability at least $1 - \delta$: the balanced value is of order $n^{-1/2}$.

Proof.

□